

Solutions Manual for:

Linear State–Space Control Systems

Robert L. Williams II and Douglas A. Lawrence

Ohio University

submitted to Wiley December, 2006

Table of Contents

NUMERICAL EXERCISES SOLUTIONS	4
CHAPTER 1.....	4
CHAPTER 2.....	5
CHAPTER 3.....	8
CHAPTER 4.....	12
CHAPTER 5.....	16
CHAPTER 6.....	18
CHAPTER 7.....	22
CHAPTER 8.....	24
CHAPTER 9.....	26
ANALYTICAL EXERCISES SOLUTIONS.....	30
CHAPTER 1.....	30
CHAPTER 2.....	38
CHAPTER 3.....	46
CHAPTER 4.....	59
CHAPTER 5.....	66
CHAPTER 6.....	70
CHAPTER 7.....	75
CHAPTER 8.....	79
CHAPTER 9.....	87
CONTINUING MATLAB EXERCISES SOLUTIONS.....	91
CONTINUING MATLAB EXERCISE 1	91
CONTINUING MATLAB EXERCISE 2	95
CONTINUING MATLAB EXERCISE 3	99
CONTINUING MATLAB EXERCISE 4	103
CONTINUING EXERCISES SOLUTIONS	108
CONTINUING EXERCISE 1 – THREE-MASS TRANSLATIONAL MECHANICAL SYSTEM.....	109
<i>Open-Loop System</i>	109
<i>Open-Loop Response</i>	109
<i>Controllability & Observability</i>	110
<i>Canonical Realizations</i>	110
<i>Stability Analysis</i>	112
<i>Dynamic Shaping</i>	112
<i>Controller Design</i>	113
<i>Observer Design</i>	115
CONTINUING EXERCISE 2 – INVERTED PENDULUM	117
<i>Open-Loop System</i>	117
<i>Open-Loop Response</i>	117
<i>Controllability & Observability</i>	118
<i>Canonical Realizations</i>	118
<i>Stability Analysis</i>	119
<i>Dynamic Shaping</i>	120
<i>Controller Design</i>	120

<i>LQR Design</i>	121
CONTINUING EXERCISE 3 – ROBOT JOINT/LINK CONTROL.....	123
<i>Open-Loop System</i>	123
<i>Open-Loop Response</i>	124
<i>Controllability & Observability</i>	124
<i>Canonical Realizations</i>	124
<i>Stability Analysis</i>	125
<i>Dynamic Shaping</i>	126
<i>Controller Design</i>	126
<i>Observer Design</i>	127
CONTINUING EXERCISE 4 – BALL/BEAM SYSTEM.....	129
<i>Open-Loop System</i>	129
<i>Open-Loop Response</i>	130
<i>Controllability & Observability</i>	130
<i>Canonical Realizations</i>	130
<i>Stability Analysis</i>	131
<i>Dynamic Shaping</i>	132
<i>Controller Design</i>	133
<i>Observer Design</i>	133
<i>LQR Design</i>	134
CONTINUING EXERCISE 5 – PROOF-MASS ACTUATOR SYSTEM.....	136
<i>Open-Loop System</i>	136
<i>Open-Loop Response</i>	137
<i>Controllability & Observability</i>	137
<i>Canonical Realizations</i>	137
<i>Minimality</i>	138
<i>Stability Analysis</i>	138
<i>Dynamic Shaping</i>	139
<i>Controller Design</i>	140
<i>Observer Design</i>	141

Numerical Exercises Solutions

Chapter 1

NE1.1a

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad D = 0$$

NE1.1b

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 3 & 1 \end{bmatrix} \quad D = 0$$

NE1.1c

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -8 & -4 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 10 & 0 & 0 \end{bmatrix} \quad D = 0$$

NE1.1d

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -66 & -44 & -11 & -10 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 6 & 4 & 1 & 0 \end{bmatrix} \quad D = 0$$

NE1.2a

$$A = \begin{bmatrix} -2 \end{bmatrix} \quad B = \begin{bmatrix} 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 \end{bmatrix} \quad D = 0$$

NE1.2b

$$A = \begin{bmatrix} 0 & 1 \\ -10 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad D = 0$$

NE1.2c

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -3 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \quad D = 0$$

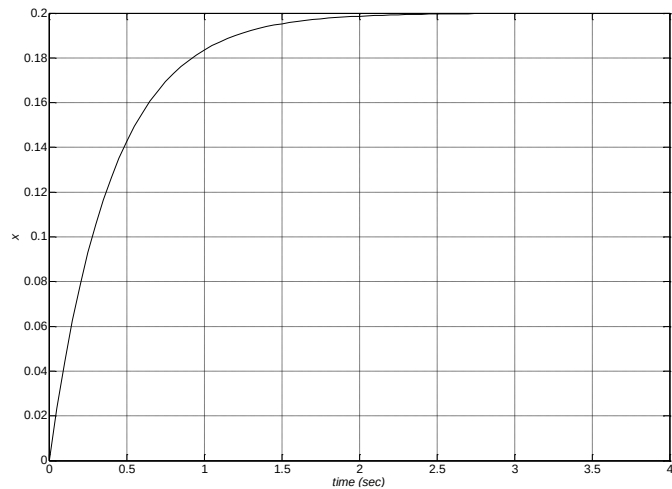
NE1.2d

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -10 & -5 & 10 & 0 \\ 0 & 0 & 0 & 1 \\ 5 & 0 & -5 & -0.5 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 0.5 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Chapter 2

NE2.1

$$x(t) = 1/5 - 1/5 \exp(-5/2 t) \quad \text{time constant } \tau = 2/5$$



NE2.2

a. Transfer function:

$$\frac{2s + 7}{s^2 + 7s + 12}$$

$$s^2 + 7s + 12$$

b. Transfer function:

$$\frac{1}{s^2 + 2s + 3}$$

$$s^2 + 2s + 3$$

c. Transfer function:

$$\frac{1}{s^2 + 12s + 2}$$

$$s^2 + 12s + 2$$

d. Transfer function:

$$\frac{9s^2 + 38s - 2}{s^2 - 5s - 2}$$

$$s^2 - 5s - 2$$

NE2.3

a. characteristic polynomial: [1 3 2]
eigenvalues: -2, -1

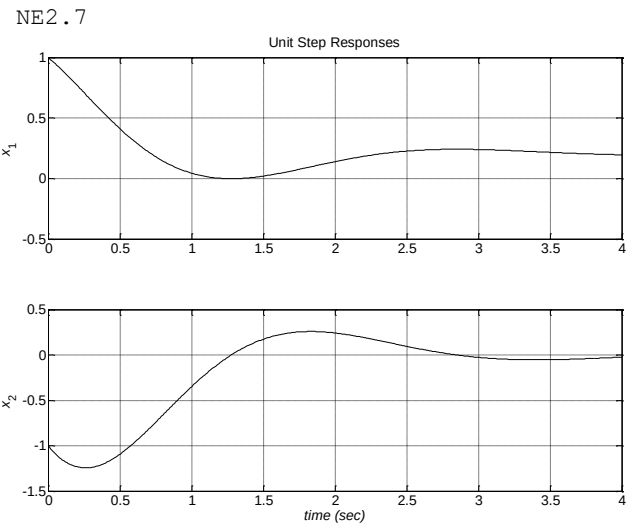
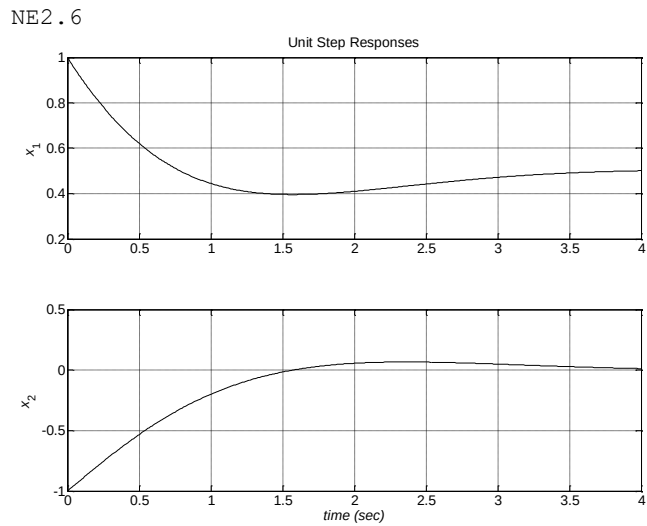
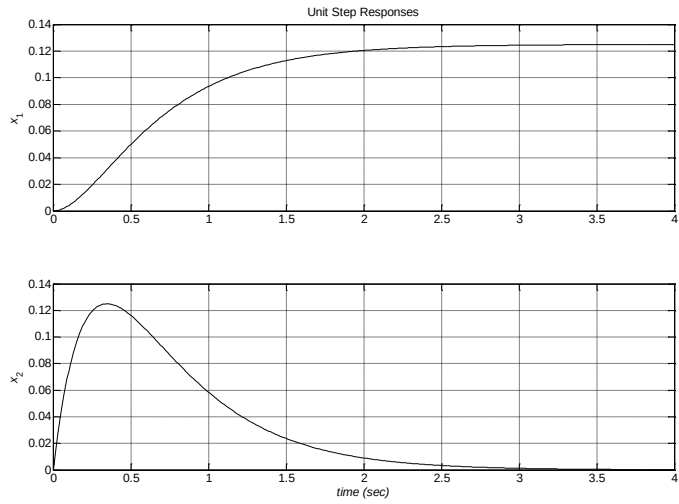
b. characteristic polynomial: [1 20 10]
eigenvalues: -0.5132, -19.4868

c. characteristic polynomial: [1 0 10]
eigenvalues: 0 + 3.1623i, 0 - 3.1623i

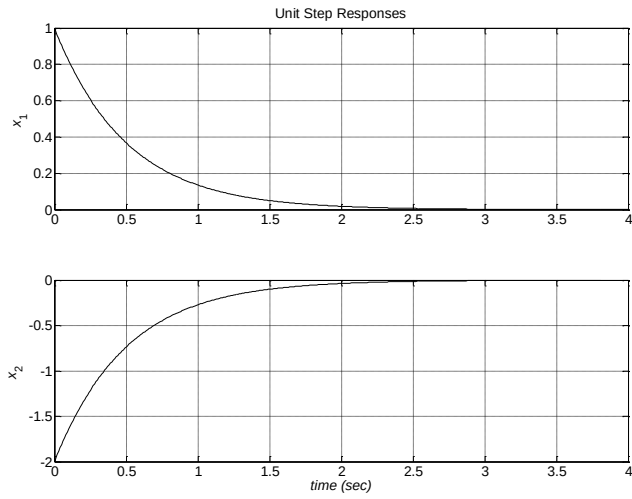
d. characteristic polynomial: [1 20 0]
eigenvalues: 0, -20

NE2.4
 $\phi(4) =$
 $\begin{bmatrix} 0.1294 & 0.0113 \\ -0.0677 & -0.0059 \end{bmatrix}$
 $x(4) =$
 $\begin{bmatrix} 0.2702 \\ -0.1412 \end{bmatrix}$

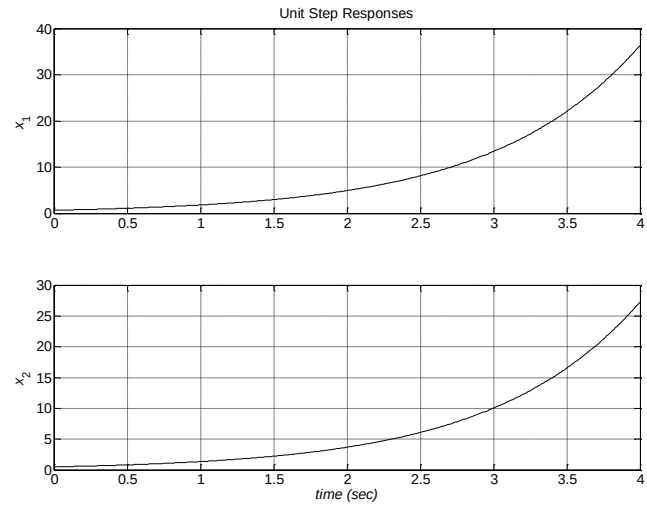
NE2.5
 $x_1(t) = 1/8 + 1/8 \exp(-4t) - 1/4 \exp(-2t)$



NE2.8



NE2.9



NE2.10

a. $A_d =$

$$\begin{bmatrix} -0.8769 & 0 \\ 0 & -9.1231 \end{bmatrix}$$

b. $A_d =$

$$\begin{bmatrix} -1.3589 & 0 \\ 0 & 7.3589 \end{bmatrix}$$

c. $A_d =$

$$\begin{bmatrix} -0.5000 + 3.1225i & 0 \\ 0 & -0.5000 - 3.1225i \end{bmatrix}$$

d. $A_d =$

$$\begin{bmatrix} 3.1623 & 0 \\ 0 & -3.1623 \end{bmatrix}$$

Chapter 3

NE3.1a

P =

$$\begin{pmatrix} 1 & -4 \\ 1 & -5 \end{pmatrix}$$

determinant of P -1

System is fully state-controllable

NE3.1b

P =

$$\begin{pmatrix} 1 & -4 \\ 0 & 0 \end{pmatrix}$$

determinant of P 0

System is NOT fully state-controllable

NE3.1c

P =

$$\begin{pmatrix} 1 & -20 \\ 2 & -3 \end{pmatrix}$$

determinant of P 37

System is fully state-controllable

NE3.1d

P =

$$\begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$$

determinant of P -1

System is fully state-controllable

NE3.1e

P =

$$\begin{pmatrix} 1 & 2 \\ -1 & -2 \end{pmatrix}$$

determinant of P 0

System is NOT fully state-controllable

NE3.2a

Ac =

$$\begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix}$$
 Bc =

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
 Cc =

$$\begin{bmatrix} 9 & 2 \end{bmatrix}$$

NE3.2b

We cannot find CCF from the formula since the transformation matrix is singular.

NE3.2c

Ac =

$$\begin{bmatrix} 0.0000 & 1.0000 \\ -10.0000 & -2.0000 \end{bmatrix}$$
 Bc =

$$\begin{bmatrix} 0.0000 \\ 1.0000 \end{bmatrix}$$
 Cc =

$$\begin{bmatrix} 1 & 2 \end{bmatrix}$$

NE3.2d

Ac =

$$\begin{bmatrix} 0.0000 & 1.0000 \\ -10.0000 & -2.0000 \end{bmatrix}$$
 Bc =

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
 Cc =

$$\begin{bmatrix} 1.0000 & 2.0000 \end{bmatrix}$$

NE3.2e

We cannot find CCF from the formula since the transformation matrix is singular.

NE3.3

Apply the Popov-Belevitch-Hautus Rank Test for Controllability

NE3.3a

The eigenvalues of A are: -5 and -4

For the eigenvalue -5

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

has rank 2

For the eigenvalue -4

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

has rank 2

System is fully state-controllable

NE3.3b

The eigenvalues of A are: -5 and -4

For the eigenvalue -5

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

has rank 1

For the eigenvalue -4

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

has rank 2

System is NOT fully state-controllable

NE3.3c

The eigenvalues of A are: -1+3i and -1-3i

For the eigenvalue -1+3i

$$\begin{bmatrix} -1.0000 + 3.0000i & 10.0000 & 1.0000 \\ -1.0000 & 1.0000 + 3.0000i & 2.0000 \end{bmatrix}$$

has rank 2

For the eigenvalue -1-3i

$$\begin{bmatrix} -1.0000 - 3.0000i & 10.0000 & 1.0000 \\ -1.0000 & 1.0000 - 3.0000i & 2.0000 \end{bmatrix}$$

has rank 2

System is fully state-controllable

NE3.3d

The eigenvalues of A are: -1+3i and -1-3i

For the eigenvalue -1+3i

$$\begin{bmatrix} -1.0000 + 3.0000i & -1.0000 & 0 \\ 10.0000 & 1.0000 + 3.0000i & 1.0000 \end{bmatrix}$$

has rank 2

For the eigenvalue $-1-3i$

$$\begin{array}{ccc} -1.0000 - 3.0000i & -1.0000 & 0 \\ 10.0000 & 1.0000 - 3.0000i & 1.0000 \end{array}$$

has rank 2

System is fully state-controllable

NE3.3e

The eigenvalues of A are: 1 and 2

For the eigenvalue 1

$$\begin{array}{ccc} -1 & 0 & 1 \\ 1 & 0 & -1 \end{array}$$

has rank 1

For the eigenvalue 2

$$\begin{array}{ccc} 0 & 0 & 1 \\ 1 & 1 & -1 \end{array}$$

has rank 2

System is NOT fully state-controllable

Chapter 4

NE4.1a

$Q =$

$$\begin{bmatrix} 1 & 1 \\ -4 & -5 \end{bmatrix}$$

determinant of Q -1

System is fully state-observable

NE4.1b

$Q =$

$$\begin{bmatrix} 1 & 0 \\ -4 & 0 \end{bmatrix}$$

determinant of Q 0

System is NOT fully state-observable

NE4.1c

$Q =$

$$\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

determinant of Q -1

System is fully state-observable

NE4.1d

$Q =$

$$\begin{bmatrix} 1 & 2 \\ -20 & -3 \end{bmatrix}$$

determinant of Q 37

System is fully state-observable

NE4.1e

$Q =$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

determinant of Q 0

System is NOT fully state-observable

NE4.2a

$A_o = \begin{bmatrix} 0 & -20 \\ 1 & -9 \end{bmatrix}$
 $B_o = \begin{bmatrix} 9 \\ 2 \end{bmatrix}$
 $C_o = \begin{bmatrix} 0 & 1 \end{bmatrix}$

NE4.2b

We cannot find OCF from the formula since the transformation matrix is singular.

NE4.2c

$A_o = \begin{bmatrix} 0 & -10 \\ 1 & -2 \end{bmatrix}$
 $B_o = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
 $C_o = \begin{bmatrix} 0 & 1 \end{bmatrix}$

NE4.2d

$A_o = \begin{bmatrix} 0.0000 & -10.0000 \\ 1.0000 & -2.0000 \end{bmatrix}$
 $B_o = \begin{bmatrix} 1.0000 \\ 2.0000 \end{bmatrix}$
 $C_o = \begin{bmatrix} 0.0000 & 1.0000 \end{bmatrix}$

NE4.2e

We cannot find OCF from the formula since the transformation matrix is singular.

NE4.3

Apply the Popov-Belevitch-Hautus Rank Test for Observability

NE4.3a

The eigenvalues of A are: -5 and -4

For the eigenvalue -5

$$\begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 0 & 0 \end{bmatrix}$$

has rank 2

For the eigenvalue -4

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

has rank 2

System is fully observable

NE4.3b

The eigenvalues of A are: -5 and -4

For the eigenvalue -5

$$\begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 0 \end{bmatrix}$$

has rank 1

For the eigenvalue -4

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

has rank 2

System is NOT fully observable

NE4.3c

The eigenvalues of A are: -1+3i and -1-3i

For the eigenvalue -1+3i

$$\begin{bmatrix} 0 & 1.0000 \\ -1.0000 + 3.0000i & 10.0000 \\ -1.0000 & 1.0000 + 3.0000i \end{bmatrix}$$

has rank 2

For the eigenvalue -1-3i

$$\begin{bmatrix} 0 & 1.0000 \\ -1.0000 - 3.0000i & 10.0000 \\ -1.0000 & 1.0000 - 3.0000i \end{bmatrix}$$

has rank 2

System is fully observable

NE4.3d

The eigenvalues of A are: $-1+3i$ and $-1-3i$

For the eigenvalue $-1+3i$

1.0000	2.0000
-1.0000 + 3.0000i	-1.0000
10.0000	1.0000 + 3.0000i

has rank 2

For the eigenvalue $-1-3i$

1.0000	2.0000
-1.0000 - 3.0000i	-1.0000
10.0000	1.0000 - 3.0000i

has rank 2

System is fully observable

NE4.3e

The eigenvalues of A are: 1 and 2

For the eigenvalue 1

1	1
-1	0
1	0

has rank 2

For the eigenvalue 2

1	1
0	0
1	1

has rank 1

System is NOT fully observable

Chapter 5

NE5.1a

Transfer function:

$$\frac{s + 1}{s^2 + 3s + 2}$$

1 state removed.

$$\begin{aligned} a &= \begin{bmatrix} x1 \\ x1 & -2 \end{bmatrix} \\ b &= \begin{bmatrix} u1 \\ x1 & 0.7071 \end{bmatrix} \\ c &= \begin{bmatrix} x1 \\ y1 & 1.414 \end{bmatrix} \\ d &= \begin{bmatrix} u1 \\ y1 & 0 \end{bmatrix} \end{aligned}$$

NE5.1b

Transfer function:

$$\frac{s^2 + 4s + 5}{s^3 + 7s^2 + 17s + 15}$$

2 states removed.

$$\begin{aligned} a &= \begin{bmatrix} x1 \\ x1 & -3 \end{bmatrix} \\ b &= \begin{bmatrix} u1 \\ x1 & 0.1543 \end{bmatrix} \\ c &= \begin{bmatrix} x1 \\ y1 & 6.481 \end{bmatrix} \\ d &= \begin{bmatrix} u1 \\ y1 & 0 \end{bmatrix} \end{aligned}$$

NE5.1c

Transfer function:

$$\frac{s + 3}{s^3 + 7s^2 + 17s + 15}$$

1 state removed.

$$\begin{aligned} a &= \begin{bmatrix} x1 & x2 \\ x1 & 1.398 & -5.077 \\ x2 & 2.471 & -5.398 \end{bmatrix} \\ b &= \begin{bmatrix} u1 \\ x1 & 0.1943 \\ x2 & 0.2686 \end{bmatrix} \\ c &= \begin{bmatrix} x1 & x2 \\ y1 & -2.562 & 1.854 \end{bmatrix} \\ d &= \begin{bmatrix} u1 \\ y1 & 0 \end{bmatrix} \end{aligned}$$

NE5.1d

Transfer function:

$$s^2 + 8s + 25$$

$$s^4 + 15s^3 + 91s^2 + 255s + 250$$

2 states removed.

a =

	x1	x2
x1	0.6813	1.215
x2	-12.53	-7.681

b =

	u1
x1	-0.001309
x2	0.03991

c =

	x1	x2
y1	26.25	0.8612

d =

	u1
y1	0

NE5.1e

Transfer function:

$$2s^2 + 9s + 24$$

$$s^4 + 15s^3 + 91s^2 + 255s + 250$$

a =

	x1	x2	x3	x4
x1	0	0	0	-250
x2	1	0	0	-255
x3	0	1	0	-91
x4	0	0	1	-15

b =

	u1
x1	24
x2	9
x3	2
x4	0

c =

	x1	x2	x3	x4
y1	0	0	0	1

d =

	u1
y1	0

NE 5.1e looks like OCF of NE5.1d; however, two numbers have changed and there are no longer any common factors of the transfer function numerator and denominator. This system is already minimal.

Chapter 6

NE6.1a & NE6.3a

Eigenvalues:

-2 + 3.16i

-2 - 3.16i

Lyapunov matrix P:

2.02 0.04

0.04 0.13

Sylvester's Criterion:

2.02

0.27

System is asymptotically stable

NE6.1b & NE6.3b

Eigenvalues:

2 + 3.16i

2 - 3.16i

Lyapunov matrix P:

-2.02 0.04

0.04 -0.13

Sylvester's Criterion:

-2.02

0.27

System is unstable

NE6.1c & NE6.3c

Eigenvalues:

0

-4

??? Error using ==> lyap

Solution does not exist or is not unique.

MATLAB function lyap fails.

System is marginally stable due to zero pole (and no positive poles).

NE6.1d & NE6.3d

Eigenvalues:

0 + 3.74i

0 - 3.74i

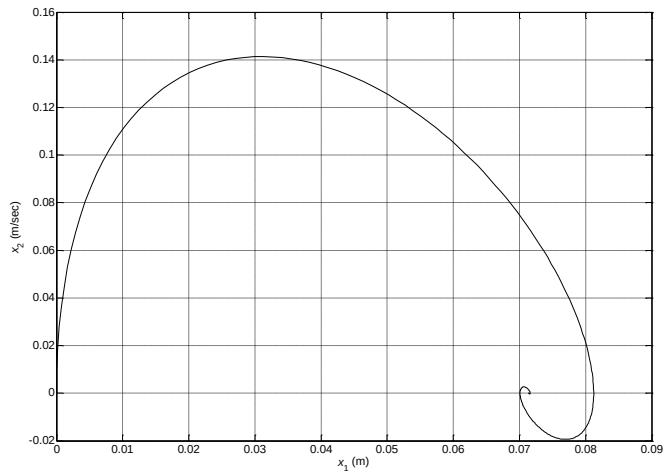
??? Error using ==> lyap

Solution does not exist or is not unique.

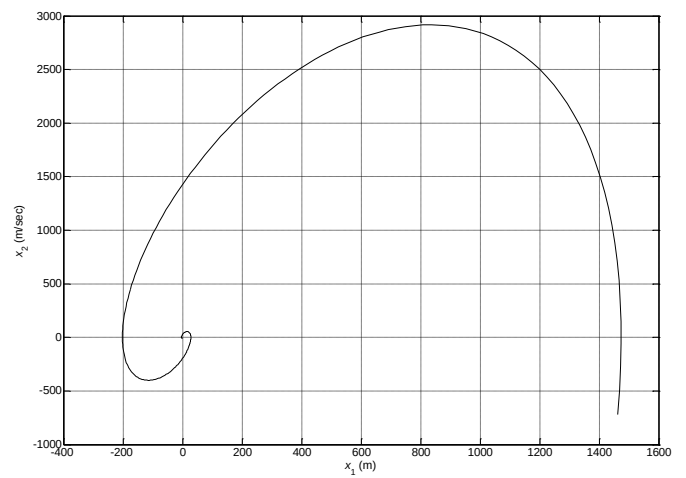
MATLAB function lyap fails.

System is marginally stable due to zero real poles (and no positive real poles).

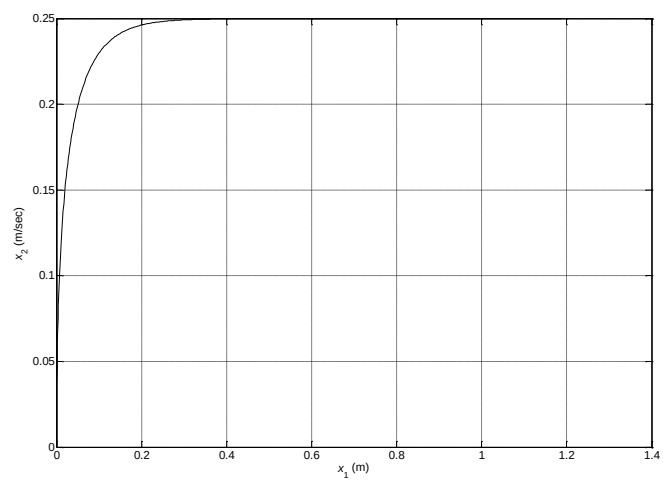
NE6.2a STABLE



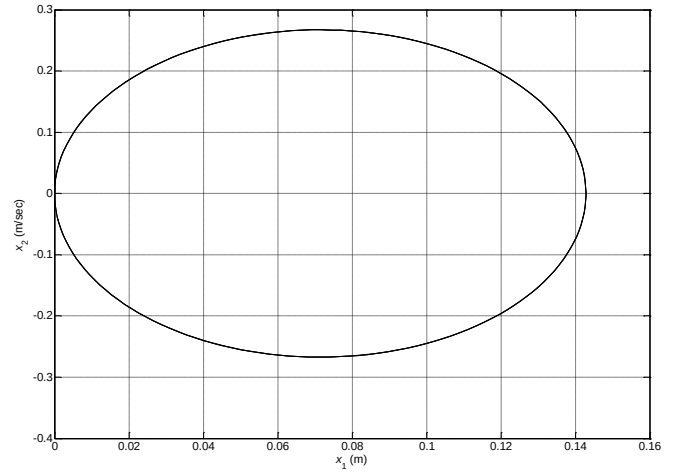
NE6.2b UNSTABLE



NE6.2c marginally stable



NE6.2d marginally stable



Note: the book did not give enough information for NE 6.2a-d; so for all cases we assumed:

$$B = [0; 1]$$

$$C = [1 \ 0]$$

$$D = [0]$$

NE 6.4 Given

$$H(s) = \frac{s^2 - s - 2}{s^3 + 2s^2 - 4s - 8} = \frac{(s-2)(s+1)}{(s-2)(s+2)^2} = \frac{(s+1)}{(s+2)^2}$$

a. The system is bounded-input, bounded-output stable because the impulse response $h(t) = (1-t)e^{-2t}$ satisfies

$$\int_0^{\infty} |h(\tau)| d\tau < \int_0^{\infty} (1+\tau)e^{-2\tau} d\tau = \frac{3}{4} < \infty$$

b. The three-dimensional controller canonical form realization is specified by

$$A_{\text{CCF}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 8 & 4 & -2 \end{bmatrix} \quad B_{\text{CCF}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C_{\text{CCF}} = [-2 \quad -1 \quad 1]$$

The observability matrix

$$Q = \begin{bmatrix} C_{\text{CCF}} \\ C_{\text{CCF}} A_{\text{CCF}} \\ C_{\text{CCF}} A_{\text{CCF}}^2 \end{bmatrix} = \begin{bmatrix} -2 & -1 & 1 \\ 8 & 2 & -3 \\ -24 & -4 & 8 \end{bmatrix}$$

has $|Q| = 0$, so this realization is not observable. The eigenvalues of A_{CCF} are $-2, -2, 2$ and so this realization is not asymptotically stable.

c. A second-order minimal realization (in controller canonical form) is specified by

$$A_{\text{CCF}} = \begin{bmatrix} 0 & 1 \\ -4 & -4 \end{bmatrix} \quad B_{\text{CCF}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad C_{\text{CCF}} = [1 \quad 1]$$

NE 6.5 Given

$$H(s) = \frac{s^2 + s - 2}{s^3 + 2s^2 - 4s - 8} = \frac{(s-1)(s+2)}{(s-2)(s+2)^2} = \frac{(s-1)}{(s-2)(s+2)}$$

a. The system is not bounded-input, bounded-output stable because the zero-state response to a unit step input is

$$y(t) = \frac{1}{4} - \frac{3}{8}e^{-2t} + \frac{1}{8}e^{2t}$$

which is unbounded.

b. The three-dimensional observer canonical form realization is specified by

$$A_{\text{CCF}} = \begin{bmatrix} 0 & 0 & 8 \\ 1 & 0 & 4 \\ 0 & 1 & -2 \end{bmatrix} \quad B_{\text{CCF}} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \quad C_{\text{CCF}} = [0 \ 0 \ 1]$$

The controllability matrix

$$P = \begin{bmatrix} B_{\text{OCF}} & A_{\text{OCF}}B_{\text{OCF}} & A_{\text{OCF}}^2B_{\text{OCF}} \end{bmatrix} = \begin{bmatrix} -2 & 8 & -8 \\ 1 & 2 & 4 \\ 1 & -1 & 4 \end{bmatrix}$$

has $|P| = 0$, so this realization is not controllable. The eigenvalues of A_{OCF} are $-2, -2, 2$ and so this realization is not asymptotically stable.

c. A second-order minimal realization (in observer canonical form) is specified by

$$A_{\text{OCF}} = \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix} \quad B_{\text{OCF}} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad C_{\text{OCF}} = [0 \ 1]$$

Chapter 7

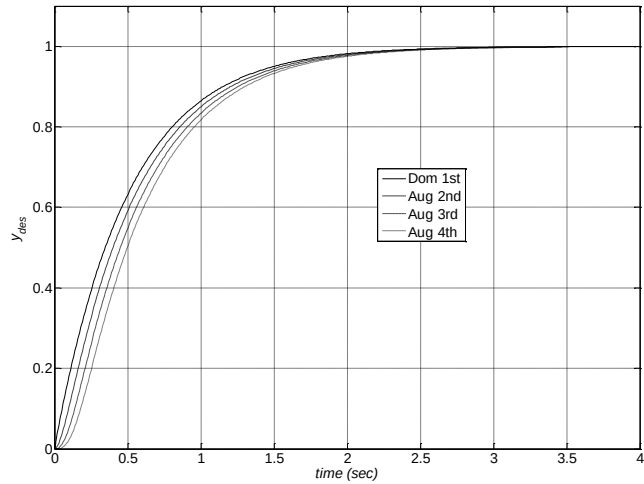
NE7.1a (one possible solution)

DomPole = -2

Poles2 = -2 -20

Poles3 = -2 -20 -21

Poles4 = -2 -20 -21 -22



NE7.1b (one possible solution)

zeta = -0.6671

wn = -1.4990

den2 = 1.0000 2.0000 2.2469

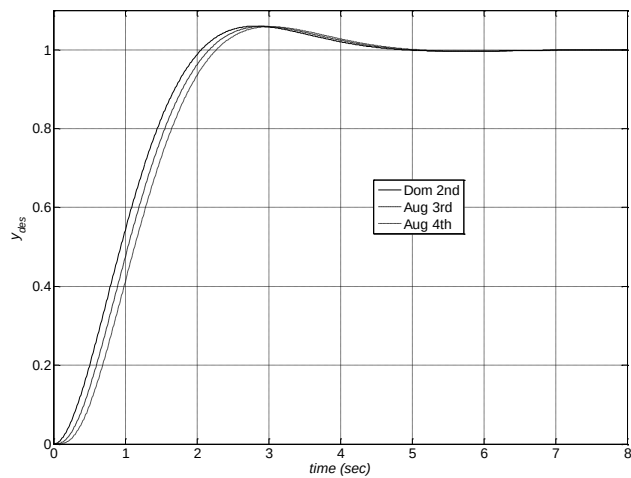
DomPoles =

-1 + 1.12i

-1 - 1.12i

Poles3 = -1-1.12i -1+1.12i -10

Poles4 = -1-1.12i -1+1.12i -10 -11



NE7.1c

ITAE2 = 1 7 25

Poles2i =

-3.5000 + 3.5707i

-3.5000 - 3.5707i

ITAE3 = 1 8.75 53.75 125

Poles3i =

-2.6048 + 5.3405i

-2.6048 - 5.3405i

-3.5405

ITAE4 = 1 10.5 85 337.5 625

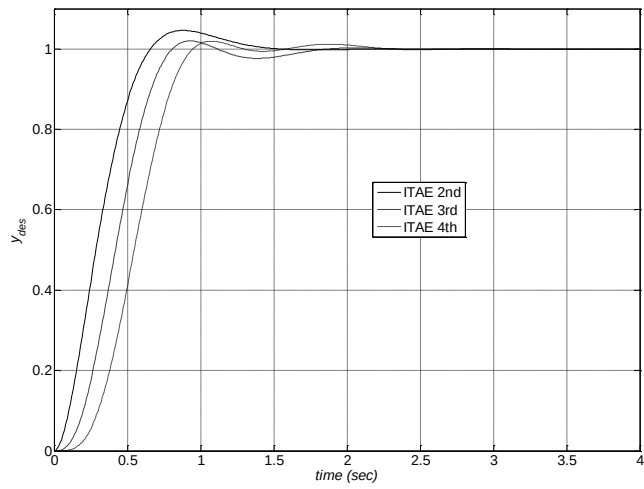
Poles4i =

-2.1199 + 6.3150i

-2.1199 - 6.3150i

-3.1301 + 2.0707i

-3.1301 - 2.0707i



NE7.2a & NE7.3a

K = 3.33 -4.33

NE7.2b & NE7.3b

K = 14 1

NE7.2c & NE7.3c

K = 14 9

NE7.2d & NE7.3d

K = 12 130

Chapter 8

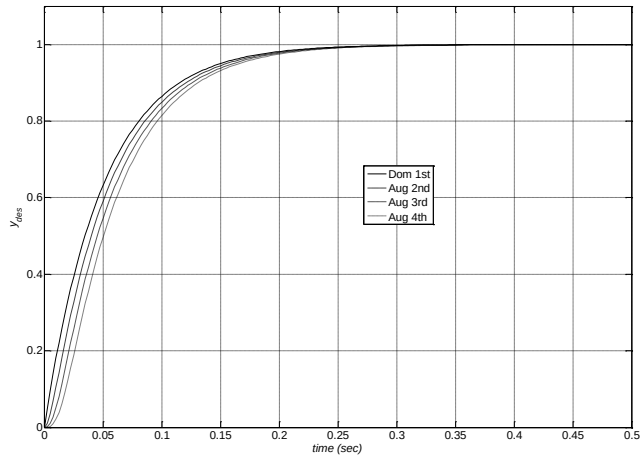
NE8.1a (one possible solution)

DomPole0 = -20

Poles20 = -20 -200

Poles30 = -20 -200 -201

Poles40 = -20 -200 -201 -202



NE8.1b (one possible solution)

DomPoles0 =

-10.0000 +11.1665i

-10.0000 -11.1665i

Poles3 =

1.0e+002 *

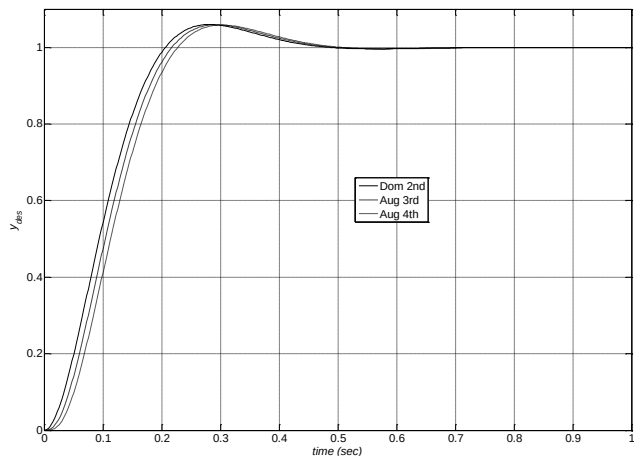
-0.1000 - 0.1117i -0.1000 + 0.1117i -1.0000

Poles4 =

1.0e+002 *

-0.1000 - 0.1117i -0.1000 + 0.1117i -1.0000

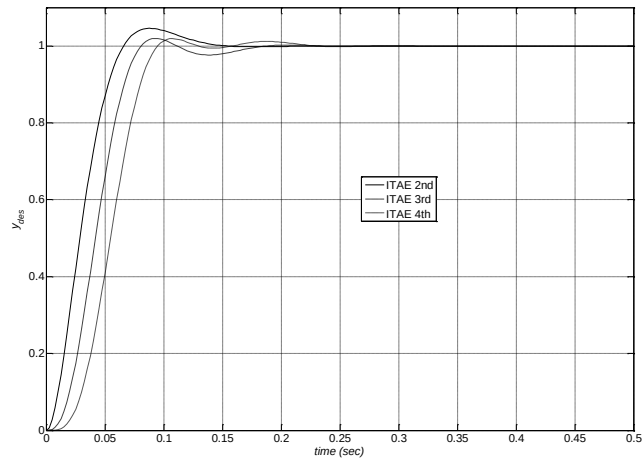
-1.1000




```
NE8.1c
Poles2iO =
  -35.0000 +35.7071i
  -35.0000 -35.7071i
```

```
Poles3iO =
  -26.0475 +53.4051i
  -26.0475 -53.4051i
  -35.4050
```

```
Poles4iO =
  -21.1991 +63.1496i
  -21.1991 -63.1496i
  -31.3009 +20.7069i
  -31.3009 -20.7069i
```



```
NE8.2a & NE8.3a
L =
  420.33
 -385.33
```

```
NE8.2b & NE8.3b
L =
  82
 1338
```

```
NE8.2c & NE8.3c
L =
  90
 1994
```

```
NE8.2d & NE8.3d
L =
  208
  30
```

Chapter 9

NE9.1a

KLQR =

0.4169	-0.0093
-0.0093	0.1234

NE9.1b

KLQR =

1.1032	0.0828
0.0828	0.0725

NE9.1c

KLQR =

6.5669	0.0828
0.0828	1.0796

NE9.1d

KLQR =

21.9	240.3
240.3	2694.9

NE9.2ai

KLQR =

0.3724	-0.0153
-0.0153	0.1221

NE9.2aii

KLQR =

0.4505	-0.0053
-0.0053	0.1241

NE9.2bi

KLQR =

1.1023	0.0822
0.0822	0.0721

NE9.2bii

KLQR =

1.1037	0.0830
0.0830	0.0727

NE9.2ci

KLQR =

4.7036	0.0822
0.0822	0.7630

NE9.2cii

KLQR =

9.2263	0.0830
0.0830	1.5271

NE9.2di

KLQR =

11.1	123.5
123.5	1426.8

NE9.2dii

KLQR =

43.5	472.3
472.3	5198.0

NE 9.3 The parameter values are the same as in Example 9.5 except that now $R = \rho^2$ for variable ρ and we fix $\sigma = 1$ yielding the performance index

$$J = \frac{1}{2} \int_0^1 [x^2(t) + \rho^2 u^2(t)] dt + \frac{1}{2} x^2(1)$$

The Hamiltonian matrix is given by

$$H = \begin{bmatrix} 0 & -\frac{1}{\rho^2} \\ -1 & 0 \end{bmatrix}$$

from which we compute the matrix exponential

$$e^{Ht} = \begin{bmatrix} \frac{1}{2}(e^{-t/\rho} + e^{t/\rho}) & \frac{1}{2\rho}(e^{-t/\rho} - e^{t/\rho}) \\ \frac{\rho}{2}(e^{-t/\rho} - e^{t/\rho}) & \frac{1}{2}(e^{-t/\rho} + e^{t/\rho}) \end{bmatrix}$$

This yields the

$$\begin{aligned} \begin{bmatrix} X(t) \\ \Lambda(t) \end{bmatrix} &= \begin{bmatrix} \frac{1}{2}(e^{-(t-1)/\rho} + e^{(t-1)/\rho}) & \frac{1}{2\rho}(e^{-(t-1)/\rho} - e^{(t-1)/\rho}) \\ \frac{\rho}{2}(e^{-(t-1)/\rho} - e^{(t-1)/\rho}) & \frac{1}{2}(e^{-(t-1)/\rho} + e^{(t-1)/\rho}) \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2\rho} [(\rho+1)e^{-(t-1)/\rho} + (\rho-1)e^{(t-1)/\rho}] \\ \frac{1}{2} [(\rho+1)e^{-(t-1)/\rho} - (\rho-1)e^{(t-1)/\rho}] \end{bmatrix} \end{aligned}$$

from which we construct

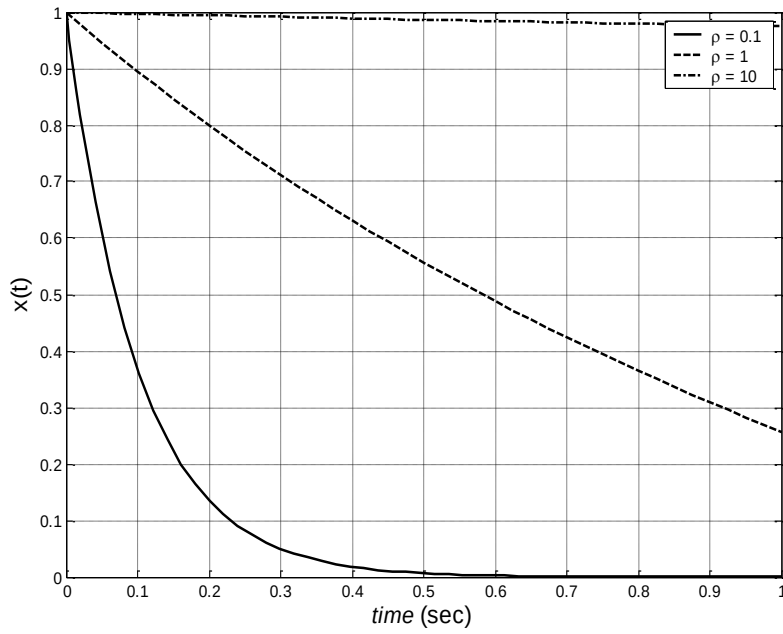
$$P(t) = \frac{\Lambda(t)}{X(t)} = \rho \frac{(\rho+1)e^{-(t-1)/\rho} - (\rho-1)e^{(t-1)/\rho}}{(\rho+1)e^{-(t-1)/\rho} + (\rho-1)e^{(t-1)/\rho}} = \rho \frac{(\rho+1) - (\rho-1)e^{2(t-1)/\rho}}{(\rho+1) + (\rho-1)e^{2(t-1)/\rho}}$$

The associated feedback gain is $K(t) = -\frac{1}{\rho^2} P(t)$, yielding the time-varying closed-loop state equation

$$\dot{x}(t) = -\frac{1}{\rho^2} P(t)x(t) \quad x(0) = x_0$$

The closed-loop state response for the initial state $x(0)=1$ and $\rho=0.1, 1$, and 10 is shown below.

Regulation performance improves with decreasing ρ corresponding to a smaller penalty on control effort.



Closed-loop state response for NE9.3.

The similarity transformation matrix

$$T = \begin{bmatrix} 1 & 1 \\ -\rho & \rho \end{bmatrix}$$

yields

$$\begin{aligned} T^{-1}HT &= \frac{1}{2\rho} \begin{bmatrix} \rho & 1 \\ \rho & -1 \end{bmatrix} \begin{bmatrix} 0 & -\frac{1}{\rho^2} \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ \rho & -\rho \end{bmatrix} \\ &= \begin{bmatrix} -\frac{1}{\rho} & 0 \\ 0 & \frac{1}{\rho} \end{bmatrix} \end{aligned}$$

from which we identify $J^- = -\frac{1}{\rho}$ and $J^+ = \frac{1}{\rho}$. With $S = \sigma = 1$,

$$M = \frac{\rho - 1}{\rho + 1}$$

from which we compute

$$P(t) = \rho \frac{1 - \frac{\rho-1}{\rho+1} e^{2(t-1)/\rho}}{1 + \frac{\rho-1}{\rho+1} e^{2(t-1)/\rho}} = \rho \frac{(\rho+1) - (\rho-1)e^{2(t-1)}}{(\rho+1) + (\rho-1)e^{2(t-1)}}$$

which agrees with the result obtained earlier.

For arbitrary final time

$$P(t) = \rho \frac{(\rho+1) - (\rho-1)e^{2(t-t_f)}}{(\rho+1) + (\rho-1)e^{2(t-t_f)}}$$

which approaches the limit

$$\bar{P} = \lim_{t_f \rightarrow \infty} P(t) = \rho \frac{(\rho+1)}{(\rho+1)} = \rho$$

The algebraic Riccati equation

$$1 - \frac{1}{\rho^2} \bar{P}^2 = 0$$

has two solutions, $\bar{P} = \pm \rho$. Of these, only $\bar{P} = \rho$ is positive (definite) for which the state feedback gain

$K_{\text{LQR}} = \frac{1}{\rho^2} (1)(\rho) = \frac{1}{\rho}$ yields the asymptotically stable closed-loop state equation

$$\dot{x}(t) = -\frac{1}{\rho} x(t)$$

whose regulation performance improves with decreasing ρ , which corresponds to greater bandwidth, equivalently, a smaller time constant.

Analytical Exercises Solutions

Chapter 1

AE1.1 From the specified dimensions of A , B has n rows and C has m columns. From the specified dimensions of H , G has p rows and F has q columns. From the upper left block of

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{bmatrix}$$

E has m rows and B has p columns. Consequently, F has m rows and D has p columns. Let k denote the number of rows of C and D , which is not constrained by the other matrix dimensions. Also, let r denote the number of rows of E and G , which is similarly unconstrained. Taking stock:

$$A \text{ is } n \times m \quad B \text{ is } n \times p \quad C \text{ is } k \times m \quad D \text{ is } k \times p$$

$$E \text{ is } m \times r \quad F \text{ is } m \times q \quad G \text{ is } p \times r \quad H \text{ is } p \times q$$

AE1.2 Let A be $n \times n$ and use proof-by-induction on n . For $n=1$, $A=a_{11}$ and the cofactor of a_{11} is $(-1)^{1+1}|B|$. Expanding the determinant about either the first row or column yields

$$\begin{vmatrix} A & 0 \\ 0 & B \end{vmatrix} = \begin{vmatrix} a_{11} & 0 \\ 0 & B \end{vmatrix} = a_{11}(-1)^{1+1}|B| = a_{11}|B| = |A||B|$$

Suppose the identity holds for $n > 1$. For the $(n+1)$ -dimensional case, expanding the determinant about the i th row for $1 \leq i \leq n+1$ gives

$$\begin{vmatrix} A & 0 \\ 0 & B \end{vmatrix} = \sum_{j=1}^{n+1} a_{ij}(-1)^{i+j} M_{ij}$$

Each minor is the determinant of a block diagonal matrix whose diagonal blocks are the $n \times n$ matrix obtained from A by deleting the i th row and j th column along with B in its entirety. Using the induction hypothesis

$$M_{ij} = M_{ij}^A |B|$$

in which M_{ij}^A is the minor of the element a_{ij} for just the matrix A . Thus

$$\begin{vmatrix} A & 0 \\ 0 & B \end{vmatrix} = \sum_{j=1}^{n+1} a_{ij} (-1)^{i+j} (M_{ij}^A |B|) = \left(\sum_{j=1}^{n+1} a_{ij} (-1)^{i+j} M_{ij}^A \right) |B| = |A| |B|$$

AE1.3 If B is singular then $|B| = 0$ and there exists a nonzero vector v such that $Bv = 0$. Then

$$\begin{bmatrix} A & 0 \\ C & B \end{bmatrix} \begin{bmatrix} 0 \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ Bv \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

implying that

$$\begin{vmatrix} A & 0 \\ C & B \end{vmatrix} = 0 = |A| \cdot |B|$$

and so the identity holds if B is singular. If B is nonsingular, the factorization

$$\begin{bmatrix} A & 0 \\ C & B \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \begin{bmatrix} I & 0 \\ B^{-1}C & I \end{bmatrix}$$

and a basic determinant property give

$$\begin{vmatrix} A & 0 \\ C & B \end{vmatrix} = \begin{vmatrix} A & 0 \\ 0 & B \end{vmatrix} \begin{vmatrix} I & 0 \\ B^{-1}C & I \end{vmatrix}$$

The rightmost factor is the determinant of a lower triangular matrix with ones on the main diagonal and therefore equals one (see AE1.6). Applying AE1.2 to the leftmost factor establishes the desired identity.

AE1.4 The factorization

$$\begin{bmatrix} A & D \\ C & B \end{bmatrix} = \begin{bmatrix} A & 0 \\ C & B - CA^{-1}D \end{bmatrix} \begin{bmatrix} I & A^{-1}D \\ 0 & I \end{bmatrix}$$

and AE1.3 yield

$$\begin{vmatrix} A & D \\ C & B \end{vmatrix} = \begin{vmatrix} A & 0 \\ C & B - CA^{-1}D \end{vmatrix} \begin{vmatrix} I & A^{-1}D \\ 0 & I \end{vmatrix} = |A| |B - CA^{-1}D|$$

where we have again used AE1.6 to conclude that the determinant of the upper triangular factor equals one.

AE1.5 First observe that, applying AE1.4,

$$\begin{vmatrix} I_m & Y \\ X & I_n \end{vmatrix} = |I_m| |I_n - XI_m^{-1}Y| = |I_n - XY|$$

and

$$\begin{vmatrix} I_n & X \\ Y & I_m \end{vmatrix} = |I_n| |I_m - YI_n^{-1}X| = |I_m - YX|$$

Next,

$$\tilde{I} = \begin{bmatrix} 0 & I_n \\ I_m & 0 \end{bmatrix} = \begin{bmatrix} 0 & I_m \\ I_n & 0 \end{bmatrix}$$

gives

$$\begin{bmatrix} 0 & I_n \\ I_m & 0 \end{bmatrix} \begin{bmatrix} I_m & Y \\ X & I_n \end{bmatrix} \begin{bmatrix} 0 & I_m \\ I_n & 0 \end{bmatrix} = \begin{bmatrix} I_n & X \\ Y & I_m \end{bmatrix}$$

This, together with $\tilde{I} \tilde{I} = I_{n+m}$ from which $|\tilde{I}| |\tilde{I}| = 1$, yields

$$|I_n - XY| = \begin{vmatrix} I_m & Y \\ X & I_n \end{vmatrix} = \begin{vmatrix} I_n & X \\ Y & I_m \end{vmatrix} = |I_m - YX|$$

When $m=1$, $I_m - YX$ is a scalar quantity and $|I_m - YX| = I_m - YX$. For $n > 1$, this identity provides a computational short cut, especially when n is large.

AE1.6 Let A be an upper triangular $n \times n$ matrix and use proof-by-induction on n . For $n = 2$,

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}(0) = a_{11}a_{22}$$

Suppose the determinant identity holds for $n > 2$. For the $n+1$ -dimensional case,

$$|A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} & a_{1n+1} \\ 0 & a_{22} & \cdots & a_{2n} & a_{2n+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn} & a_{nn+1} \\ \hline 0 & 0 & \cdots & 0 & a_{n+1n+1} \end{vmatrix} = a_{n+1n+1}(-1)^{(n+1)+(n+1)} \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix}$$

The rightmost factor is the determinant of an upper triangular $n \times n$ matrix. Thus, invoking the induction hypothesis and using the fact that $(n+1)+(n+1) = 2n+2$ is even for any integer n , yields

$$|A| = a_{n+1n+1}(+1)(a_{11}a_{22} \cdots a_{nn}) = a_{11}a_{22} \cdots a_{nn}a_{n+1n+1}$$

AE1.7 It suffices to check that the expression asserted to be the inverse satisfies

$$[A + BCD][A + BCD]^{-1} = I_n :$$

$$\begin{aligned} [A + BCD][A + BCD]^{-1} &= [A + BCD] \left[A^{-1} - A^{-1}B[C^{-1} + DA^{-1}B]^{-1}DA^{-1} \right] \\ &= I_n + BCDA^{-1} - B[C^{-1} + DA^{-1}B]^{-1}DA^{-1} - BCDA^{-1}B[C^{-1} + DA^{-1}B]^{-1}DA^{-1} \\ &= I_n + BCDA^{-1} - B(I_m + CDA^{-1}B)[C^{-1} + DA^{-1}B]^{-1}DA^{-1} \\ &= I_n + BCDA^{-1} - BC(C^{-1} + DA^{-1}B)[C^{-1} + DA^{-1}B]^{-1}DA^{-1} \\ &= I_n + BCDA^{-1} - BCDA^{-1} \\ &= I_n \end{aligned}$$

For $m=1$ and $C=1$

$$[C^{-1} + DA^{-1}B]^{-1} = \frac{1}{1 + DA^{-1}B}$$

and therefore

$$[A + BCD]^{-1} = A^{-1} - \frac{A^{-1}BDA^{-1}}{1 + DA^{-1}B}$$

AE1.8 The identity $X - XYX = X - XYX$ can be factored to give

$$X(I_m - YX) = (I_n - XY)X$$

Cross-multiplying by the inverses (when they exist) yields the desired identity.

AE1.9 Using block partitioned multiplication (as illustrated in AE1.1)

$$\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \begin{bmatrix} A^{-1} & 0 \\ 0 & B^{-1} \end{bmatrix} = \begin{bmatrix} AA^{-1} & 0 \\ 0 & BB^{-1} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

from which the identity follows

AE1.10 Partitioning the inverse as

$$\begin{bmatrix} A & 0 \\ C & B \end{bmatrix}^{-1} = \begin{bmatrix} E & F \\ G & H \end{bmatrix}$$

the identity

$$\begin{bmatrix} A & 0 \\ C & B \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} AE & AF \\ CE + BG & CF + BH \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

implies $E = A^{-1}$ and $F = 0$. Substituting into the second block row implies $G = -B^{-1}CA^{-1}$ and

$H = B^{-1}$. Thus

$$\begin{bmatrix} A & 0 \\ C & B \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & 0 \\ -B^{-1}CA^{-1} & B^{-1} \end{bmatrix}$$

and a similar analysis reveals

$$\begin{bmatrix} A & D \\ 0 & B \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & -A^{-1}DB^{-1} \\ 0 & B^{-1} \end{bmatrix}$$

AE1.11 Consider the product

$$\begin{bmatrix} A & D \\ C & B \end{bmatrix} \begin{bmatrix} A^{-1} + E\Delta^{-1}F & -E\Delta^{-1} \\ -\Delta^{-1}F & \Delta^{-1} \end{bmatrix} = \begin{bmatrix} A(A^{-1} + E\Delta^{-1}F) - D\Delta^{-1}F & -AE\Delta^{-1} + D\Delta^{-1} \\ C(A^{-1} + E\Delta^{-1}F) - B\Delta^{-1}F & -CE\Delta^{-1} + B\Delta^{-1} \end{bmatrix}$$

in which $\Delta = B - CA^{-1}D$, $E = A^{-1}D$, and $F = CA^{-1}$. The upper left block reduces to

$$A(A^{-1} + E\Delta^{-1}F) - D\Delta^{-1}F = AA^{-1} + [A(A^{-1}D) - D]\Delta^{-1}F = I$$

The upper right block reduces to

$$-AE\Delta^{-1} + D\Delta^{-1} = -A(A^{-1}D)\Delta^{-1} + D\Delta^{-1} = -D\Delta^{-1} + D\Delta^{-1} = 0$$

The lower left block reduces to

$$\begin{aligned} C(A^{-1} + E\Delta^{-1}F) - B\Delta^{-1}F &= CA^{-1} - [B - C(A^{-1}D)]\Delta^{-1}F \\ &= CA^{-1} - \Delta\Delta^{-1}(CA^{-1}) \\ &= 0 \end{aligned}$$

Finally, the lower right block reduces to

$$\begin{aligned} -CE\Delta^{-1} + B\Delta^{-1} &= [B - C(A^{-1}D)]\Delta^{-1} \\ &= \Delta\Delta^{-1} \\ &= I \end{aligned}$$

Thus,

$$\begin{bmatrix} A & D \\ C & B \end{bmatrix} \begin{bmatrix} A^{-1} + E\Delta^{-1}F & -E\Delta^{-1} \\ -\Delta^{-1}F & \Delta^{-1} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

which verifies the identity.

AE1.12 First, direct computations show that for $\lambda \neq 0$

$$J_2^{-1}(\lambda) = \begin{bmatrix} \frac{1}{\lambda} & -\frac{1}{\lambda^2} \\ 0 & \frac{1}{\lambda} \end{bmatrix} \quad \text{and} \quad J_3^{-1}(\lambda) = \begin{bmatrix} \frac{1}{\lambda} & -\frac{1}{\lambda^2} & \frac{1}{\lambda^3} \\ 0 & \frac{1}{\lambda} & -\frac{1}{\lambda^2} \\ 0 & 0 & \frac{1}{\lambda} \end{bmatrix}$$

In general,

$$J_k^{-1}(\lambda) = \begin{bmatrix} \frac{1}{\lambda} & -\frac{1}{\lambda^2} & \frac{1}{\lambda^3} & \cdots & (-1)^{k+1} \frac{1}{\lambda^k} \\ 0 & \frac{1}{\lambda} & -\frac{1}{\lambda^2} & \cdots & (-1)^k \frac{1}{\lambda^{k-1}} \\ 0 & 0 & \frac{1}{\lambda} & \cdots & (-1)^{k-1} \frac{1}{\lambda^{k-2}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{\lambda} \end{bmatrix}$$

That is, the inverse is upper triangular with (i,j) th element on or above the main diagonal given by

$$(-1)^{i+j} \frac{1}{\lambda^{j-i+1}}. \text{ This identity can be verified by checking that } J_k(\lambda) J_k^{-1}(\lambda) = I_k.$$

AE1.13 By definition, $A^{-1}\mathbb{S}$ is a subset of $\mathbb{R}^n$. To show that it is a *subspace*, it suffices to check that for any two vectors $x_1, x_2 \in A^{-1}\mathbb{S}$ and any two scalars $\alpha_1, \alpha_2 \in \mathbb{R}$, the linear combination $\alpha_1 x_1 + \alpha_2 x_2$ is also an element of $A^{-1}\mathbb{S}$. By definition, $Ax_1 \in \mathbb{S}$ and $Ax_2 \in \mathbb{S}$. Because $\mathbb{S}$ is a subspace of $\mathbb{R}^m$, $\alpha_1(Ax_1) + \alpha_2(Ax_2) \in \mathbb{S}$. By linearity of the transformation A , $A(\alpha_1 x_1 + \alpha_2 x_2) = \alpha_1(Ax_1) + \alpha_2(Ax_2) \in \mathbb{S}$. Thus, by definition, $\alpha_1 x_1 + \alpha_2 x_2 \in A^{-1}\mathbb{S}$. It follows that $A^{-1}\mathbb{S}$ is a subspace of $\mathbb{R}^n$.

AE1.14 By definition, any induced matrix norm satisfies $\|Az\| \leq \|A\|\|z\|$ for any vector z in the domain of A . In particular $\|ABx\| \leq \|A\|\|Bx\|$ for any vector x in the domain of B . It follows that

$$\|AB\| = \sup_{x \neq 0} \frac{\|ABx\|}{\|x\|} \leq \sup_{x \neq 0} \frac{\|A\|\|Bx\|}{\|x\|} = \|A\| \sup_{x \neq 0} \frac{\|Bx\|}{\|x\|} = \|A\|\|B\|$$

AE1.15 It is implicitly assumed here that the same vector norm is used on both the domain and range spaces of A . The corresponding induced matrix norm clearly satisfies

$$\|I\| = \sup_{x \neq 0} \frac{\|Ix\|}{\|x\|} = \sup_{x \neq 0} \frac{\|x\|}{\|x\|} = 1$$

Using AE1.14, $1 = \|I\| = \|AA^{-1}\| \leq \|A\|\|A^{-1}\|$ from which

$$\|A^{-1}\| \geq \frac{1}{\|A\|}$$

AE1.16 For any eigenvalue $\lambda_i \in \sigma(A)$ and v_i an associated (right) eigenvector,

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} \geq \frac{\|Av_i\|}{\|v_i\|} = \frac{\|\lambda_i v_i\|}{\|v_i\|} = \frac{|\lambda_i| \|v_i\|}{\|v_i\|} = |\lambda_i|$$

It follows that $\|A\| \geq \max_{\lambda_i \in \sigma(A)} |\lambda_i| = \rho(A)$. Note that this result is independent of the induced matrix norm as long as same vector norm is used on both the domain and range spaces of A .

Chapter 2

AE2.1 There are several ways to attack this problem.

Method 1 It suffices to check that

$$X(t) = e^{At} + \int_0^t e^{A(t-\tau)} B e^{(A+B)\tau} d\tau$$

satisfies $X(0) = I$ and $\dot{X}(t) = X(t)(A+B)$. Clearly

$$X(0) = e^{A0} + \int_0^0 e^{A(0-\tau)} B e^{(A+B)\tau} d\tau = I + 0 = I$$

For the second identity, the change of variable $\sigma = t - \tau$ in the integral term yields

$$X(t) = e^{At} + \int_0^t e^{A\sigma} B e^{(A+B)(t-\sigma)} d\sigma$$

Differentiating using the Leibniz rule

$$\begin{aligned} \dot{X}(t) &= e^{At} A + e^{At} B(1) - B e^{(A+B)t}(0) + \int_0^t e^{A\sigma} B e^{(A+B)(t-\sigma)} (A+B) d\sigma \\ &= \left[e^{At} + \int_0^t e^{A\sigma} B e^{(A+B)(t-\sigma)} d\sigma \right] (A+B) \\ &= X(t)(A+B) \end{aligned}$$

and the conclusion follows.

Method 2 The fundamental theorem of calculus gives

$$\begin{aligned} e^{(A+B)t} - e^{At} &= e^{A(t-\tau)} e^{(A+B)\tau} \Big|_{\tau=t} - e^{A(t-\tau)} e^{(A+B)\tau} \Big|_{\tau=0} \\ &= \int_0^t \frac{d}{d\tau} \left[e^{A(t-\tau)} e^{(A+B)\tau} \right] d\tau \\ &= \int_0^t \left[-e^{A(t-\tau)} A e^{(A+B)\tau} + e^{A(t-\tau)} (A+B) e^{(A+B)\tau} \right] d\tau \\ &= \int_0^t e^{A(t-\tau)} B e^{(A+B)\tau} d\tau \end{aligned}$$

from which the identity is apparent.

Method 3 For another approach, let

$$Z(t) = e^{(A+B)t} - e^{At} - \int_0^t e^{A(t-\tau)} B e^{(A+B)\tau} d\tau$$

from which

$$\begin{aligned}
 \dot{Z}(t) &= (A+B)e^{(A+B)t} - Ae^{At} - \left[Be^{(A+B)t} + \int_0^t Ae^{A(t-\tau)} Be^{(A+B)\tau} d\tau \right] \\
 &= A \left[e^{(A+B)t} - e^{At} - \int_0^t e^{A(t-\tau)} Be^{(A+B)\tau} d\tau \right] \\
 &= AZ(t)
 \end{aligned}$$

this along with $Z(0) = I - I - 0 = 0$ implies $Z(t) = e^{At}Z(0) \equiv 0$ and the conclusion follows.

Method 4 Apply AE1.7 to the Laplace transform of $e^{(A+B)t}$ to yield

$$\begin{aligned}
 [sI - (A+B)]^{-1} &= [(sI - A) + (-B)(I)(I)]^{-1} \\
 &= (sI - A)^{-1} + (sI - A)^{-1}B \left[I - (sI - A)^{-1}B \right]^{-1} (sI - A)^{-1} \\
 &= (sI - A)^{-1} + (sI - A)^{-1}B \left[(sI - A)(I - (sI - A)^{-1}B) \right]^{-1} \\
 &= (sI - A)^{-1} + (sI - A)^{-1}B [sI - (A+B)]^{-1}
 \end{aligned}$$

Taking inverse Laplace transforms using the linearity and time-domain convolution properties yields the desired result.

AE2.2 It suffices to check that

$$X(t) = e^{\gamma t} e^{At}$$

satisfies $X(0) = I$ and $\dot{X}(t) = (\gamma I + A)X(t)$. Clearly $X(0) = e^{\gamma 0} e^{A0} = (1)(I) = I$. Next, the product rule gives

$$\begin{aligned}
 \dot{X}(t) &= (\gamma e^{\gamma t})e^{At} + e^{\gamma t}(Ae^{At}) \\
 &= (\gamma I + A)e^{\gamma t}e^{At} \\
 &= (\gamma I + A)X(t)
 \end{aligned}$$

Alternatively, since the matrices γI and A commute and $e^{(\gamma I)t} = e^{\gamma t} I$, we have from Proposition 2.1 that

$$e^{(\gamma I + A)t} = (e^{(\gamma I)t})(e^{At}) = (e^{\gamma t} I)(e^{At}) = e^{\gamma t} e^{At}$$

AE2.3 We have from Proposition 2.1 that

$$\left[e^{At} \right]^{-1} = e^{-At} = e^{A^T t} = \left[e^{At} \right]^T$$

AE2.4 It suffices to check that

$$X(t) = \begin{bmatrix} e^{A_{11}t} & \int_0^t e^{A_{11}(t-\tau)} A_{12} e^{A_{22}\tau} d\tau \\ 0 & e^{A_{22}t} \end{bmatrix}$$

satisfies $X(0) = I$ and $\dot{X}(t) = AX(t)$ with

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$$

First,

$$X(0) = \begin{bmatrix} e^{A_{11}0} & \int_0^0 e^{A_{11}(0-\tau)} A_{12} e^{A_{22}\tau} d\tau \\ 0 & e^{A_{22}0} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

Next,

$$\begin{aligned} \dot{X}(t) &= \begin{bmatrix} A_{11}e^{A_{11}t} & A_{12}e^{A_{22}t} + \int_0^t A_{11}e^{A_{11}(t-\tau)} A_{12}e^{A_{22}\tau} d\tau \\ 0 & A_{22}e^{A_{22}t} \end{bmatrix} \\ &= \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} e^{A_{11}t} & \int_0^t e^{A_{11}(t-\tau)} A_{12}e^{A_{22}\tau} d\tau \\ 0 & e^{A_{22}t} \end{bmatrix} \\ &= AX(t) \end{aligned}$$

and the result follows.

AE2.5 The fundamental theorem of calculus yields

$$\begin{aligned}
 e^{At} - I &= e^{A\tau} \Big|_{\tau=t} - e^{A\tau} \Big|_{\tau=0} \\
 &= \int_0^t \frac{d}{d\tau} [e^{A\tau}] d\tau \\
 &= \int_0^t A e^{A\tau} d\tau \\
 &= A \int_0^t e^{A\tau} d\tau
 \end{aligned}$$

from which the identity follows. For A nonsingular

$$\int_0^t e^{A\tau} d\tau = A^{-1} [e^{At} - I]$$

AE2.6 First,

$$W(t_0, t_0) = \int_{t_0}^{t_0} e^{A(t-\tau)} Q e^{A^T(t-\tau)} d\tau = 0$$

Next, applying the Leibniz rule

$$\begin{aligned}
 \dot{W}(t, t_0) &= e^{A0} Q e^{A^T 0} (1) - e^{A(t-t_0)} Q e^{A^T(t-t_0)} (0) + \int_{t_0}^t [A e^{A(t-\tau)} Q e^{A^T(t-\tau)} + e^{A(t-\tau)} Q e^{A^T(t-\tau)} A^T] d\tau \\
 &= Q + A \left[\int_{t_0}^t e^{A(t-\tau)} Q e^{A^T(t-\tau)} d\tau \right] + \left[\int_{t_0}^t e^{A(t-\tau)} Q e^{A^T(t-\tau)} d\tau \right] A^T \\
 &= AW(t, t_0) + W(t, t_0)A^T + Q
 \end{aligned}$$

AE2.7 A direct calculation gives

$$\begin{aligned}
 \begin{bmatrix} s^2 & s & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ a_2 & 1 & 0 \\ a_1 & a_2 & 1 \end{bmatrix} \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ a_0 & a_1 & s+a_2 \end{bmatrix} &= \begin{bmatrix} s^2 + a_2s + a_1 & s + a_2 & 1 \end{bmatrix} \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ a_0 & a_1 & s+a_2 \end{bmatrix} \\
 &= \begin{bmatrix} s^3 + a_2s^2 + a_1s + a_0 & 0 & 0 \end{bmatrix} \\
 &= (s^3 + a_2s^2 + a_1s + a_0) C
 \end{aligned}$$

It follows that

$$C(sI - A)^{-1} \begin{bmatrix} 1 & 0 & 0 \\ a_2 & 1 & 0 \\ a_1 & a_2 & 1 \end{bmatrix}^{-1} = \frac{1}{s^3 + a_2s^2 + a_1s + a_0} \begin{bmatrix} s^2 & s & 1 \end{bmatrix}$$

from which

$$\begin{aligned} C(sI - A)^{-1}B &= \frac{1}{s^3 + a_2s^2 + a_1s + a_0} \begin{bmatrix} s^2 & s & 1 \end{bmatrix} \begin{bmatrix} b_2 \\ b_1 \\ b_0 \end{bmatrix} \\ &= \frac{b_2s^2 + b_1s + b_0}{s^3 + a_2s^2 + a_1s + a_0} \\ &= H(s) \end{aligned}$$

AE2.8 A direct calculation gives

$$\begin{aligned} \begin{bmatrix} s & 0 & a_0 \\ -1 & s & a_1 \\ 0 & -1 & s+a_2 \end{bmatrix} \begin{bmatrix} 1 & a_2 & a_1 \\ 0 & 1 & a_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s^2 \\ s \\ 1 \end{bmatrix} &= \begin{bmatrix} s & 0 & a_0 \\ -1 & s & a_1 \\ 0 & -1 & s+a_2 \end{bmatrix} \begin{bmatrix} s^2 + a_2s + a_1 \\ s + a_2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} s^3 + a_2s^2 + a_1s + a_0 \\ 0 \\ 0 \end{bmatrix} \\ &= (s^3 + a_2s^2 + a_1s + a_0) B \end{aligned}$$

It follows that

$$\begin{bmatrix} 1 & a_2 & a_1 \\ 0 & 1 & a_2 \\ 0 & 0 & 1 \end{bmatrix}^{-1} (sI - A)^{-1}B = \frac{1}{s^3 + a_2s^2 + a_1s + a_0} \begin{bmatrix} s^2 \\ s \\ 1 \end{bmatrix}$$

from which

$$\begin{aligned}
C(sI - A)^{-1}B &= \frac{1}{s^3 + a_2s^2 + a_1s + a_0} \begin{bmatrix} b_2 & b_1 & b_0 \end{bmatrix} \begin{bmatrix} s^2 \\ s \\ 1 \end{bmatrix} \\
&= \frac{b_2s^2 + b_1s + b_0}{s^3 + a_2s^2 + a_1s + a_0} \\
&= H(s)
\end{aligned}$$

AE2.9 From $H(s) = C(sI - A)^{-1}B + D$ we have

$$\begin{aligned}
H^T(-s) &= \left[C(-sI - A)^{-1}B + D \right]^T \\
&= B^T(-sI - A)^{-T}C^T + D^T \\
&= B^T(-sI - A^T)^{-1}C^T + D^T \\
&= B^T \left(sI - (-A^T) \right)^{-1} (-C^T) + D^T
\end{aligned}$$

in which the last line is the transfer function associated with the dual state equation.

AE2.10 Because z_0 is not an eigenvalue of A , $(z_0I - A)$ is nonsingular and we have from AE1.4

$$0 = \begin{vmatrix} z_0I - A & -B \\ C & D \end{vmatrix} = |z_0I - A| C(z_0I - A)^{-1}B + D = |z_0I - A| H(z_0)$$

Moreover, $|z_0I - A| \neq 0$ implies that $H(z_0) = 0$, i.e., z_0 is a zero of the transfer function $H(s)$. Next,

there exists

$$\begin{bmatrix} x_0 \\ u_0 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{R}^{n+1} \quad \text{such that} \quad \begin{bmatrix} z_0I - A & -B \\ C & D \end{bmatrix} \begin{bmatrix} x_0 \\ u_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

which can be decomposed into

$$(z_0I - A)x_0 - Bu_0 = 0 \quad \text{and} \quad Cx_0 + Du_0 = 0$$

We claim that for $x(0) = x_0$ and $u(t) = u_0 e^{z_0 t}$, the state equation solution is $x(t) = x_0 e^{z_0 t}$. Clearly,

$x(0) = x_0$. Also,

$$\begin{aligned}\dot{x}(t) - Ax(t) - Bu(t) &= z_0 e^{z_0 t} x_0 - e^{z_0 t} Ax_0 - e^{z_0 t} Bu_0 \\ &= e^{z_0 t} [(z_0 I - A)x_0 - Bu_0] \\ &= 0\end{aligned}$$

for all $t \geq 0$, which verifies the claim. Finally,

$$\begin{aligned}y(t) &= Cx(t) + u(t) \\ &= e^{z_0 t} Cx_0 + e^{z_0 t} Du_0 \\ &= e^{z_0 t} [Cx_0 + Du_0] \\ &= 0\end{aligned}$$

for all $t \geq 0$.

AE2.11 For $v(t) = y(t) = Cx(t) + Du(t)$, the difference $z(t) - x(t)$ satisfies

$$\begin{aligned}\dot{z}(t) - \dot{x}(t) &= \left[(A - BD^{-1}C)z(t) + BD^{-1}(Cx(t) + Du(t)) \right] - [Ax(t) + Bu(t)] \\ &= (A - BD^{-1}C)(z(t) - x(t))\end{aligned}$$

Then $z_0 = x_0$ implies that

$$z(t) - x(t) = e^{(A - BD^{-1}C)t} (z_0 - x_0) = 0$$

for all $t \geq 0$. This gives

$$\begin{aligned}w(t) - u(t) &= -D^{-1}Cz(t) + D^{-1}(Cx(t) + Du(t)) - u(t) \\ &= -D^{-1}C(z(t) - x(t)) \\ &= 0\end{aligned}$$

for all $t \geq 0$ from which the desired conclusion follows.

AE2.12 For

$$v(t) = \dot{y}(t) = C\dot{x}(t) = CAx(t) + CBu(t)$$

the difference $z(t) - x(t)$ satisfies

$$\begin{aligned} \dot{z}(t) - \dot{x}(t) &= \left[\left(A - B(CB)^{-1}CA \right) z(t) + B(CB)^{-1} \left(CAx(t) + CBu(t) \right) \right] - \left[Ax(t) + Bu(t) \right] \\ &= \left(A - B(CB)^{-1}CA \right) (z(t) - x(t)) \end{aligned}$$

Then $z_0 = x_0$ implies that

$$z(t) - x(t) = e^{\left(A - B(CB)^{-1}CA \right)t} (z_0 - x_0) = 0$$

for all $t \geq 0$. This gives

$$\begin{aligned} w(t) - u(t) &= -(CB)^{-1}CAz(t) + (CB)^{-1} \left(CAx(t) + DBu(t) \right) - u(t) \\ &= -(CB)^{-1}CA(z(t) - x(t)) \\ &= 0 \end{aligned}$$

for all $t \geq 0$ from which the desired conclusion follows.

Chapter 3

AE 3.1

a. For $n = 2$

$$|\lambda I - A| = \begin{vmatrix} \lambda & -1 \\ a_0 & \lambda + a_1 \end{vmatrix} = \lambda(\lambda + a_1) - (-1)a_0 = \lambda^2 + a_1\lambda + a_0$$

Suppose the identity holds for $n > 2$. For the $(n+1)$ -dimensional case, expand $|\lambda I - A|$ about the first column:

$$\begin{aligned} |\lambda I - A| &= \begin{vmatrix} \lambda & -1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 0 \\ 0 & 0 & 0 & \cdots & \lambda & -1 \\ a_0 & a_1 & a_2 & \cdots & a_{n-1} & \lambda + a_n \end{vmatrix} \\ &= \lambda(-1)^{1+1} \begin{vmatrix} \lambda & -1 & \cdots & 0 & 0 \\ 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \lambda & -1 \\ a_1 & a_2 & \cdots & a_{n-1} & \lambda + a_n \end{vmatrix} + a_0(-1)^{(n+1)+1} \begin{vmatrix} -1 & 0 & \cdots & 0 & 0 \\ \lambda & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & 0 \\ 0 & 0 & \cdots & \lambda & -1 \end{vmatrix} \end{aligned}$$

Invoking the induction hypothesis, the first determinant on the right hand side (noting the index shift in the elements in the bottom row) is

$$\lambda^n + a_n\lambda^{n-1} + a_{n-1}\lambda^{n-2} + \cdots + a_2\lambda + a_1$$

The second determinant on the right hand side is that of a lower triangular $n \times n$ matrix having -1 's on the main diagonal. Thus

$$\begin{aligned} |\lambda I - A| &= \lambda(-1)^2(\lambda^n + a_n\lambda^{n-1} + a_{n-1}\lambda^{n-2} + \cdots + a_2\lambda + a_1) + a_0(-1)^{(n+1)+1}(-1)^n \\ &= \lambda^{n+1} + a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_2\lambda^2 + a_1\lambda + a_0 \end{aligned}$$

which completes the induction argument.

b. For λ_i an eigenvalue of A , the vector

$$v_i = \begin{bmatrix} 1 \\ \lambda_i \\ \lambda_i^2 \\ \vdots \\ \lambda_i^{n-2} \\ \lambda_i^{n-1} \end{bmatrix}$$

is nonzero due to the 1 in the first component. By definition, λ_i is a root of the characteristic polynomial established in part a from which

$$\lambda_i^n = -a_0 - a_1\lambda_i - a_2\lambda_i^2 - \cdots - a_{n-2}\lambda_i^{n-2} - a_{n-1}\lambda_i^{n-1}$$

A direct calculation gives

$$\begin{aligned} \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix} \begin{bmatrix} 1 \\ \lambda_i \\ \lambda_i^2 \\ \vdots \\ \lambda_i^{n-2} \\ \lambda_i^{n-1} \end{bmatrix} &= \begin{bmatrix} \lambda_i \\ \lambda_i^2 \\ \lambda_i^3 \\ \vdots \\ \lambda_i^{n-1} \\ -a_0 - a_1\lambda_i - a_2\lambda_i^2 - \cdots - a_{n-2}\lambda_i^{n-2} - a_{n-1}\lambda_i^{n-1} \end{bmatrix} \\ &= \begin{bmatrix} \lambda_i \\ \lambda_i^2 \\ \lambda_i^3 \\ \vdots \\ \lambda_i^{n-1} \\ \lambda_i^n \end{bmatrix} \\ &= \lambda_i \begin{bmatrix} 1 \\ \lambda_i \\ \lambda_i^2 \\ \vdots \\ \lambda_i^{n-2} \\ \lambda_i^{n-1} \end{bmatrix} \end{aligned}$$

so that $Av_i = \lambda_i v_i$.

c. In order to show that $n_i = \text{nullity}(\lambda_i I - A) = 1$ for each distinct eigenvalue λ_i , by Sylvester's law of nullity it is equivalent to show that $\text{rank}(\lambda_i I - A) = n - 1$. Because $(\lambda_i I - A)$ is singular, $\text{rank}(\lambda_i I - A) < n$. The $(n-1) \times (n-1)$ submatrix of $(\lambda_i I - A)$ obtained by deleting the last row and first column is lower triangular having -1 's on the main diagonal and is hence nonsingular. It follows that $\text{rank}(\lambda_i I - A) = n - 1$ and thus $\text{nullity}(\lambda_i I - A) = 1$.

AE3.2 The controllability Gramian

$$W(t_0, t_f) = \int_{t_0}^{t_f} e^{A(t_0-\tau)} B B^T e^{A^T(t_0-\tau)} d\tau$$

clearly satisfies $W(t_f, t_f) = 0$. Applying the Liebniz rule yields

$$\begin{aligned} \frac{d}{dt} W(t, t_f) &= e^{A(t_0-t_f)} B B^T e^{A^T(t_0-t_f)}(0) - B B^T(1) + \int_t^{t_f} \left[A e^{A(t-\tau)} B B^T e^{A^T(t-\tau)} + e^{A(t-\tau)} B B^T e^{A^T(t-\tau)} A^T \right] d\tau \\ &= A W(t, t_f) + W(t, t_f) A^T - B B^T \end{aligned}$$

from which the identity follows.

AE3.3 Let

$$X(t) = \begin{bmatrix} e^{At} & e^{At} W(0, t) \\ 0 & e^{-A^T t} \end{bmatrix}$$

Using $W(0, 0) = 0$, we have

$$X(0) = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

Using

$$\begin{aligned} \frac{d}{dt} W(0, t) &= e^{-At} B B^T e^{-A^T t}(1) - B B^T(0) + \int_0^t \frac{d}{dt} \left[e^{-A\tau} B B^T e^{-A^T \tau} \right] d\tau \\ &= e^{-At} B B^T e^{-A^T t} \end{aligned}$$

we find

$$\dot{X}(t) = \begin{bmatrix} Ae^{At} & Ae^{At}W(0,t) + e^{At}(e^{-At}BB^Te^{-A^Tt}) \\ 0 & -A^Te^{-A^Tt} \end{bmatrix} = \begin{bmatrix} Ae^{At} & Ae^{At}W(0,t) + BB^Te^{-A^Tt} \\ 0 & -A^Te^{-A^Tt} \end{bmatrix}$$

to which we compare

$$\begin{bmatrix} A & BB^T \\ 0 & -A^T \end{bmatrix} X(t) = \begin{bmatrix} A & BB^T \\ 0 & -A^T \end{bmatrix} \begin{bmatrix} e^{At} & e^{At}W(0,t) \\ 0 & e^{-A^Tt} \end{bmatrix} = \begin{bmatrix} Ae^{At} & Ae^{At}W(0,t) + BB^Te^{-A^Tt} \\ 0 & -A^Te^{-A^Tt} \end{bmatrix}$$

from which the result follows.

AE 3.4 There are two implications to prove:

If the pair (A, B) is controllable, then $W_R(t_0, t_f)$ is nonsingular for any t_0 and any finite $t_f > t_0$.

If $W_R(t_0, t_f)$ is nonsingular for any t_0 and any finite $t_f > t_0$, then the pair (A, B) is controllable.

The first implication can be verified by a contradiction argument. Suppose the pair (A, B) is controllable, but $W_R(t_0, t_f)$ is singular for some t_0 and finite $t_f > t_0$. Consequently, there is a nonzero vector $x_f \in \mathbb{R}^n$ that satisfies

$$0 = x_f^T W_R(t_0, t_f) x_f = \int_{t_0}^{t_f} \|B^T e^{A(t_f - \tau)} x_f\|^2 d\tau$$

which, as argued in the proof of Lemma 3.3, implies that $x_f^T e^{A(t_f - \tau)} B = 0$ for all $\tau \in [t_0, t_f]$. On the other hand, controllability implies that this x_f is reachable from the origin at time t_f using a piecewise continuous input signal $u(t)$ defined on $[t_0, t_f]$. That is,

$$x_f = \int_{t_0}^{t_f} e^{A(t_f - \tau)} B u(\tau) d\tau$$

This yields

$$\|x_f\|^2 = x_f^T x_f = \int_{t_0}^{t_f} (x_f^T e^{A(t_f-\tau)} B) u(\tau) d\tau = 0$$

which contradicts the assumption that x_f is a nonzero vector. This contradiction indicates that

$W_R(t_0, t_f)$ must be nonsingular for any t_0 and any finite $t_f > t_0$.

For the second implication, assume for arbitrarily selected t_0 and finite $t_f > t_0$ that $W_R(t_0, t_f)$ is

nonsingular. For any $x_f \in \mathbb{R}^n$, define the input signal $u(t)$ on $[t_0, t_f]$

$$u(t) = B^T e^{A^T(t_f-t)} W_R^{-1}(t_0, t_f) x_f$$

For $x(t_0) = 0$

$$\begin{aligned} x(t_f) &= e^{A(t_f-t_0)} x(t_0) + \int_{t_0}^{t_f} e^{A(t_f-\tau)} B u(\tau) d\tau \\ &= \int_{t_0}^{t_f} e^{A(t_f-\tau)} B B^T e^{A^T(t_f-\tau)} d\tau W_R^{-1}(t_0, t_f) x_f \\ &= W_R(t_0, t_f) W_R^{-1}(t_0, t_f) x_f \\ &= x_f \end{aligned}$$

thereby demonstrating that x_f is a reachable state. Since x_f was arbitrary, the pair (A, B) is controllable. Note that this construction addresses the second part of the problem.

AE 3.5 Assume $B \neq 0 \in \mathbb{R}^n$ to avoid a trivial situation. Scanning the columns of the controllability matrix from left to right, let $r \geq 1$ denote the least integer such that $\{B, AB, A^2B, \dots, A^{r-1}B\}$ is a linearly independent set and $A^r B$ is linearly dependent on these vectors. If it should happen that each successive column $A^k B$, $k > r$ is also linearly dependent on the set $\{B, AB, A^2B, \dots, A^{r-1}B\}$, then it follows that $r = q$, yielding the desired conclusion.

That $A^k B$ is linearly dependent on the set $\{B, AB, A^2 B, \dots, A^{r-1} B\}$ for every $k > r$ can be established by induction on k . By definition of r ,

$$A^r B = c_0 B + c_1 AB + c_2 A^2 B + \dots + c_{r-1} A^{r-1} B$$

for scalars $c_0, c_1, c_2, \dots, c_{r-1}$. For $k = r + 1$

$$\begin{aligned} A^{r+1} B &= A(A^r B) \\ &= A(c_0 B + c_1 AB + c_2 A^2 B + \dots + c_{r-1} A^{r-1} B) \\ &= c_0 AB + c_1 A^2 B + c_2 A^3 B + \dots + c_{r-1} A^r B \\ &= c_0 AB + c_1 A^2 B + c_2 A^3 B + \dots + c_{r-1} (c_0 B + c_1 AB + c_2 A^2 B + \dots + c_{r-1} A^{r-1} B) \\ &= c_{r-1} c_0 B + (c_0 + c_{r-1} c_1) AB + (c_1 + c_{r-1} c_2) A^2 B + \dots + (c_{r-2} + c_{r-1}^2) A^{r-1} B \end{aligned}$$

indicating that $A^{r+1} B$ is linearly dependent on the set $\{B, AB, A^2 B, \dots, A^{r-1} B\}$. Suppose for arbitrary

$k > r$

$$A^k B = d_0 B + d_1 AB + d_2 A^2 B + \dots + d_{r-1} A^{r-1} B$$

for scalars $d_0, d_1, d_2, \dots, d_{r-1}$. Then

$$\begin{aligned} A^{k+1} B &= A(A^k B) \\ &= A(d_0 B + d_1 AB + d_2 A^2 B + \dots + d_{r-1} A^{r-1} B) \\ &= d_0 AB + d_1 A^2 B + d_2 A^3 B + \dots + d_{r-1} A^r B \\ &= d_0 AB + d_1 A^2 B + d_2 A^3 B + \dots + d_{r-1} (c_0 B + c_1 AB + c_2 A^2 B + \dots + c_{r-1} A^{r-1} B) \\ &= d_{r-1} c_0 B + (d_0 + d_{r-1} c_1) AB + (d_1 + d_{r-1} c_2) A^2 B + \dots + (d_{r-2} + d_{r-1} c_{r-1}) A^{r-1} B \end{aligned}$$

indicating that $A^{k+1} B$ is linearly dependent on the set $\{B, AB, A^2 B, \dots, A^{r-1} B\}$ which concludes the

induction argument.

AE 3.6 Because A is an $n \times n$ Jordan block matrix, A has one distinct eigenvalue μ with algebraic multiplicity n . Using the Popov-Belevetch-Hautus rank test for controllability, it equivalent to show that

$$\text{rank}[\mu I - A \quad B] = n \text{ if and only if } b_n \neq 0$$

The $n \times (n+1)$ matrix

$$[\mu I - A \quad B] = \left[\begin{array}{ccccc|c} 0 & -1 & 0 & \cdots & 0 & b_1 \\ 0 & 0 & -1 & \cdots & 0 & b_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & b_{n-1} \\ 0 & 0 & 0 & \cdots & 0 & b_n \end{array} \right]$$

has rank full row rank n if and only if the $n \times n$ submatrix

$$\left[\begin{array}{ccccc} -1 & 0 & \cdots & 0 & b_1 \\ 0 & -1 & \cdots & 0 & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & b_{n-1} \\ 0 & 0 & \cdots & 0 & b_n \end{array} \right]$$

is nonsingular. Due to its upper triangular structure, this matrix has determinant equal to $(-1)^{n-1}b_n$

which is nonzero if and only if $b_n \neq 0$.

AE 3.7 Clearly if $C_1 = C_2$ then $h_1(t) = C_1 e^{At} B = C_2 e^{At} B = h_2(t)$. For the converse statement, suppose

$h_1(t)$ and $h_2(t)$ are identical for all $t \geq 0$. It follows that

$$\begin{aligned} 0 &= \left. \frac{d^k}{dt^k} (h_1(t) - h_2(t)) \right|_{t=0} = (C_1 - C_2) \left. \frac{d^k}{dt^k} (e^{At} B) \right|_{t=0} = (C_1 - C_2) A^k e^{At} B \Big|_{t=0} \\ &= (C_1 - C_2) A^k B \end{aligned}$$

for $k = 0, \dots, n-1$. In other words

$$(C_1 - C_2) \begin{bmatrix} B & AB & A^2 B & \cdots & A^{n-1} B \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

Because the state equation is controllable, the controllability matrix has full row rank and so $C_1 - C_2 = 0$, or $C_1 = C_2$.

AE3.8 The state equation

$$\begin{aligned} \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} &= \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u(t) \\ y(t) &= [C_1 \quad C_2] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \end{aligned}$$

satisfies

$$\begin{aligned} [C_1 \quad C_2] \begin{bmatrix} sI - A_1 & 0 \\ 0 & sI - A_2 \end{bmatrix}^{-1} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} &= [C_1 \quad C_2] \begin{bmatrix} (sI - A_1)^{-1} & 0 \\ 0 & (sI - A_2)^{-1} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} \\ &= C_1 (sI - A_1)^{-1} B_1 + C_2 (sI - A_2)^{-1} B_2 \\ &= H_1(s) + H_2(s) \end{aligned}$$

and is therefore a realization of the parallel interconnection. Let $(\bar{A}, \bar{B})$ denote the coefficient matrices in this parallel realization. There are two implications to prove:

If the parallel realization is controllable, then A_1 and A_2 have no common eigenvalues.

If A_1 and A_2 have no common eigenvalues, then the parallel realization is controllable.

To prove the contrapositive of the first implication, suppose A_1 and A_2 have a common eigenvalue, λ .

Then there exist associated left eigenvectors w_1, w_2 for A_1, A_2 , respectively, so that

$$w_i \neq 0 \quad w_i^* A_i = \lambda w_i^* \quad i=1, 2$$

Since (A_1, B_1) and (A_2, B_2) are assumed to be controllable,

$$\alpha_1 \triangleq w_1^* B_1 \neq 0 \quad \text{and} \quad \alpha_2 \triangleq w_2^* B_2 \neq 0$$

We can then rescale w_2 to obtain $\hat{w}_2 = -\left(\frac{\alpha_1}{\alpha_2}\right)^* w_2$ which is also a left eigenvector of A_2 associated with the eigenvalue λ . From this we define

$$\bar{w} = \begin{bmatrix} w_1 \\ \hat{w}_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

which satisfies

$$\begin{aligned} \bar{w}^* \bar{A} &= \begin{bmatrix} w_1^* & \hat{w}_2^* \end{bmatrix} \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} = \begin{bmatrix} w_1^* A_1 & \hat{w}_2^* A_2 \end{bmatrix} = \begin{bmatrix} \lambda w_1^* & \lambda \hat{w}_2^* \end{bmatrix} = \lambda \begin{bmatrix} w_1^* & \hat{w}_2^* \end{bmatrix} \\ &= \lambda \bar{w}^* \end{aligned}$$

and

$$\begin{aligned} \bar{w}^* \bar{B} &= \begin{bmatrix} w_1^* & \hat{w}_2^* \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = w_1^* B_1 + \hat{w}_2^* B_2 = \alpha_1 - \left(\frac{\alpha_1}{\alpha_2} \right) \alpha_2 \\ &= 0 \end{aligned}$$

Thus, by the Popov-Belevitch-Hautus eigenvector test, the pair $(\bar{A}, \bar{B})$ is not controllable.

To prove the contrapositive of the second implication, suppose the pair $(\bar{A}, \bar{B})$ is not controllable. By the Popov-Belevitch-Hautus eigenvector test, there is an eigenvalue λ of $\bar{A}$ and a corresponding left eigenvector $\bar{w}$ satisfying

$$\bar{w} \neq 0 \quad \bar{w}^* \bar{A} = \lambda \bar{w}^* \quad \bar{w}^* \bar{B} = 0$$

Partitioning

$$\bar{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

we have

$$w_1^* A_1 = \lambda w_1^* \quad w_2^* A_2 = \lambda w_2^*$$

If we can show that both $w_1 \neq 0$ and $w_2 \neq 0$, then w_1 and w_2 are left eigenvectors for A_1 and A_2 , respectively, corresponding to the same eigenvalue λ , implying that A_1 and A_2 have a common eigenvalue. If $w_1 = 0$ then $w_2 \neq 0$ because $\bar{w} \neq 0$ as it is a left eigenvector of $\bar{A}$. In this case,

$$w_2 \neq 0 \quad w_2^* A_2 = \lambda w_2^* \quad 0 = \bar{w}^* \bar{B} = \begin{bmatrix} 0 & w_2^* \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = w_2^* B_2$$

which contradicts the assumed controllability of the pair (A_2, B_2) . Similarly, the assumption that $w_2 = 0$ contradicts the controllability of the pair (A_1, B_1) . Thus both $w_1 \neq 0$ and $w_2 \neq 0$, so that λ is an eigenvalue common to both A_1 and A_2 .

AE3.9 The state equation

$$\begin{aligned} \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} &= \begin{bmatrix} A_1 & B_1 C_2 \\ 0 & A_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ B_2 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} C_1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \end{aligned}$$

satisfies

$$\begin{aligned} \begin{bmatrix} C_1 & 0 \end{bmatrix} \begin{bmatrix} sI - A_1 & -B_1 C_2 \\ 0 & sI - A_2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ B_2 \end{bmatrix} &= \begin{bmatrix} C_1 & 0 \end{bmatrix} \begin{bmatrix} (sI - A_1)^{-1} & (sI - A_1)^{-1} B_1 C_2 (sI - A_2)^{-1} \\ 0 & (sI - A_2)^{-1} \end{bmatrix} \begin{bmatrix} 0 \\ B_2 \end{bmatrix} \\ &= C_1 (sI - A_1)^{-1} B_1 C_2 (sI - A_2)^{-1} B_2 \\ &= H_1(s) H_2(s) \end{aligned}$$

and is therefore a realization of the series interconnection. Let $(\bar{A}, \bar{B})$ denote the coefficient matrices in this series realization. There are two implications to prove:

If the series realization is controllable, no eigenvalue of A_1 is a zero of $H_2(s)$.

If no eigenvalue of A_1 is a zero of $H_2(s)$, then the series realization is controllable.

To prove the contrapositive of the first implication, suppose λ is an eigenvalue of A_1 for which $H_2(\lambda) = 0$. Let w_1 denote a left eigenvector of A_1 associated with λ so that by the assumed controllability of the pair (A_1, B_1) , $w_1^* B_1 \neq 0$. Next, with

$$H_2(s) = \frac{b_2(s)}{a_2(s)} = \frac{C_2 \text{adj}(sI - A_2) B_2}{|sI - A_2|}$$

the determinant identity of AE1.4 yields (recall AE2.11)

$$\begin{vmatrix} sI - A_2 & -B_2 \\ C_2 & 0 \end{vmatrix} = |sI - A_2| H_2(s) = b_2(s)$$

so that $H_2(\lambda) = 0$ implies

$$\begin{bmatrix} \lambda I - A_2 & -B_2 \\ C_2 & 0 \end{bmatrix}$$

is singular and therefore has linearly dependent rows. Consequently, there exists

$$\begin{bmatrix} w_2 \\ z_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ such that } \begin{bmatrix} w_2^* & z_2^* \end{bmatrix} \begin{bmatrix} \lambda I - A_2 & -B_2 \\ C_2 & 0 \end{bmatrix} = \begin{bmatrix} w_2^*(\lambda I - A_2) + z_2^* C_2 & -w_2^* B_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

Necessarily $z_2 \neq 0 \in \mathbb{C}$, for otherwise the preceding identity would contradict the assumed controllability of the pair (A_2, B_2) . Consequently, w_2 and z_2 can be scaled so that $z_2^* = -w_1^* B_1$. Next, using the preceding identity

$$\begin{aligned} \begin{bmatrix} w_1^* & w_2^* \end{bmatrix} \begin{bmatrix} A_1 & B_1 C_2 \\ 0 & A_2 \end{bmatrix} &= \begin{bmatrix} w_1^* A_1 & w_1^* B_1 C_2 + w_2^* A_2 \end{bmatrix} \\ &= \begin{bmatrix} w_1^* A_1 & -z_2^* C_2 + w_2^* A_2 \end{bmatrix} \\ &= \begin{bmatrix} \lambda w_1^* & \lambda w_2^* \end{bmatrix} \\ &= \lambda \begin{bmatrix} w_1^* & w_2^* \end{bmatrix} \end{aligned}$$

and

$$\begin{aligned} \begin{bmatrix} w_1^* & w_2^* \end{bmatrix} \begin{bmatrix} 0 \\ B_2 \end{bmatrix} &= w_2^* B_2 \\ &= 0 \end{aligned}$$

from which we conclude that the series realization is not controllable.

To prove the contrapositive of the second implication, suppose the series realization is not controllable,

so there exists $\lambda \in \mathbb{C}$ and

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ such that } \begin{bmatrix} w_1^* & w_2^* \end{bmatrix} \begin{bmatrix} A_1 & B_1 C_2 \\ 0 & A_2 \end{bmatrix} = \lambda \begin{bmatrix} w_1^* & w_2^* \end{bmatrix} \text{ and } \begin{bmatrix} w_1^* & w_2^* \end{bmatrix} \begin{bmatrix} 0 \\ B_2 \end{bmatrix} = 0$$

Observe that $w_1^* \neq 0$, for otherwise $w_2^* A_2 = \lambda w_2^*$ and $w_2^* B_2 = 0$, thereby contradicting the assumed controllability of the pair (A_2, B_2) . Thus, λ is an eigenvalue of A_1 with associated left eigenvector w_1 . It follows from the assumed controllability of the pair (A_1, B_1) that $w_1^* B_1 \neq 0$. With $z_2^* = -w_1^* B_1 \neq 0$, we have

$$\begin{bmatrix} w_2^* & z_2^* \end{bmatrix} \begin{bmatrix} \lambda I - A_2 & -B_2 \\ C_2 & 0 \end{bmatrix} = \begin{bmatrix} w_2^* (\lambda I - A_2) + z_2^* C_2 & -w_2^* B_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

indicating that

$$\begin{bmatrix} \lambda I - A_2 & -B_2 \\ C_2 & 0 \end{bmatrix}$$

is singular. The determinant identity used earlier indicates that $b_2(s)|_{s=\lambda} = 0$, so that λ is also a zero of $H_2(s)$.

AE3.10 There are two implications to prove:

If the pair (A, B) is controllable, then the only square matrix X that satisfies $AX = XA$ and $XB = 0$ is $X = 0$.

If the only square matrix X that satisfies $AX = XA$ and $XB = 0$ is $X = 0$, then the pair (A, B) is controllable.

To prove the contrapositive of the first implication, suppose there is a nonzero square matrix X that satisfies $AX = XA$ and $XB = 0$. A straightforward induction argument shows that $A^k X = X A^k$ for any integer $k \geq 0$ from which

$$X A^k B = A^k X B = 0 \quad \text{for all } k \geq 0$$

Because X is a nonzero matrix, at least one of its rows is a nonzero row vector, denoted r^T . Thus

$r^T A^k B = 0$ for all $k \geq 0$ from which

$$r^T \begin{bmatrix} B & AB & A^2 B & \cdots & A^{n-1} B \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

thereby indicating that the controllability matrix has less than full row rank which implies that the pair (A, B) is not controllable.

To prove the contrapositive of the second implication, suppose that the pair (A, B) is not controllable so there exists a left eigenvector w of A corresponding to an eigenvalue λ with $w^* B = 0$. Let v denote an associated right eigenvector from which we construct

$$X = v w^*$$

This so-called outer product yields a square matrix that, because v and w are nonzero vectors, is nonzero. Next, compare

$$AX = (Av)w^* = (\lambda v)w^* = \lambda X \quad \text{and} \quad XA = v(w^* A) = v(\lambda w^*) = \lambda(vw^*) = \lambda X$$

so that $AX = XA$. Also

$$XB = v(w^* B) = 0$$

We have shown that there exists a nonzero square matrix X that satisfies $AX = XA$ and $XB = 0$.

Chapter 4

AE4.1 There are two implications to prove:

If the n –dimensional controller canonical form realization of $H(s)$ is observable, then $H(s)$ has no pole-zero cancellations.

If $H(s)$ has no pole-zero cancellations, then the n –dimensional controller canonical form realization of $H(s)$ is observable.

To prove the contrapositive of the first implication, suppose $H(s)$ has a pole-zero cancellation. Let λ denote a common root of the numerator and denominator polynomials. By AE3.1 part a, the n th-order denominator polynomial is the characteristic polynomial of the $n \times n$ matrix A_{CCF} , hence λ is an eigenvalue of A_{CCF} for which, from AE3.1 part b, a corresponding right eigenvector is

$$\mathbf{v} = \begin{bmatrix} 1 \\ \lambda \\ \lambda^2 \\ \vdots \\ \lambda^{n-2} \\ \lambda^{n-1} \end{bmatrix}$$

Because λ is also a zero of $H(s)$

$$\begin{aligned} 0 &= b_0 + b_1\lambda + b_2\lambda^2 + \cdots + b_{n-2}\lambda^{n-2} + b_{n-1}\lambda^{n-1} \\ &= \begin{bmatrix} b_0 & b_1 & b_2 & \cdots & b_{n-2} & b_{n-1} \end{bmatrix} \begin{bmatrix} 1 \\ \lambda \\ \lambda^2 \\ \vdots \\ \lambda^{n-2} \\ \lambda^{n-1} \end{bmatrix} \\ &= C_{\text{CCF}} \mathbf{v} \end{aligned}$$

We conclude from the Popov-Belevitch-Hautus eigenvector test that the n –dimensional controller canonical form realization of $H(s)$ is not observable.

To prove the contrapositive of the second implication, suppose the n -dimensional controller canonical form realization of $H(s)$ is not observable. From the Popov-Belevitch-Hautus eigenvector test, there exists an eigenvalue λ of A_{CCF} and a corresponding right eigenvector v for which $C_{\text{CCF}}v=0$. As argued earlier, λ is also a pole of $H(s)$. By AE3.1 part c, the geometric multiplicity of λ is one, so any corresponding right eigenvector necessarily is a nonzero scalar multiple of

$$v = \begin{bmatrix} 1 \\ \lambda \\ \lambda^2 \\ \vdots \\ \lambda^{n-2} \\ \lambda^{n-1} \end{bmatrix}$$

and so we can assume v has this form without loss of generality. From an earlier identity, $C_{\text{CCF}}v=0$ implies that λ is a zero of $H(s)$. Thus $H(s)$ has a pole-zero cancellation.

AE4.2 The observability Gramian

$$M(t_0, t_f) = \int_{t_0}^{t_f} e^{A^T(\tau-t_0)} C^T C e^{A(\tau-t_0)} d\tau$$

clearly satisfies $M(t_f, t_f) = 0$. Applying the Liebniz rule yields

$$\begin{aligned} \frac{d}{dt} M(t, t_f) &= e^{A^T(t_f-t)} C^T C e^{A(t_f-t)} (0) - C^T C (1) + \int_t^{t_f} \left[(-A^T) e^{A^T(\tau-t)} C^T C e^{A(\tau-t)} + e^{A^T(\tau-t)} C^T C e^{A(\tau-t)} (-A) \right] d\tau \\ &= -A^T M(t, t_f) - M(t, t_f) A - C^T C \end{aligned}$$

from which the identity follows.

AE4.3 Let

$$X(t) = \begin{bmatrix} e^{At} & 0 \\ -e^{-A^T t} M(0,t) & e^{-A^T t} \end{bmatrix}$$

Using $M(0,0)=0$, we have

$$X(0) = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

Using

$$\begin{aligned} \frac{d}{dt} M(0,t) &= e^{A^T t} C^T C e^{At} (1) - C^T C(0) + \int_0^t \frac{d}{d\tau} \left[e^{A^T \tau} C^T C e^{A\tau} \right] d\tau \\ &= e^{A^T t} C^T C e^{At} \end{aligned}$$

we find

$$\dot{X}(t) = \begin{bmatrix} Ae^{At} & 0 \\ A^T e^{-A^T t} M(0,t) - e^{-A^T t} (e^{A^T t} C^T C e^{At}) & -A^T e^{-A^T t} \end{bmatrix} = \begin{bmatrix} Ae^{At} & 0 \\ A^T e^{-A^T t} M(0,t) - C^T C e^{At} & -A^T e^{-A^T t} \end{bmatrix}$$

to which we compare

$$\begin{aligned} \begin{bmatrix} A & 0 \\ -C^T C & -A^T \end{bmatrix} X(t) &= \begin{bmatrix} A & 0 \\ -C^T C & -A^T \end{bmatrix} \begin{bmatrix} e^{At} & 0 \\ -e^{-A^T t} M(0,t) & e^{-A^T t} \end{bmatrix} \\ &= \begin{bmatrix} Ae^{At} & 0 \\ -C^T C e^{At} + A^T e^{-A^T t} M(0,t) & -A^T e^{-A^T t} \end{bmatrix} \end{aligned}$$

from which the result follows.

AE4.4 Because A is an $n \times n$ Jordan block matrix, A has one distinct eigenvalue μ with algebraic multiplicity n . Using the Popov-Belevetch-Hautus rank test for observability, it equivalent to show that

$$\text{rank} \begin{bmatrix} C \\ \mu I - A \end{bmatrix} = n \quad \text{if and only if} \quad c_1 \neq 0$$

The $(n+1) \times n$ matrix

$$\begin{bmatrix} C \\ \mu I - A \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 & \cdots & c_n \\ 0 & -1 & 0 & \cdots & 0 \\ 0 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

has rank full column rank n if and only if the $n \times n$ submatrix

$$\begin{bmatrix} c_1 & c_2 & c_3 & \cdots & c_n \\ 0 & -1 & 0 & \cdots & 0 \\ 0 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -1 \end{bmatrix}$$

is nonsingular. Due to its upper triangular structure, this matrix has determinant equal to $c_1(-1)^{n-1}$ which is nonzero if and only if $c_1 \neq 0$.

AE4.5 There are two implications to prove:

If the pair (A, C) is observable, then the only square matrix X that satisfies $AX = XA$ and $CX = 0$ is $X = 0$.

If the only square matrix X that satisfies $AX = XA$ and $CX = 0$ is $X = 0$, then the pair (A, C) is observable.

To prove the contrapositive of the first implication, suppose there is a nonzero square matrix X that satisfies $AX = XA$ and $CX = 0$. A straightforward induction argument shows that $A^k X = X A^k$ for any integer $k \geq 0$ from which

$$CA^k X = C X A^k = 0 \quad \text{for all } k \geq 0$$

Because X is a nonzero matrix, at least one of its columns is a nonzero column vector, denoted c .

Thus $CA^k c = 0$ for all $k \geq 0$ from which

$$\begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} c = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

thereby indicating that the observability matrix has less than full column rank which implies that the pair (A, C) is not observable.

To prove the contrapositive of the second implication, suppose that the pair (A, C) is not observable so there exists a right eigenvector v of A corresponding to an eigenvalue λ with $Cv = 0$. Let w denote an associated left eigenvector from which we construct

$$X = vw^*$$

This so-called outer product yields a square matrix that, because v and w are nonzero vectors, is nonzero. Next, compare

$$AX = (Av)w^* = (\lambda v)w^* = \lambda X \quad \text{and} \quad XA = v(w^*A) = v(\lambda w^*) = \lambda(vw^*) = \lambda X$$

so that $AX = XA$. Also

$$CX = (Cv)w^* = 0$$

We have shown that there exists a nonzero square matrix X that satisfies $AX = XA$ and $CX = 0$.

AE4.6 Observe that

$$\begin{bmatrix} HC \\ \lambda I - (A - LC) \end{bmatrix} = \begin{bmatrix} H & 0 \\ L & I \end{bmatrix} \begin{bmatrix} C \\ \lambda I - A \end{bmatrix}$$

Since the left factor on the righthand side is nonsingular and matrix rank is unaffected by pre- and postmultiplication by nonsingular matrices, we conclude that

$$\text{rank} \begin{bmatrix} HC \\ \lambda I - (A - LC) \end{bmatrix} = \text{rank} \begin{bmatrix} C \\ \lambda I - A \end{bmatrix} \text{ for all } \lambda \in \mathbb{C}$$

Thus, by the Popov-Belevitch-Hautus rank test for observability, the pair (A, C) is observable if and only if the pair $(A - LC, HC)$ is observable for any $n \times p$ matrix L and any nonsingular $p \times p$ matrix H .

AE4.7 Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \text{ and } C = [C_1 \ 0]$$

with C_1 nonsingular. There are two implications to prove:

If the pair (A, C) is observable, then the pair (A_{22}, A_{12}) is observable.

If the pair (A_{22}, A_{12}) is observable, then the pair (A, C) is observable.

To prove the contrapositive of the first implication, suppose that the pair (A_{22}, A_{12}) is not observable so there exists an eigenvalue λ of A_{22} and a corresponding right eigenvector v_2 for which $A_{12}v_2 = 0$.

Thus,

$$\begin{bmatrix} 0 \\ v_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

satisfies

$$\begin{aligned} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} 0 \\ v_2 \end{bmatrix} &= \begin{bmatrix} A_{12}v_2 \\ A_{22}v_2 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ \lambda v_2 \end{bmatrix} \quad \text{and} \\ &= \lambda \begin{bmatrix} 0 \\ v_2 \end{bmatrix} \end{aligned} \qquad [C_1 \ 0] \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = 0$$

which indicates that the pair (A, C) is not observable.

To prove the contrapositive of the second implication, suppose that the pair (A, C) is not observable.

Hence there exists eigenvalue λ of A and a corresponding right eigenvector partitioned as

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

such that

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \text{ and } \begin{bmatrix} C_1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = C_1 v_1 = 0$$

Under the assumption that C_1 is nonsingular, $C_1 v_1 = 0$ implies $v_1 = 0$. Thus necessarily $v_2 \neq 0$ and the preceding eigenvalue-eigenvector relationship becomes

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = \begin{bmatrix} A_{12} v_2 \\ A_{22} v_2 \end{bmatrix} = \lambda \begin{bmatrix} 0 \\ v_2 \end{bmatrix}$$

from which $A_{22} v_2 = \lambda v_2$ and $A_{12} v_2 = 0$. It follows that the pair (A_{22}, A_{12}) is not observable.

Chapter 5

AE5.1 Minimality of each state equation is equivalent to joint controllability and observability.

Because matrix rank is unaffected by pre- and postmultiplication by nonsingular matrices, the identity

$$\begin{bmatrix} \lambda I - (A - BKC) & B \end{bmatrix} = \begin{bmatrix} \lambda I - A & B \end{bmatrix} \begin{bmatrix} I & 0 \\ KC & I \end{bmatrix}$$

implies that

$$\text{rank} \begin{bmatrix} \lambda I - (A - BKC) & B \end{bmatrix} = \text{rank} \begin{bmatrix} \lambda I - A & B \end{bmatrix} \quad \text{for all } \lambda \in \mathbb{C}$$

and the identity

$$\begin{bmatrix} C \\ \lambda I - (A - BKC) \end{bmatrix} = \begin{bmatrix} I & 0 \\ BK & I \end{bmatrix} \begin{bmatrix} C \\ \lambda I - A \end{bmatrix}$$

implies that

$$\text{rank} \begin{bmatrix} C \\ \lambda I - (A - BKC) \end{bmatrix} = \text{rank} \begin{bmatrix} C \\ \lambda I - A \end{bmatrix} \quad \text{for all } \lambda \in \mathbb{C}$$

It follows that the realization (A, B, C) is both controllable and observable if and only if the realization

$(A - BKC, B, C)$ is both controllable and observable for any gain matrix K .

AE 5.2 There are two implications to prove:

If the state equation is minimal, then A and $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ have no eigenvalue in common.

If A and $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ have no eigenvalue in common, then the state equation is minimal.

To prove the contrapositive of the first implication, suppose that A and

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

have a common eigenvalue $\lambda \in \mathbb{C}$. Let

$$\begin{bmatrix} v \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{C}^{n+1}$$

satisfy

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} v \\ z \end{bmatrix} = \lambda \begin{bmatrix} v \\ z \end{bmatrix}$$

which can be decomposed into $Av + Bz = \lambda v$ and $Cv + Dz = \lambda z$. There are two cases to examine:

$z = 0 \in \mathbb{C}$ and $z \neq 0 \in \mathbb{C}$. If $z = 0$, then $v \neq 0 \in \mathbb{C}^n$ and

$$Av = \lambda v \quad Cv = 0$$

which indicates that the state equation is not observable. If $z \neq 0$, then with $w \in \mathbb{C}^n$ a left eigenvector of A associated with the eigenvalue λ , the relationship

$$w^*(\lambda v) = w^*(Av + Bz) = (w^*A)v + (w^*B)z = \lambda w^*v + (w^*B)z$$

implies that $w^*B = 0$, indicating that the state equation is not controllable. We conclude that the state equation cannot be both controllable and observable and is therefore not minimal.

To prove the contrapositive of the second implication, suppose that the state equation is not minimal, so it is either not controllable, not observable, or both. If it is not controllable then there is an eigenvalue λ of A and a corresponding left eigenvector $w \in \mathbb{C}^n$ for which $w^*B = 0$. This gives

$$\begin{bmatrix} w^* & 0 \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} w^*A & w^*B \end{bmatrix} = \begin{bmatrix} \lambda w^* & 0 \end{bmatrix} = \lambda \begin{bmatrix} w^* & 0 \end{bmatrix}$$

which indicates that λ is also an eigenvalue of

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Alternatively, if the state equation is not observable, then there is an eigenvalue λ of A and a corresponding right eigenvector $v \in \mathbb{C}^n$ for which $Cv = 0$. This yields

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} v \\ 0 \end{bmatrix} = \begin{bmatrix} Av \\ Cv \end{bmatrix} = \begin{bmatrix} \lambda v \\ 0 \end{bmatrix} = \lambda \begin{bmatrix} v \\ 0 \end{bmatrix}$$

which indicates that λ is also an eigenvalue of

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Thus, in either case A and $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ have a common eigenvalue.

AE5.3 To prove the contrapositive, suppose A and the product BC do commute. It follows that for any integer $k \geq 0$, A^k and BC commute. Consider the product of the $n \times n$ controllability matrix and the $n \times n$ observability matrix

$$\begin{aligned} PQ &= \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \\ &= BC + ABCA + A^2BCA^2 + \cdots + A^{n-1}BCA^{n-1} \\ &= BC + A^2BC + A^4BC + \cdots + A^{2(n-1)}BC \end{aligned}$$

Since C is $1 \times n$ with $n \geq 2$, nullity $C = n - \text{rank } C \geq n - 1 \geq 1$. Hence there exists a nonzero vector

$x \in \mathbb{R}^n$ satisfying $Cx = 0$. It follows that

$$\begin{aligned}
PQx &= (BC + A^2BC + A^4BC + \cdots + A^{2(n-1)}BC)x \\
&= B(Cx) + A^2B(Cx) + A^4B(Cx) + \cdots + A^{2(n-1)}B(Cx) \\
&= 0
\end{aligned}$$

which indicates that the $n \times n$ product PQ is singular, implying the same for at least one of the factors.

This implies that the realization cannot be both controllable and observable and is therefore not minimal.

Chapter 6

AE6.1 With $\|\cdot\|$ denoting the Euclidean norm, along any trajectory of the homogeneous linear state equation

$$\begin{aligned}
 \frac{d}{dt}\|x(t)\|^2 &= \frac{d}{dt}x^T(t)x(t) \\
 &= \dot{x}^T(t)x(t) + x^T(t)\dot{x}(t) \\
 &= x^T(t)A^T x(t) + x^T(t)Ax(t) \\
 &= x^T(t)(A^T + A)x(t) \\
 &= 0
 \end{aligned}$$

and so $\|x(t)\| = \|x_0\|$ for all $t \geq 0$. It follows that homogeneous state equation (more precisely, the equilibrium state $\tilde{x} = 0 \in \mathbb{R}^n$) is stable (take $\gamma = 1$ in Definition 6.2) but not asymptotically stable.

AE6.2 The identity clearly holds at $t = 0$. From

$$\begin{aligned}
 \frac{d}{dt} \left[e^{A^T t} P e^{A t} + \int_0^t e^{A^T \tau} Q e^{A \tau} d\tau \right] &= e^{A^T t} A^T P e^{A t} + e^{A^T t} P A e^{A t} + e^{A^T t} Q e^{A t} \\
 &= e^{A^T t} (A^T P + P A + Q) e^{A t} \\
 &= 0
 \end{aligned}$$

we conclude that $e^{A^T t} P e^{A t} + \int_0^t e^{A^T \tau} Q e^{A \tau} d\tau$ is constant and therefore must equal P for all $t \geq 0$.

AE6.3 The eigenvalues of A have real part less than $-\mu$ if and only if the eigenvalues of $\mu I + A$ have negative real part. Upon re-writing the matrix equation as

$$(\mu I + A)^T P + P(\mu I + A) = -Q$$

the desired conclusion follows from Theorem 6.4.

AE6.4 With $\|\cdot\|$ denoting the Euclidean norm, along any trajectory of the perturbed homogeneous linear state equation

$$\begin{aligned}
 \frac{d}{dt} \left[x^T(t) P x(t) \right] &= \dot{x}^T(t) P x(t) + x^T(t) P \dot{x}(t) \\
 &= x^T(t) (A + \Delta A)^T P x(t) + x^T(t) P (A + \Delta A) x(t) \\
 &= x^T(t) (A^T P + P A) x(t) + 2 x^T(t) P \Delta A x(t) \\
 &= -\|x(t)\|^2 + 2 x^T(t) P \Delta A x(t)
 \end{aligned}$$

For a symmetric positive definite matrix, the spectral norm induced by the Euclidean vector norm satisfies $\|P\| = \lambda_{\max}(P)$. The Cauchy-Schwarz inequality applied pointwise in t gives

$$\begin{aligned}
 x^T(t) P \Delta A x(t) &= (P x(t))^T (\Delta A x(t)) \\
 &\leq \|P x(t)\| \|\Delta A x(t)\| \\
 &\leq \|P\| \|\Delta A\| \|x(t)\|^2 \\
 &= \lambda_{\max}(P) \|\Delta A\| \|x(t)\|^2
 \end{aligned}$$

The spectral norm bound

$$\|\Delta A\| < \frac{1}{2\lambda_{\max}(P)}$$

yields $\varepsilon \triangleq 1 - 2\lambda_{\max}(P)\|\Delta A\| > 0$ from which

$$\frac{d}{dt} \left[x^T(t) P x(t) \right] \leq -\varepsilon \|x(t)\|^2$$

Following similar steps as in the Exponential Stability discussion of Section 6.1, this bound implies that the perturbed homogeneous linear state equation is exponentially, hence asymptotically, stable.

AE 6.5 Define

$$W = \int_0^{\infty} e^{At} B B^T e^{A^T t} dt$$

As argued in the proof of Theorem 6.2, this improper integral is well defined because A has negative real-part eigenvalues. Also, W is symmetric by inspection. To show that W is positive definite, consider for any $x \in \mathbb{R}^n$

$$\begin{aligned} x^T W x &= \int_0^{\infty} (x^T e^{At} B)(B^T e^{A^T t} x) dt \\ &= \int_0^{\infty} \|B^T e^{A^T t} x\|^2 dt \end{aligned}$$

Clearly, $x^T W x \geq 0$ with equality if and only if $B^T e^{A^T t} x = 0 \in \mathbb{R}^m$ for all $t \geq 0$. Repeated differentiation and subsequent evaluation at $t = 0$ yields

$$0 = \left. \frac{d^k}{dt^k} B^T e^{A^T t} x \right|_{t=0} = B^T (A^T)^k x \quad \text{for } k = 0, \dots, n-1$$

Taking transposes gives

$$x^T \begin{bmatrix} B & AB & A^2 B & \cdots & A^{n-1} B \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

Under the assumption that the pair (A, B) is controllable, we must have $x = 0$. Thus $x^T W x \geq 0$ with equality if and only if $x = 0$, so that W is positive definite. Finally, W satisfies the Lyapunov matrix equation because

$$\begin{aligned} AW + WA^T &= \int_0^{\infty} \left[A e^{At} B B^T e^{A^T t} + e^{At} B B^T e^{A^T t} A^T \right] dt \\ &= \int_0^{\infty} \frac{d}{dt} \left[e^{At} B B^T e^{A^T t} \right] dt \\ &= \left. e^{At} B B^T e^{A^T t} \right|_0^{\infty} \\ &= 0 - B B^T \\ &= -B B^T \end{aligned}$$

and is in fact the unique solution.

AE6.6 Let λ be any eigenvalue of A and $w \in \mathbb{C}^n$ be a corresponding left eigenvector. Premultiplying the Lyapunov matrix equation by w^* and postmultiplying by w yields

$$\begin{aligned}
 -w^* B B^T w &= w^* [A W + W A^T] w \\
 &= w^* A W w + w^* W A^T w \\
 &= \lambda w^* W w + \bar{\lambda} w^* W w \\
 &= (\lambda + \bar{\lambda}) w^* W w \\
 &= 2 \operatorname{Re}(\lambda) w^* W w
 \end{aligned}$$

Since $w \neq 0$ and W is a positive definite matrix, $w^* W w > 0$. Under the assumption that the pair (A, B) is controllable, the Popov-Belevitch-Hautus eigenvector test ensures that $w^* B \neq 0$ from which $w^* B B^T w = \|B^T w\|^2 > 0$. Consequently,

$$\operatorname{Re}(\lambda) = -\frac{1}{2} \frac{w^* B B^T w}{w^* W w} < 0$$

from which we conclude that every eigenvalue of A necessarily has a strictly negative real part.

AE6.7 The input $u(t) = e^{-\lambda t}$, $t \geq 0$ has Laplace transform $U(s) = \frac{1}{s + \lambda}$. By the convolution property of the Laplace transform, the zero-state response

$$y(t) = \int_0^t h(t-\tau)u(\tau)d\tau = \int_0^t h(t-\tau)e^{-\lambda\tau}d\tau$$

has Laplace transform

$$\int_0^\infty y(t)e^{-st}dt = Y(s) = H(s)U(s) = H(s)\frac{1}{s + \lambda}$$

whose region of convergence is

$$\{s \in \mathbb{C} \mid \operatorname{Re}(s) > \max\{-\lambda, \operatorname{Re}(p_1), \dots, \operatorname{Re}(p_n)\}\}$$

in which $p_1, \dots, p_n$ denote the poles of $H(s)$. For a bounded-input, bounded-output stable state equation, each pole has negative real-part. Thus for $\lambda > 0$, any $\mu > 0$ lies in the region of convergence so that

$$\int_0^\infty y(t)e^{-\mu t}dt = Y(\mu) = H(\mu)\frac{1}{\mu + \lambda}$$

is well defined. If the state equation is not bounded-input, bounded-output stable, this relationship still holds for any $\lambda > 0$ provided that $\mu > \max\{\operatorname{Re}(p_1), \dots, \operatorname{Re}(p_n)\}$.

Chapter 7

AE7.1 For the $(n+1)$ –dimensional state equation, set

$$A_e = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} \quad B_e = \begin{bmatrix} B \\ 0 \end{bmatrix}$$

A straightforward induction argument shows that

$$A_e^k = \begin{bmatrix} A^k & 0 \\ CA^{k-1} & 0 \end{bmatrix} \quad \text{for all } k \geq 1$$

from which we construct the $(n+1) \times (n+1)$ controllability matrix

$$\begin{aligned} P_e &= \begin{bmatrix} B_e & A_e B_e & A_e^2 B_e & \cdots & A_e^{n-1} B_e & A_e^n B_e \end{bmatrix} \\ &= \begin{bmatrix} B & AB & A^2 B & \cdots & A^{n-1} B & A^n B \\ 0 & CB & CAB & \cdots & CA^{n-2} B & CA^{n-1} B \end{bmatrix} \end{aligned}$$

The key observation is that P_e can be factored as

$$\begin{aligned} P_e &= \begin{bmatrix} A & B \\ C & 0 \end{bmatrix} \begin{bmatrix} 0 & B & AB & \cdots & A^{n-2} B & A^{n-1} B \\ 1 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} A & B \\ C & 0 \end{bmatrix} \begin{bmatrix} 0 & P \\ 1 & 0 \end{bmatrix} \end{aligned}$$

Thus, P_e is nonsingular if and only if each factor is nonsingular and the second factor is nonsingular if and only if the $n \times n$ controllability matrix P is nonsingular. It follows that the pair (A_e, B_e) is controllable if and only if

$$\begin{bmatrix} A & B \\ C & 0 \end{bmatrix}$$

is nonsingular and the pair (A, B) is controllable.

AE7.2 From Chapter 3, for any finite $t_f > 0$ the controllability Gramian

$$W(0, t_f) = \int_0^{t_f} e^{-At} BB^T e^{-A^T t} dt$$

is positive definite and hence nonsingular. In addition

$$\begin{aligned} AW(0, t_f) + W(0, t_f)A^T &= \int_0^{t_f} \left[Ae^{-At} BB^T e^{-A^T t} + e^{-At} BB^T e^{-A^T t} A^T \right] dt \\ &= - \int_0^{t_f} \frac{d}{dt} \left[e^{-At} BB^T e^{-A^T t} \right] dt \\ &= -e^{-At} BB^T e^{-A^T t} \Big|_0^{t_f} \\ &= -e^{-At_f} BB^T e^{-A^T t_f} + BB^T \end{aligned}$$

With $K = B^T W^{-1}(0, t_f)$, $W(0, t_f)$ satisfies the Lyapunov matrix equation

$$\begin{aligned} (A - BK)W(0, t_f) + W(0, t_f)(A - BK)^T &= AW(0, t_f) + W(0, t_f)A^T - 2BB^T \\ &= - \left(e^{-At_f} BB^T e^{-A^T t_f} + BB^T \right) \end{aligned}$$

Let λ be any eigenvalue of $A - BK$ and $w \in \mathbb{C}^n$ be a corresponding left eigenvector. Premultiplying the Lyapunov matrix equation by w^* and postmultiplying by w yields

$$\begin{aligned} -w^* \left(e^{-At_f} BB^T e^{-A^T t_f} + BB^T \right) w &= w^* \left[(A - BK)W(0, t_f) + W(0, t_f)(A - BK)^T \right] w \\ &= w^* (A - BK)W(0, t_f)w + w^* W(0, t_f)(A - BK)^T w \\ &= \lambda w^* W(0, t_f)w + \bar{\lambda} w^* W(0, t_f)w \\ &= (\lambda + \bar{\lambda}) w^* W(0, t_f)w \\ &= 2 \operatorname{Re}(\lambda) w^* W(0, t_f)w \end{aligned}$$

Since $w \neq 0$ and $W(0, t_f)$ is a positive definite matrix, $w^* W(0, t_f)w > 0$. Under the assumption that the pair (A, B) is controllable, so is the pair $(A - BK, B)$ for any gain matrix K as guaranteed by Theorem 3.9. Consequently, the Popov-Belevitch-Hautus eigenvector test ensures that $w^* B \neq 0$ from which

$w^* BB^T w = \|B^T w\|^2 > 0$. Therefore,

$$w^* \left(e^{-At_f} BB^T e^{-A^T t_f} + BB^T \right) w = w^* e^{-At_f} BB^T e^{-A^T t_f} w + w^* BB^T w \geq w^* BB^T w > 0$$

and as a consequence

$$\operatorname{Re}(\lambda) = -\frac{1}{2} \frac{w^* \left(e^{-At_f} B B^T e^{-A^T t_f} + B B^T \right) w}{w^* W(0, t_f) w} \leq -\frac{1}{2} \frac{w^* B B^T w}{w^* W(0, t_f) w} < 0$$

from which we conclude that every eigenvalue of $A - BK$ necessarily has a strictly negative real part.

Hence, the closed-loop state equation

$$\dot{x}(t) = (A - BK)x(t)$$

is asymptotically stable.

AE7.3 P satisfies the Lyapunov matrix equation

$$\begin{aligned} (A - BK)^T P + P(A - BK) &= A^T P + P A - 2 P B B^T P \\ &= -(Q + P B B^T P) \end{aligned}$$

Because Q is symmetric and positive definite and $P B B^T P$ is symmetric and (at least) positive semidefinite, $Q + P B B^T P$ is symmetric and positive definite. Because P is also symmetric and positive definite, Theorem 6.4 indicates that every eigenvalue of $A - BK$ has strictly negative real part and so the closed-loop state equation

$$\dot{x}(t) = (A - BK)x(t)$$

that is asymptotically stable, thereby implying that the pair (A, B) is stabilizable.

AE7.4 First observe that

$$\begin{aligned}(sI - A + BK)^{-1} &= \left[(sI - A) \left(I + (sI - A)^{-1} BK \right) \right]^{-1} \\ &= \left[I + (sI - A)^{-1} BK \right]^{-1} (sI - A)^{-1}\end{aligned}$$

Next, upon identifying $X \leftrightarrow (sI - A)^{-1} B$ and $Y \leftrightarrow -K$, an application of the identity in AE1.8 yields

$$\left[I + (sI - A)^{-1} BK \right]^{-1} (sI - A)^{-1} B = (sI - A)^{-1} B \left[I + K (sI - A)^{-1} B \right]^{-1}$$

Thus

$$\begin{aligned}H_{\text{CL}}(s) &= C(sI - A + BK)^{-1} BG \\ &= C \left[I + (sI - A)^{-1} BK \right]^{-1} (sI - A)^{-1} BG \\ &= C(sI - A)^{-1} B \left[I + K (sI - A)^{-1} B \right]^{-1} G \\ &= H(s) \left[I + K (sI - A)^{-1} B \right]^{-1} G\end{aligned}$$

as desired.

Chapter 8

AE8.1 We argue that, beginning with an observable open-loop state equation, state feedback can yield an unobservable closed-loop state equation, and vice-versa. The first step is to show that open-loop and closed-loop transfer functions related by state feedback have the same numerator polynomial.

Consider an n – dimensional single-input, single-output state equation that is a realization of the transfer function

$$H(s) = \frac{b(s)}{a(s)}$$

with $n = \deg a(s)$ so that $a(s) = |sI - A|$ and $b(s) = C \operatorname{adj}(sI - A)B$. Given any state feedback gain K ,

let $\alpha(s) = |sI - A + BK|$ and also define the polynomial $k(s) \triangleq K \operatorname{adj}(sI - A)B$. Identifying

$X \leftrightarrow (sI - A)^{-1}B$, $Y \leftrightarrow -K$, an application of AE1.5 produces

$$\begin{aligned} \alpha(s) &= |sI - A + BK| \\ &= |sI - A| |I + (sI - A)^{-1}BK| \\ &= |sI - A| \left(1 + K(sI - A)^{-1}B \right) \\ &= a(s) \left(1 + \frac{k(s)}{a(s)} \right) \\ &= a(s) + k(s) \end{aligned}$$

so that

$$\left[I + K(sI - A)^{-1}B \right]^{-1} = \frac{1}{1 + K(sI - A)^{-1}B} = \frac{1}{1 + \frac{k(s)}{a(s)}} = \frac{a(s)}{a(s) + k(s)} = \frac{a(s)}{\alpha(s)}$$

Next, an application of AE7.4 gives

$$\begin{aligned}
H_{\text{CL}}(s) &= H(s) \left[I + K(sI - A)^{-1} B \right]^{-1} \\
&= \frac{b(s)}{a(s)} \frac{a(s)}{\alpha(s)} \\
&= \frac{b(s)}{\alpha(s)}
\end{aligned}$$

Thus, the closed-loop denominator polynomial matches the characteristic polynomial achieved by state feedback and the closed-loop numerator polynomial is the same as the open-loop numerator polynomial.

Suppose the open-loop state equation is controllable and observable, hence minimal, and consequently $H(s)$ is irreducible. Controllability implies that any desired degree $-n$ characteristic polynomial $\alpha(s)$ can be achieved by state feedback. Choosing $\alpha(s)$ to have a common root with $b(s)$, we see that $H_{\text{CL}}(s)$ is not irreducible and so the closed-loop state equation is not minimal. The closed-loop state equation is controllable because controllability is unaffected by state feedback (see Theorem 3.9) and so it cannot be observable.

Conversely, suppose the open-loop state equation is controllable but not observable, so that $H(s)$ is not irreducible. Choosing $\alpha(s)$ so that it does not have a root in common with $b(s)$, $H_{\text{CL}}(s)$ is irreducible. Thus, the closed-loop state equation is minimal and hence observable.

AE8.2 Let λ be any eigenvalue of $A - LC$ and $w \in \mathbb{C}^n$ a corresponding left eigenvector. Then

$$\begin{aligned}
\begin{bmatrix} w^* & -w^* \end{bmatrix} \begin{bmatrix} A & -BK \\ LC & A - BK - LC \end{bmatrix} &= \begin{bmatrix} w^*A - w^*LC & -w^*BK - w^*(A - BK - LC) \end{bmatrix} \\
&= \begin{bmatrix} w^*(A - LC) & -w^*(A - LC) \end{bmatrix} \\
&= \begin{bmatrix} \lambda w^* & -\lambda w^* \end{bmatrix} \\
&= \lambda \begin{bmatrix} w^* & -w^* \end{bmatrix}
\end{aligned}$$

and

$$\begin{bmatrix} w^* & -w^* \end{bmatrix} \begin{bmatrix} B \\ B \end{bmatrix} = w^*B - w^*B = 0$$

By the Popov-Belevitch-Eigenvector test, the closed-loop state equation (8.5) is not controllable.

AE8.3 We first show that the pair

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \quad B = \begin{bmatrix} I \\ 0 \end{bmatrix}$$

is controllable if and only if the pair (A_{22}, A_{21}) is controllable. First suppose that the pair (A_{22}, A_{21}) is not controllable so there exists an eigenvalue λ of A_{22} and a corresponding left eigenvector w_2 for which $w_2^* A_{21} = 0$. Thus,

$$\begin{bmatrix} 0 \\ w_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

satisfies

$$\begin{aligned} \begin{bmatrix} 0 & w_2^* \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} &= \begin{bmatrix} w_2^* A_{21} & w_2^* A_{22} \end{bmatrix} & \begin{bmatrix} 0 & w_2^* \end{bmatrix} \begin{bmatrix} I \\ 0 \end{bmatrix} &= 0 \\ &= \begin{bmatrix} 0 & \lambda w_2^* \end{bmatrix} \\ &= \lambda \begin{bmatrix} 0 & w_2^* \end{bmatrix} \end{aligned}$$

which indicates that the pair (A, B) is not controllable.

Next, suppose that the pair (A, B) is not controllable. Hence there exists eigenvalue λ of A and a corresponding left eigenvector partitioned as

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

such that

$$\begin{bmatrix} w_1^* & w_2^* \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \lambda \begin{bmatrix} w_1^* & w_2^* \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} w_1^* & w_2^* \end{bmatrix} \begin{bmatrix} I \\ 0 \end{bmatrix} = w_1^* = 0$$

The preceding eigenvalue-eigenvector relationship becomes

$$\begin{bmatrix} 0 & w_2^* \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} w_2^* A_{21} & w_2^* A_{22} \end{bmatrix} = \lambda \begin{bmatrix} 0 & w_2^* \end{bmatrix}$$

from which $w_2^* A_{22} = \lambda w_2^*$ and $w_2^* A_{21} = 0$. It follows that the pair (A_{22}, A_{21}) is not controllable.

Now, let

$$L = \begin{bmatrix} L_1 \\ L_2 \end{bmatrix} \quad \text{and} \quad K$$

locate the eigenvalues of

$$A - LC = \begin{bmatrix} A_{11} - L_1 C_1 & A_{12} - L_1 C_2 \\ A_{21} - L_2 C_1 & A_{22} - L_2 C_2 \end{bmatrix} \quad \text{and} \quad F = A_{22} - A_{21} K$$

at desired locations in the open left half of the complex plane. Consider the $(n-m)$ -dimensional compensator given by

$$\begin{aligned} \dot{z}(t) &= (F - L_2(C_2 - C_1 K))z(t) - L_2 y(t) \\ u(t) &= (A_{12} + KF - A_{11}K - (L_1 + KL_2)(C_2 - C_1 K))z(t) - (L_1 + KL_2)y(t) \end{aligned}$$

The closed-loop state equation is then

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{z}(t) \end{bmatrix} = \begin{bmatrix} A_{11} - (L_1 + KL_2)C_1 & A_{12} - (L_1 + KL_2)C_2 & A_{12} + KF - A_{11}K - (L_1 + KL_2)(C_2 - C_1 K) \\ A_{21} & A_{22} & 0 \\ -L_2 C_1 & -L_2 C_2 & F - L_2(C_2 - C_1 K) \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ z(t) \end{bmatrix}$$

$$y(t) = \begin{bmatrix} C_1 & C_2 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ z(t) \end{bmatrix}$$

in which we identify the closed-loop system dynamics matrix A_{CL} . Consider the closed-loop state coordinate transformation represented by the matrix

$$T_{\text{CL}} = \begin{bmatrix} I & 0 & -K \\ 0 & I & I \\ 0 & 0 & -I \end{bmatrix}$$

which satisfies $T_{\text{CL}}^{-1} = T_{\text{CL}}$. A tedious calculation reveals

$$T_{\text{CL}}^{-1} A_{\text{CL}} T_{\text{CL}} = \begin{bmatrix} A_{11} - L_1 C_1 & A_{12} - L_1 C_2 & 0 \\ A_{21} - L_2 C_1 & A_{22} - L_2 C_2 & 0 \\ L_2 C_1 & L_2 C_2 & F \end{bmatrix}$$

from which we conclude that the $2n - m$ closed-loop eigenvalues are those of

$$A - LC = \begin{bmatrix} A_{11} - L_1 C_1 & A_{12} - L_1 C_2 \\ A_{21} - L_2 C_1 & A_{22} - L_2 C_2 \end{bmatrix} \quad \text{and} \quad F = A_{22} - A_{21} K$$

AE8.4 The closed-loop state equation is

$$\begin{bmatrix} \dot{x}(t) \\ \dot{z}(t) \end{bmatrix} = \begin{bmatrix} A + BNC & B(M + NCH) \\ GC & F + GCH \end{bmatrix} \begin{bmatrix} x(t) \\ z(t) \end{bmatrix}$$

The state coordinate transformation matrix and its inverse are

$$\begin{bmatrix} I & H \\ 0 & I \end{bmatrix} \begin{bmatrix} I & H \\ 0 & I \end{bmatrix}^{-1} = \begin{bmatrix} I & -H \\ 0 & I \end{bmatrix}$$

Adjusting for the fact that the coordinate transformation as specified does not follow our standard convention, we compute

$$\begin{aligned} & \begin{bmatrix} I & H \\ 0 & I \end{bmatrix} \begin{bmatrix} A + BNC & B(M + NCH) \\ GC & F + GCH \end{bmatrix} \begin{bmatrix} I & -H \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} A + BNC + HGC & -(A + BNC + HGC)H + B(M + NCH) + H(F + GCH) \\ GC & F \end{bmatrix} \\ &= \begin{bmatrix} A + (BN + HG)C & -AH + BM + HF \\ GC & F \end{bmatrix} \\ &= \begin{bmatrix} A - LC & 0 \\ GC & F \end{bmatrix} \end{aligned}$$

We conclude from the block triangular structure so obtained that the $2n - m$ closed-loop eigenvalues are those of F together with those of $A - LC$.

AE 8.5 The state equation

$$\begin{aligned} \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} &= \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} C_1 & C_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \end{aligned}$$

satisfies

$$\begin{aligned} \begin{bmatrix} C_1 & C_2 \end{bmatrix} \begin{bmatrix} sI - A_1 & 0 \\ 0 & sI - A_2 \end{bmatrix}^{-1} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} &= \begin{bmatrix} C_1 & C_2 \end{bmatrix} \begin{bmatrix} (sI - A_1)^{-1} & 0 \\ 0 & (sI - A_2)^{-1} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} \\ &= C_1(sI - A_1)^{-1}B_1 + C_2(sI - A_2)^{-1}B_2 \\ &= H_1(s) + H_2(s) \end{aligned}$$

and is therefore a realization of the parallel interconnection. Let $(\bar{A}, \bar{C})$ denote the coefficient matrices in this parallel realization. There are two implications to prove:

If the parallel realization is observable, then A_1 and A_2 have no common eigenvalues.

If A_1 and A_2 have no common eigenvalues, then the parallel realization is observable.

To prove the contrapositive of the first implication, suppose A_1 and A_2 have a common eigenvalue, λ .

Then there exist associated right eigenvectors v_1, v_2 for A_1, A_2 , respectively, so that

$$v_i \neq 0 \quad A_i v_i = \lambda v_i \quad i = 1, 2$$

Since (A_1, C_1) and (A_2, C_2) are assumed to be observable,

$$\alpha_1 \triangleq C_1 v_1 \neq 0 \quad \text{and} \quad \alpha_2 \triangleq C_2 v_2 \neq 0$$

We can then rescale v_2 to obtain $\hat{v}_2 = -\frac{\alpha_1}{\alpha_2} v_2$ which is also a right eigenvector of A_2 associated with the eigenvalue λ . From this we define

$$\bar{v} = \begin{bmatrix} v_1 \\ \hat{v}_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

which satisfies

$$\begin{aligned} \bar{A}\bar{v} &= \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \begin{bmatrix} v_1 \\ \hat{v}_2 \end{bmatrix} = \begin{bmatrix} A_1 v_1 \\ A_2 \hat{v}_2 \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda \hat{v}_2 \end{bmatrix} = \lambda \begin{bmatrix} v_1 \\ \hat{v}_2 \end{bmatrix} \\ &= \lambda \bar{v} \end{aligned}$$

and

$$\begin{aligned} \bar{C}\bar{v} &= [C_1 \ C_2] \begin{bmatrix} v_1 \\ \hat{v}_2 \end{bmatrix} = C_1 v_1 + C_2 \hat{v}_2 = \alpha_1 - \left(\frac{\alpha_1}{\alpha_2} \right) \alpha_2 \\ &= 0 \end{aligned}$$

Thus, by the Popov-Belevitch-Hautus eigenvector test, the pair $(\bar{A}, \bar{C})$ is not observable.

To prove the contrapositive of the second implication, suppose the pair $(\bar{A}, \bar{C})$ is not observable. By the Popov-Belevitch-Hautus eigenvector test, there is an eigenvalue λ of $\bar{A}$ and a corresponding right eigenvector $\bar{v}$ satisfying

$$\bar{v} \neq 0 \quad \bar{A}\bar{v} = \lambda \bar{v} \quad \bar{C}\bar{v} = 0$$

Partitioning

$$\bar{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

we have

$$A_1 v_1 = \lambda v_1 \quad A_2 v_2 = \lambda v_2$$

If we can show that both $v_1 \neq 0$ and $v_2 \neq 0$, then v_1 and v_2 are right eigenvectors for A_1 and A_2 , respectively, corresponding to the same eigenvalue λ , implying that A_1 and A_2 have a common eigenvalue. If $v_1 = 0$ then $v_2 \neq 0$ because $\bar{v} \neq 0$ as it is a right eigenvector of $\bar{A}$. In this case,

$$v_2 \neq 0 \quad A_2 v_2 = \lambda v_2 \quad 0 = \bar{C} \bar{v} = [C_1 \ C_2] \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = C_2 v_2$$

which contradicts the assumed observability of the pair (A_2, C_2) . Similarly, the assumption that $v_2 = 0$ contradicts the assumed observability of the pair (A_1, C_1) . Thus both $v_1 \neq 0$ and $v_2 \neq 0$, so that λ is an eigenvalue common to both A_1 and A_2 .

Chapter 9

AE 9.1 Subtracting the algebraic Riccati equations satisfied by $\bar{P}_1$ and $\bar{P}_2$, respectively, gives

$$\begin{aligned}
 0 &= A^T (\bar{P}_1 - \bar{P}_2) + (\bar{P}_1 - \bar{P}_2) A - \bar{P}_1 B R^{-1} B^T \bar{P}_1 + \bar{P}_2 B R^{-1} B^T \bar{P}_2 \\
 &= A^T (\bar{P}_1 - \bar{P}_2) + (\bar{P}_1 - \bar{P}_2) A - \bar{P}_1 B R^{-1} B^T \bar{P}_1 + \bar{P}_2 B R^{-1} B^T \bar{P}_2 + [\bar{P}_1 B R^{-1} B^T \bar{P}_2 - \bar{P}_1 B R^{-1} B^T \bar{P}_2] \\
 &= A^T (\bar{P}_1 - \bar{P}_2) + (\bar{P}_1 - \bar{P}_2) A - \bar{P}_1 B R^{-1} B^T (\bar{P}_1 - \bar{P}_2) - (\bar{P}_1 - \bar{P}_2) B R^{-1} B^T \bar{P}_2 \\
 &= (A - B R^{-1} B^T \bar{P}_2)^T (\bar{P}_1 - \bar{P}_2) + (\bar{P}_1 - \bar{P}_2) (A - B R^{-1} B^T \bar{P}_1)
 \end{aligned}$$

Because $\bar{P}_1$ and $\bar{P}_2$ each are symmetric and positive definite, following the reasoning in the proof of Theorem 9.1, both $A - B R^{-1} B^T \bar{P}_1$ and $A - B R^{-1} B^T \bar{P}_2$ have strictly negative real-part eigenvalues. A slight modification to the sufficiency proof of Theorem 6.4 indicates that the unique solution to the above matrix equation is

$$\begin{aligned}
 \bar{P}_1 - \bar{P}_2 &= \int_0^\infty e^{(A - B R^{-1} B^T \bar{P}_2)^T t} (0) e^{(A - B R^{-1} B^T \bar{P}_1) t} dt \\
 &= 0
 \end{aligned}$$

from which we conclude that $\bar{P}_1 = \bar{P}_2$.

AE 9.2 Differentiating the identity $I = W^{-1}(t, t_f) W(t, t_f)$ gives

$$0 = \frac{d}{dt} W^{-1}(t, t_f) W(t, t_f) + W^{-1}(t, t_f) \frac{d}{dt} W(t, t_f)$$

which can be reorganized into

$$-\frac{d}{dt} W^{-1}(t, t_f) = W^{-1}(t, t_f) \frac{d}{dt} W(t, t_f) W^{-1}(t, t_f)$$

Applying AE3.2 yields

$$\begin{aligned}
 -\frac{d}{dt} W^{-1}(t, t_f) &= W^{-1}(t, t_f) [A W(t, t_f) + W(t, t_f) A^T - B B^T] W^{-1}(t, t_f) \\
 &= A^T W^{-1}(t, t_f) + W^{-1}(t, t_f) A - W^{-1}(t, t_f) B B^T W^{-1}(t, t_f)
 \end{aligned}$$

as required. Thus $W^{-1}(t, t_f)$ satisfies the differential Riccati equation with $Q=0$ and $R=I$. With these weighting matrices, Equation (9.12) coincides with Equation (9.9). With the terminal cost term replaced by the constraint $x(t_f)=x_f$, the optimal control signal for this linear quadratic regulator problem is the minimum energy control signal characterized in Section 9.3.

AE 9.3 A direct calculation gives

$$\begin{aligned}
 T^{-1}HT &= \begin{bmatrix} I & 0 \\ -\bar{P} & I \end{bmatrix} \begin{bmatrix} A & -BR^{-1}B^T \\ -Q & -A^T \end{bmatrix} \begin{bmatrix} I & 0 \\ \bar{P} & I \end{bmatrix} \\
 &= \begin{bmatrix} A - BR^{-1}B^T\bar{P} & -BR^{-1}B^T \\ -\bar{P}A - Q + \bar{P}BR^{-1}B^T\bar{P} - A^T\bar{P} & \bar{P}BR^{-1}B^T - A^T \end{bmatrix} \\
 &= \begin{bmatrix} A - BR^{-1}B^T\bar{P} & -BR^{-1}B^T \\ 0 & -(A - BR^{-1}B^T\bar{P})^T \end{bmatrix}
 \end{aligned}$$

from which $X = -BR^{-1}B^T$. The $2n$ eigenvalues of H are the n strictly negative real part eigenvalues of $A - BR^{-1}B^T\bar{P}$ along with their negatives.

AE 9.4 The solution to the optimal control problem involving the weighted performance index corresponds to the solution of the linear quadratic regulator problem with A replaced by $\alpha I + A$. Because the triple $(\alpha I + A, B, C)$ is controllable and observable if and only if the same is true for the triple (A, B, C) , Theorem 9.1 indicates that the algebraic Riccati equation

$$(\alpha I + A)^T \bar{P}_\alpha + \bar{P}_\alpha (\alpha I + A) - \bar{P}_\alpha B R^{-1} B^T \bar{P}_\alpha + C^T C = 0$$

has a unique symmetric positive definite solution which determines the optimal control

$$u(t) = -R^{-1}B^T \bar{P}_\alpha x(t)$$

The closed-loop state equation

$$\dot{x}(t) = (A - BR^{-1}B^T \bar{P}_\alpha)x(t) \quad x(0) = x_0$$

has eigenvalues with have real parts less than $-\alpha$ because $\alpha I + A - BR^{-1}B^T \bar{P}_\alpha$ has negative real part eigenvalues.

AE 9.5 Add and subtract $j\omega \bar{P}$ from the algebraic Riccati equation and rearrange to form

$$(-j\omega I - A)^T \bar{P} + \bar{P}(j\omega I - A) + \bar{P}BR^{-1}B^T \bar{P} = Q$$

Premultiply by $B^T (-j\omega I - A^T)^{-1}$, postmultiply by $(j\omega I - A)^{-1} B$, and scale by R^{-1} to obtain

$$\begin{aligned} R^{-1}B^T \bar{P}(j\omega I - A)^{-1} B + B^T (-j\omega I - A^T)^{-1} \bar{P}BR^{-1} + B^T (-j\omega I - A^T)^{-1} (\bar{P}BR^{-1})(R^{-1}B^T \bar{P})(j\omega I - A)^{-1} B \\ = B^T (-j\omega I - A^T)^{-1} QR^{-1}(j\omega I - A)^{-1} B \end{aligned}$$

Add 1 to both sides and substitute $K_{\text{LQR}} = R^{-1}B^T \bar{P}$ to write

$$\begin{aligned} 1 + K_{\text{LQR}}(j\omega I - A)^{-1} B + B^T (-j\omega I - A^T)^{-1} K_{\text{LQR}}^T + B^T (-j\omega I - A^T)^{-1} K_{\text{LQR}}^T K_{\text{LQR}}(j\omega I - A)^{-1} B \\ = 1 + \left[(j\omega I - A)^{-1} B \right]^* (QR^{-1})(j\omega I - A)^{-1} B \\ \geq 1 \quad \text{for all } -\infty < \omega < \infty \end{aligned}$$

Use the fact that $K_{\text{LQR}}(j\omega I - A)^{-1} B$ is a scalar quantity whose complex conjugate is

$B^T (-j\omega I - A^T)^{-1} K_{\text{LQR}}^T$ along with the identity

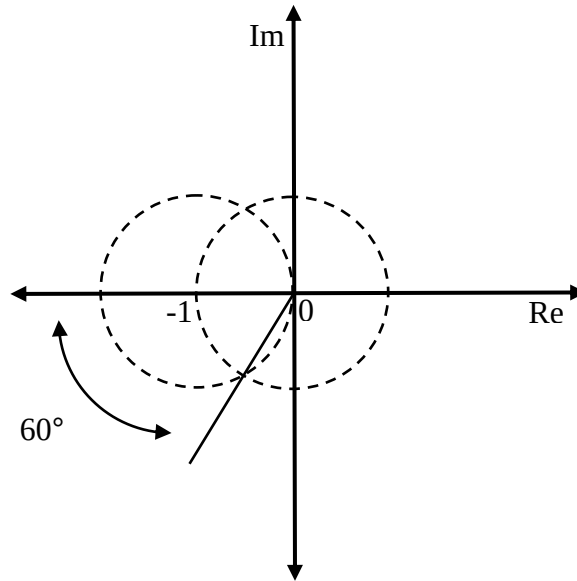
$$|1+z|^2 = (1+z)(1+\bar{z}) = 1+z+\bar{z}+\bar{z}z \quad z \in \mathbb{C}$$

to rewrite the preceding inequality as

$$\left| 1 + K_{\text{LQR}}(j\omega I - A)^{-1} B \right|^2 \geq 1 \quad \text{for all } -\infty < \omega < \infty$$

As a result, the Nyquist plot of $K_{\text{LQR}}(sI - A)^{-1} B$ never enters the circle of radius one centered at $-1 + j0$ in the complex plane. Consequently, the Nyquist curve does not cross the negative real axis between $-1 + j0$ and the origin, which corresponds to infinite gain margin. Also, from the geometry depicted

below, the Nyquist curve cannot cross the unit circle at a point corresponding to a phase margin of less than 60° .



Continuing MATLAB Exercises Solutions

Continuing MATLAB Exercise 1

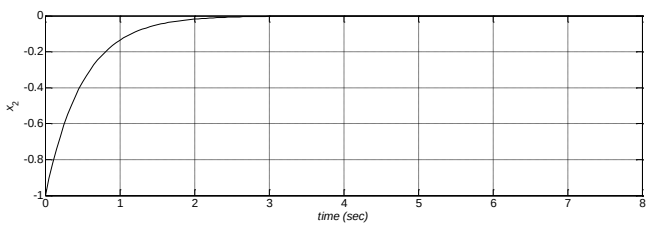
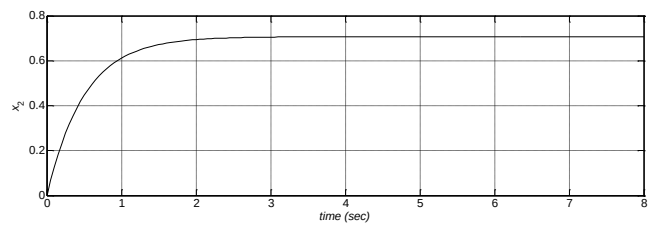
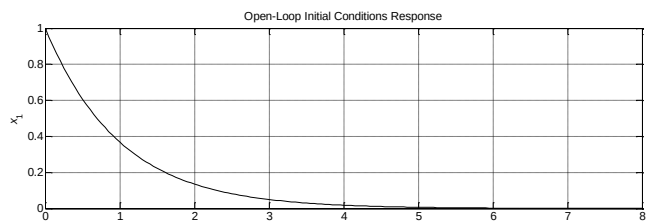
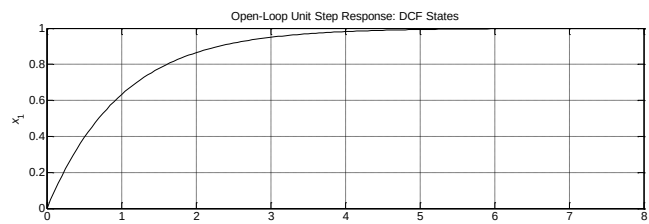
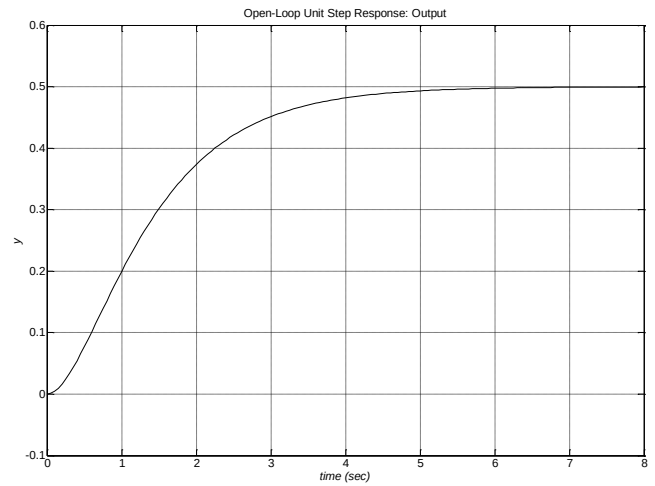
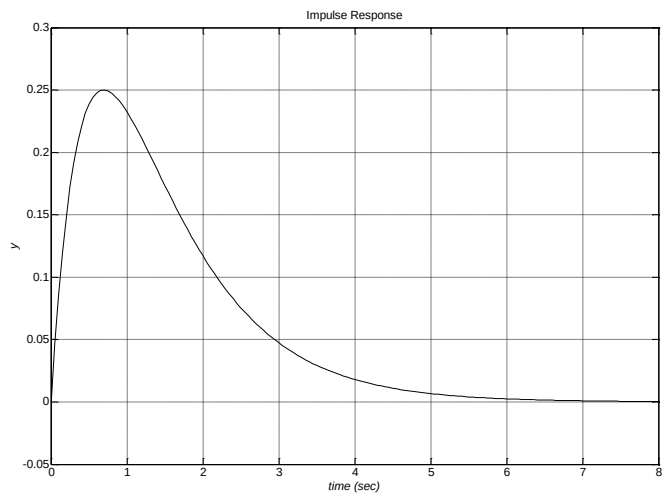
CharPoly = 1 3 2

Poles =
-2
-1

Transfer function:

1

 $s^2 + 3s + 2$



P =
1.0000 -1.0000
1.4142 -2.8284

determinant of P = -1.4142

rank of P = 2; n = 2; System is fully state-controllable

Q =

1.0000	-0.7071
-1.0000	1.4142

determinant of Q = 0.7071

rank of Q = 2; n = 2; System is fully state-observable

Tc =

2.0000	1.0000
1.4142	1.4142

Ac =

0.0000	1.0000
-2.0000	-3.0000

Bc =

0.0000
1.0000

Cc =

1.0000	0.0000
--------	--------

Dc =

0

To =

1.0000	-1.0000
1.4142	-2.8284

Ao =

0	-2.0000
1.0000	-3.0000

Bo =

1.0000
0

Co =

0.0000	1.0000
--------	--------

Do =

0

Td =

0	1
1	0

Ad =

-2	0
0	-1

Bd =

1.4142
1.0000

Cd =

-0.7071	1.0000
---------	--------

Dd =

0

P_{LYAP} =

```

0.5      0
0      0.25

```

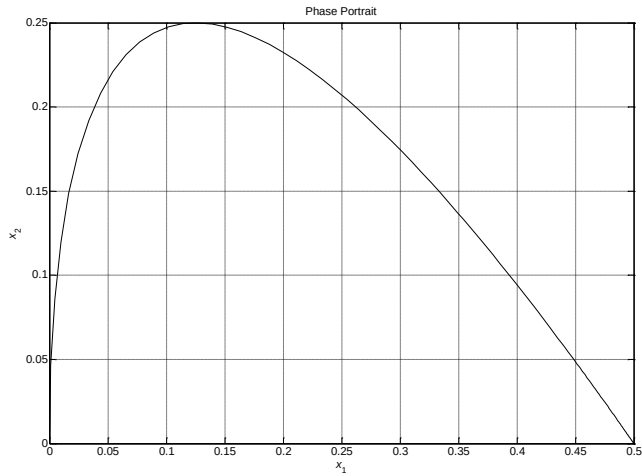
Sylvester's Criterion; the two principal minors are:

```

0.5
0.13

```

Both are positive so the system is asymptotically stable.



```

zeta =    -0.8261

```

```

wn =      -4.8421

```

```

den2 =      1.0000      8.0000     23.4461

```

```

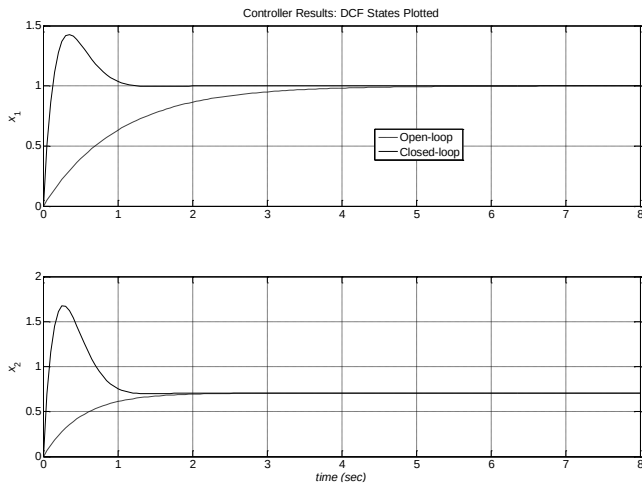
Poles2 =
-4.0000 + 2.7288i
-4.0000 - 2.7288i

```

```

K =      16.45      -8.09

```



```

PolesObs =
-40.0000 +27.2875i
-40.0000 -27.2875i

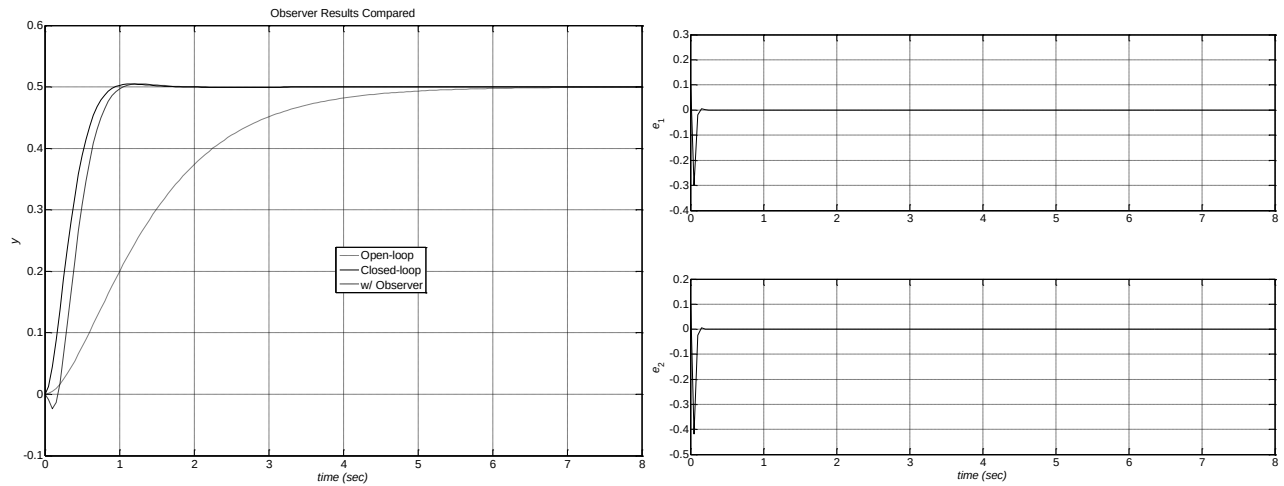
```

```

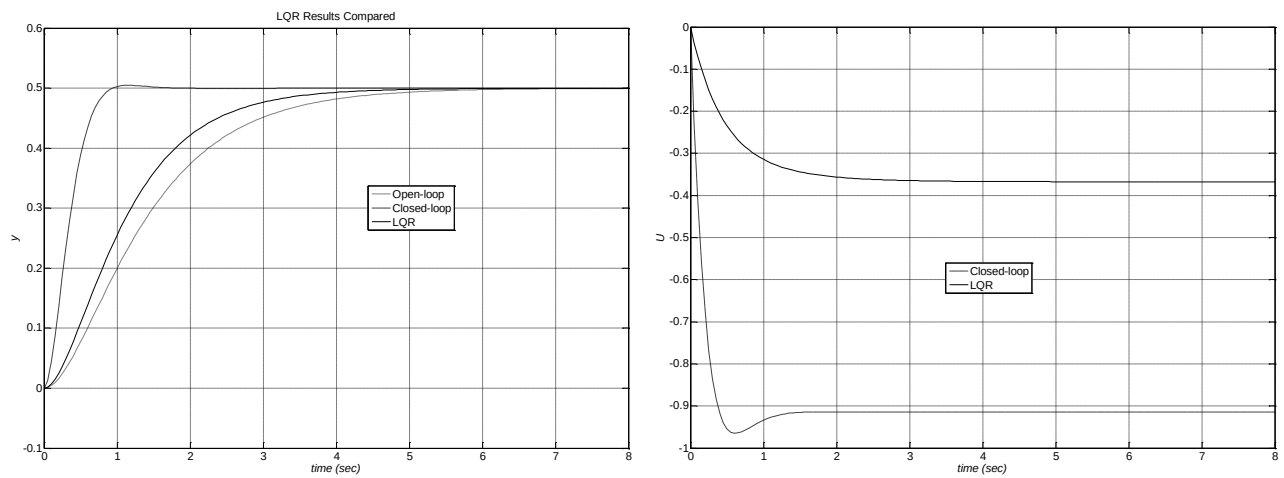
L =
2265.61
3095.16

```

Ahat =
 $1.0e+003 \cdot$
 $\begin{bmatrix} -2.2666 & 1.6020 \\ -3.0952 & 2.1866 \end{bmatrix}$



$K_{LQR} =$
 $\begin{bmatrix} 0.43 & -0.04 \\ -0.04 & 0.23 \end{bmatrix}$

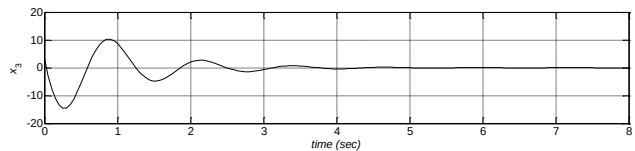
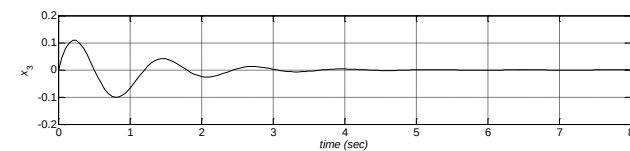
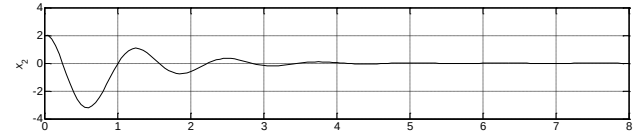
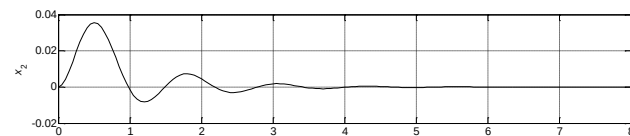
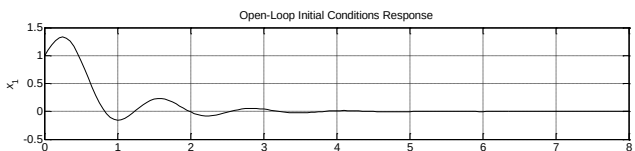
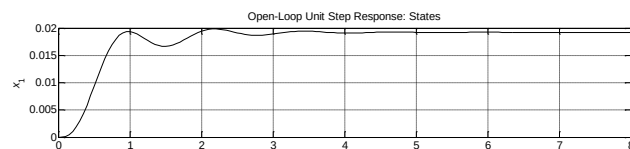
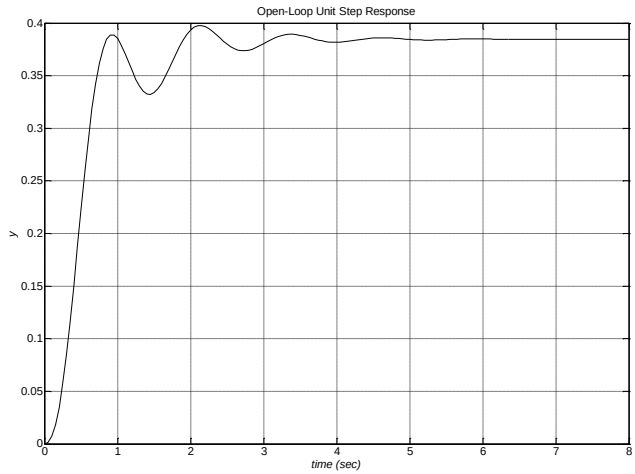
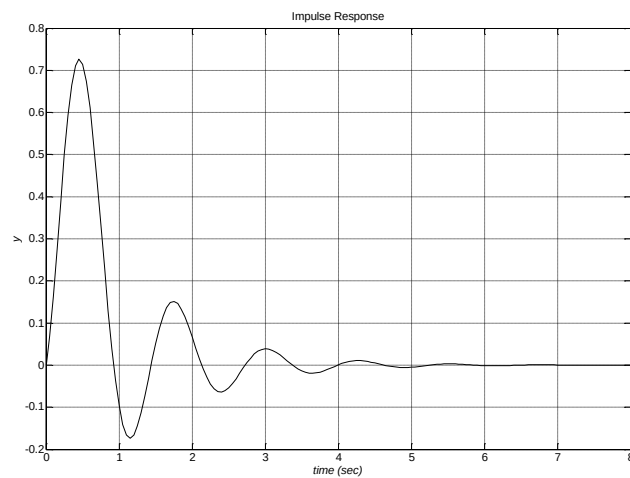


Continuing MATLAB Exercise 2

```
CharPoly =      1.0000      4.0000     30.0000     52.0000
```

```
Poles =
-1.0000 + 5.0000i
-1.0000 - 5.0000i
-2.0000
```

```
Transfer function:
          s + 20
-----
s^3 + 4 s^2 + 30 s + 52
```



```
P =
0  0  1
0  1 -4
1 -4 -14
```

```
rank of P = 3; n = 3; System is fully state-controllable
```

```
Q =
20  1  0
0  20  1
-52 -30 16
```

rank of $Q = 3$; $n = 3$; System is fully state-observable

$T_c =$

1.0000	0	0
0.0000	1.0000	0
0.0000	0.0000	1.0000

$A_c =$

0.0000	1.0000	0
0.0000	0	1.0000
-52.0000	-30.0000	-4.0000

$B_c =$

0
0
1

$C_c =$ 20 1 0

$D_c =$ 0

$T_o =$

0.0001	-0.0029	0.0576
-0.0029	0.0576	-0.1514
0.0576	-0.1514	-0.9718

$A_o =$

0.0000	0.0000	-52.0000
1.0000	0.0000	-30.0000
0.0000	1.0000	-4.0000

$B_o =$

20.0000
1.0000
0.0000

$C_o =$

0.0000	0.0000	1.0000
--------	--------	--------

$D_o =$

0

$T_m =$

4.5826	0.3525	0.1763
4.0791	0.3138	-0.8629
9.7898	6.0559	0.5805

$A_m =$

-2.0000	0	0
0	-1.0000	5.0000
0	-5.0000	-1.0000

$B_m =$

0.1763
-0.8629
0.5805

$C_m =$

3.9279	0.7340	-0.1015
--------	--------	---------

$D_m =$

0

$P_{LYAP} =$

21.82	11.92	0.01
11.92	8.5	0.41

0.01 0.41 0.23

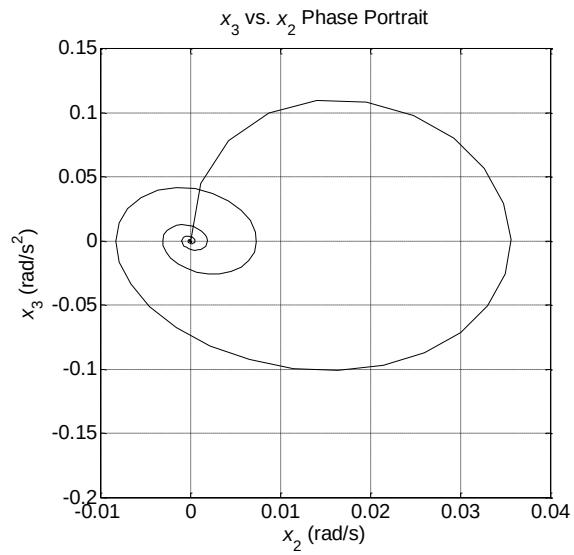
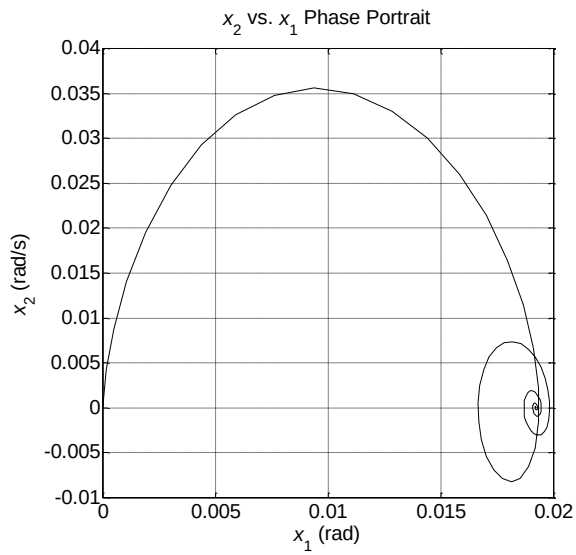
Sylvester's Criterion; the three principal minors are:

21.82

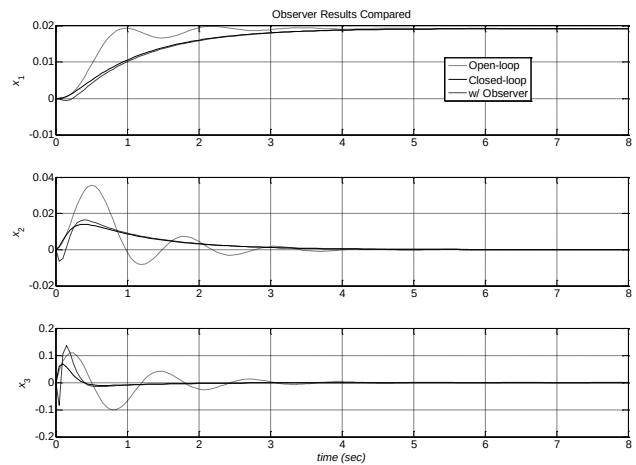
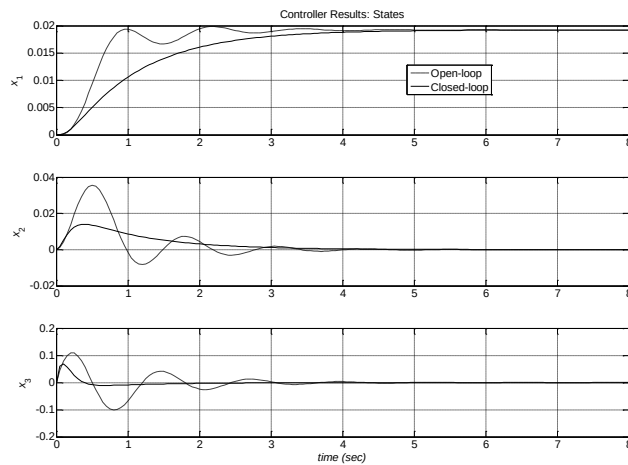
43.38

6.27

All three are positive so the system is asymptotically stable.



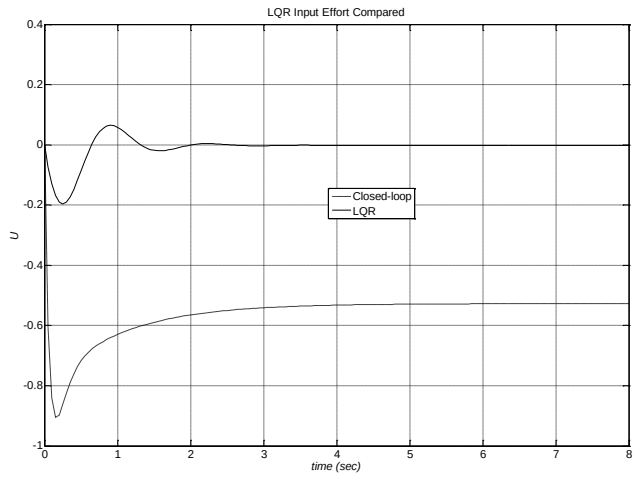
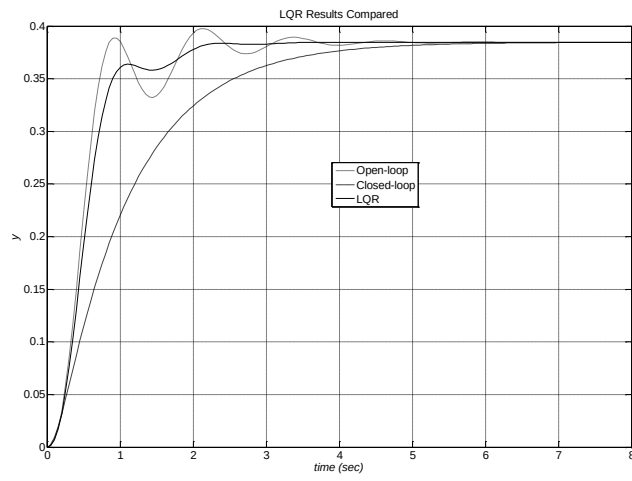
```
Poles3 =      -1   -10   -11
K       =      58   101   18
```



```
ObsPoles =      -10   -100   -110
```

```
L =
  -9.36
 403.25
4140.92
```

$K_{LQR} =$
 153.02 85.61 0.10
 85.61 65.22 2.88
 0.10 2.88 1.64



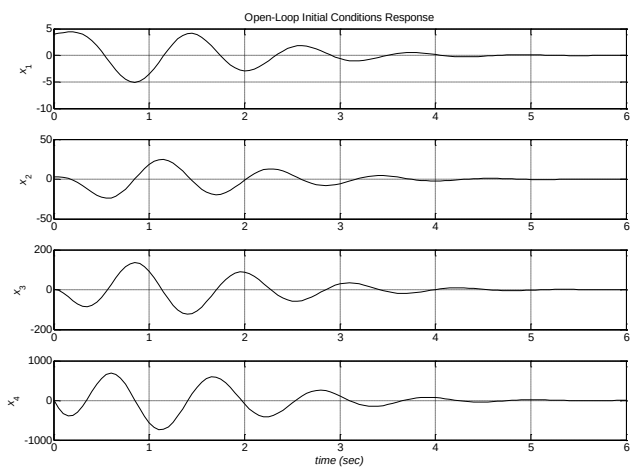
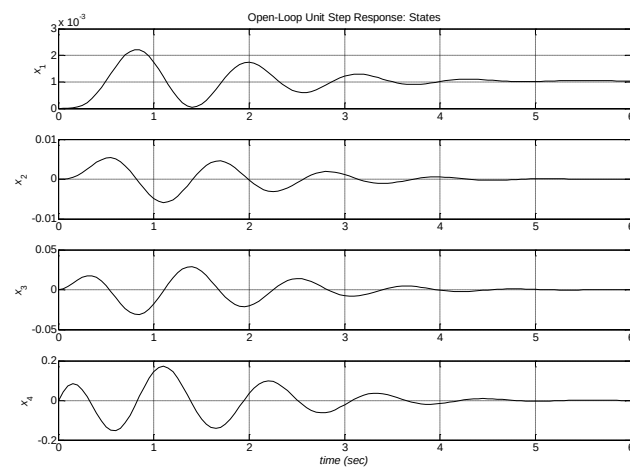
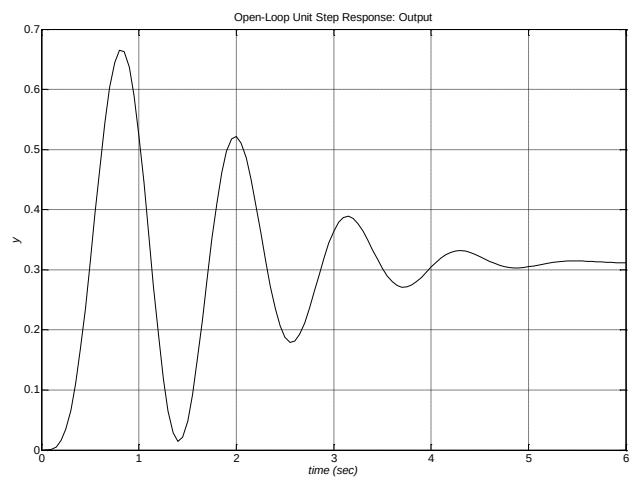
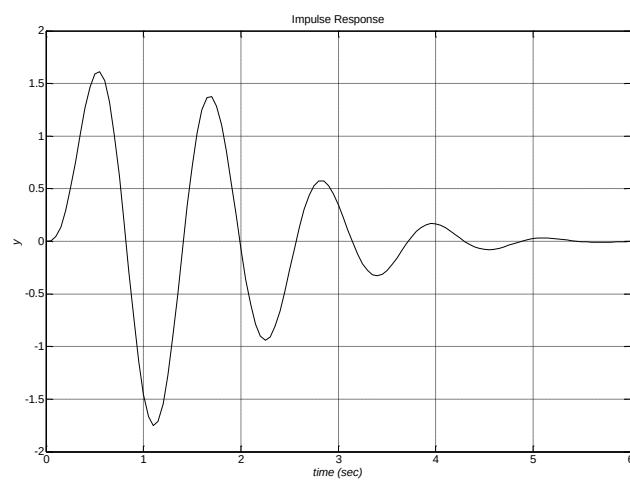
Continuing MATLAB Exercise 3

```
CharPoly = 1.0000 4.0000 67.0000 126.0000 962.0000
```

```
Poles =
-1.0000 + 6.0000i
-1.0000 - 6.0000i
-1.0000 + 5.0000i
-1.0000 - 5.0000i
```

```
Transfer function:
          300
```

```
-----
s^4 + 4 s^3 + 67 s^2 + 126 s + 962
```



```
P =
0 0 0 1
0 0 1 -4
0 1 -4 -51
1 -4 -51 346
```

```
rank of P = 4; n = 4; System is fully state-controllable
```

```

Q =
300    0    0    0
  0  300    0    0
  0    0  300    0
  0    0    0  300

```

rank of Q = 4; n = 4; System is fully state-observable

```

Tc =
  1.0000    0    0    0
  0.0000    1.0000    0    0
  0.0000    0.0000    1.0000    0
    0    0.0000    0.0000    1.0000

```

```

Ac =
  0.0000    1.0000    0    0
  0.0000    0    1.0000    0
  0.0000    0    0    1.0000
 -962.0000 -126.0000 -67.0000 -4.0000

```

```

Bc =
  0
  0
  0
  1

```

```

Cc =    300    0    0    0

```

```

Dc =    0

```

```

To =
  0    0.0000    0    0.0033
  0.0000    0.0000    0.0033 -0.0133
  0    0.0033 -0.0133 -0.1700
  0.0033 -0.0133 -0.1700    1.1533

```

```

Ao =
  0.0000    0.0000    0.0000 -962.0000
  1.0000    0.0000    0.0000 -126.0000
  0.0000    1.0000    0.0000 -67.0000
  0.0000    0.0000    1.0000 -4.0000

```

```

Bo =
300.0000
  0.0000
  0.0000
  0.0000

```

```

Co =    0    0.0000    0    1.0000

```

```

Do =    0

```

```

Tm =
-177.3217    65.4358   -0.7373    3.0414
-517.1883   -82.5166  -23.1790   -1.6436
 178.3728   -65.8236    0.7417   -2.0396
 428.0947    73.9079   14.3143    1.3721

```

```

Am =
 -1.0000    6.0000    0    0
 -6.0000   -1.0000    0    0
    0    0   -1.0000    5.0000
    0    0   -5.0000   -1.0000

```

```

Bm =
  3.0414

```

-1.6436
-2.0396
1.3721

Cm = 0.6251 1.1567 1.2386 1.8411

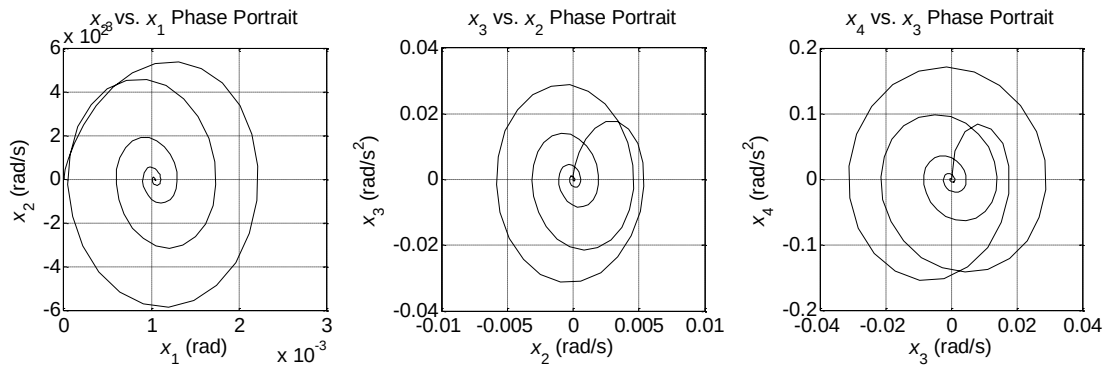
Dm = 0

P_{LYAP} =
24090.06 3154.74 908.93 0
3154.74 1182.05 219.21 25.04
908.93 219.21 51.38 3.28
0 25.04 3.28 0.94

Sylvester's Criterion; the four principal minors are:

24090.06
18523324.02
74680332.52
9651031.28

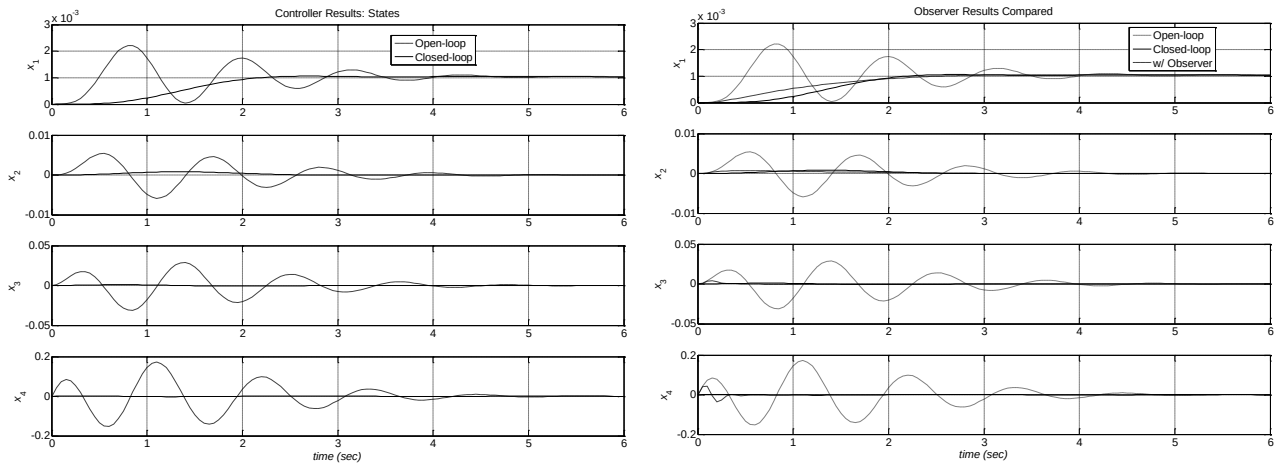
All four are positive so the system is asymptotically stable.



ITAE4 = 1.0000 4.2000 13.6000 21.6000 16.0000

Poles4i =
-0.8480 + 2.5260i
-0.8480 - 2.5260i
-1.2520 + 0.8283i
-1.2520 - 0.8283i

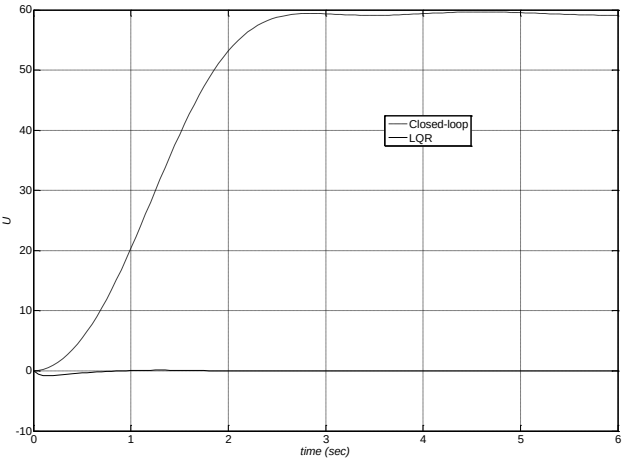
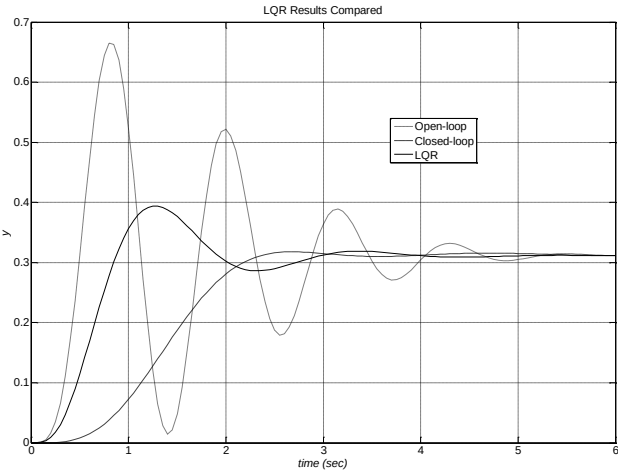
K = -946 -104.4 -53.4 0.2



```
ObsPoles =  
-8.4796 +25.2598i  
-8.4796 -25.2598i  
-12.5204 + 8.2828i  
-12.5204 - 8.2828i
```

```
L =  
0.13  
3.80  
47.88  
67.82
```

```
KLQR =  
278108.96      78035.82      17191.97      0.16  
78035.82      35829.21      8571.46      289.03  
17191.97      8571.46      2681.43      81.09  
0.16          289.03      81.09      17.87
```



Continuing MATLAB Exercise 4

```
CharPoly =      1.0000      6.0000     86.0000    176.0000    680.0000
```

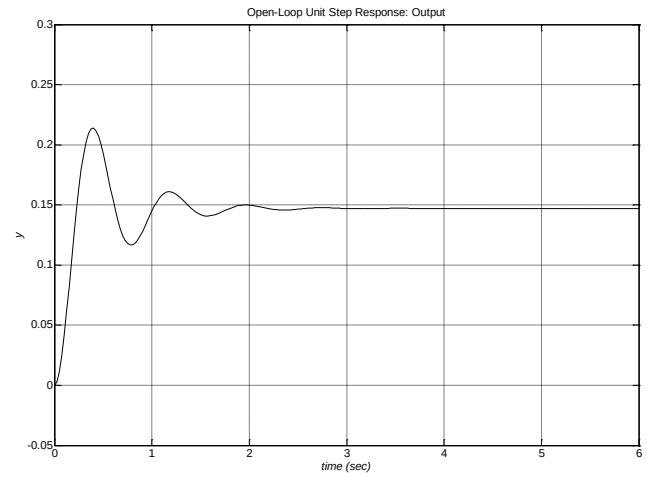
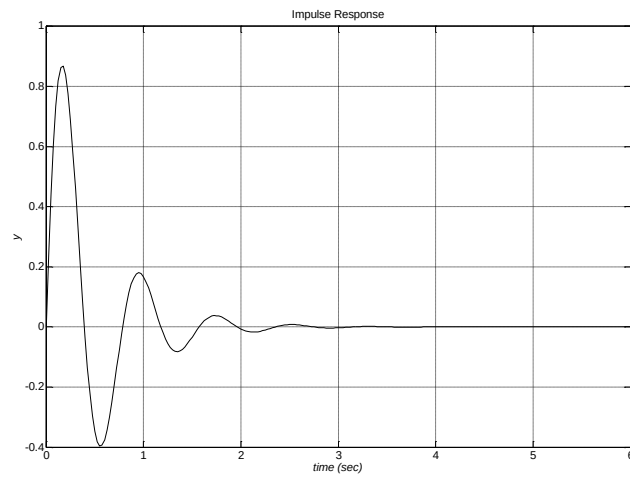
```
Poles =
  -2.0000 + 8.0000i
  -2.0000 - 8.0000i
  -1.0000 + 3.0000i
  -1.0000 - 3.0000i
```

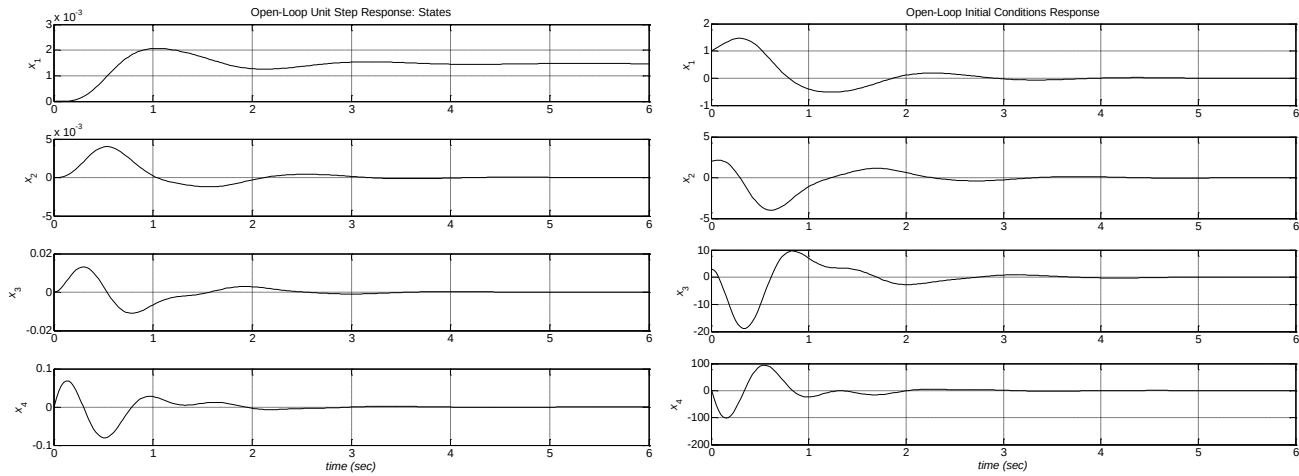
```
Transfer function:
      10 s^2 + 20 s + 100
-----
s^4 + 6 s^3 + 86 s^2 + 176 s + 680
```

```
zeros =
  -1.0000 + 3.0000i
  -1.0000 - 3.0000i
```

```
poles =
  -2.0000 + 8.0000i
  -2.0000 - 8.0000i
  -1.0000 + 3.0000i
  -1.0000 - 3.0000i
```

```
DC gain =      10
```





```
P =
0      0      0      1
0      0      1     -6
0      1     -6    -50
1     -6    -50    640
```

rank of P = 4; n = 4; System is fully state-controllable

```
Q =
100      20      10      0
0      100      20      10
-6800  -1760   -760   -40
27200   240   1680  -520
```

rank of Q = 2; n = 2; System is NOT fully state-observable
Controllable but not Observable, therefore this system is NOT MINIMAL!


```

Tc =
  1.0000      0      0      0
  0.0000      1.0000      0      0
  0.0000      0.0000      1.0000      0
  0.0000      0.0000      0.0000      1.0000

Ac =
  0.0000      1.0000      0      0
  0.0000      0      1.0000      0
  0.0000      0      0      1.0000
 -680.0000 -176.0000 -86.0000 -6.0000

Bc =
  0
  0
  0
  1

Cc = 100.0000  20.0000  10.0000      0

Dc =      0

To =
 176      86      6      1
  86      6      1      0
   6      1      0      0
   1      0      0      0

Ao =
  0      0      0 -680
  1      0      0 -176
  0      1      0 -86
  0      0      1  -6

Bo =
  1
  0
  0
  0

Co =      0      10     -40    -520

Do =      0

Tm =
 -20.7331      7.1600      0.1880      1.1307
 -101.2898 -25.6763 -11.2126 -0.5418
  21.6922  -7.4913  -0.1967 -0.1289
  36.4550  12.2976   1.1333   0.1493

Am =
 -2.0000      8.0000      0      0
 -8.0000     -2.0000      0      0
      0      0     -1.0000      3.0000
      0      0     -3.0000     -1.0000

Bm =
  1.1307
 -0.5418
 -0.1289
  0.1493

Cm =      -0.4308     -0.8991      0.0000      0.0000

Dm =      0

```

2 states removed.

```
a =
      x1      x2
x1      28.293 -18.575
x2      52.849 -32.293
```

```
b =
      u1
x1      0.047518
x2      0.087768
```

```
c =
      x1      x2
y1      -90.11  48.786
```

```
d =
      u1
y1      0
```

Transfer function:
10

s^2 + 4 s + 68

```
CharPoly2 =
  1.0000  4.0000  68.0000
```

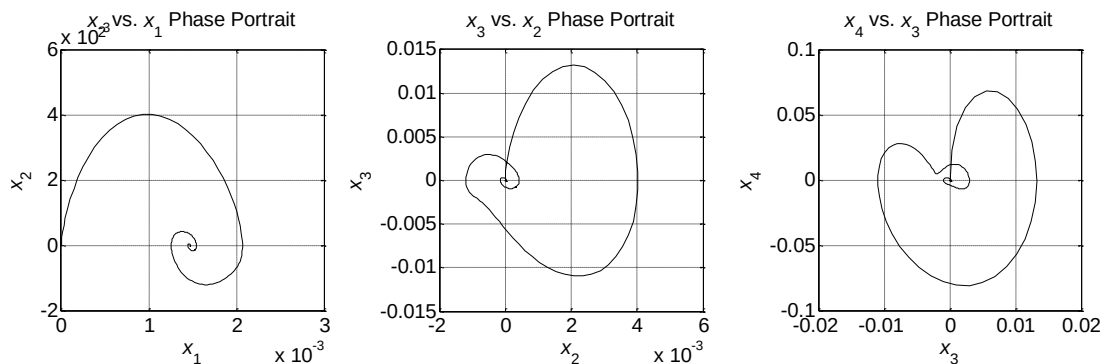
```
PolesOp2 =
-2.0000 + 8.0000i
-2.0000 - 8.0000i
```

```
PLYAP =
1193.75      308.44      108.07      0
308.44      122.71      38.5      1.76
108.07      38.5      14.63      0.45
0           1.76      0.45      0.16
```

Sylvester's Criterion; the four principal minors are:

```
1193.75
51346.98
115417.19
10051.24
```

All four are positive, so the system is asymptotically stable.



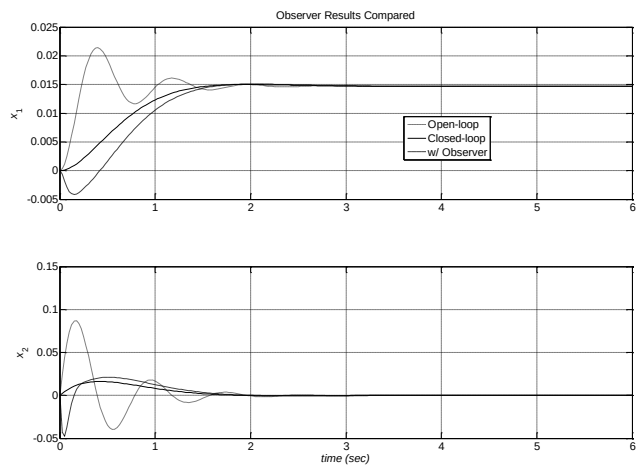
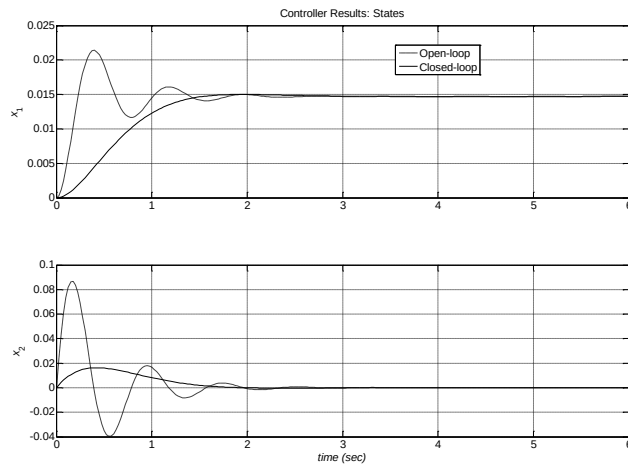
```
zeta = -0.7797
```

```
wn = -2.5651
```

```
den2 = 1.0000  4.0000  6.5796
```

```
Poles2 =
-2.0000 + 1.6061i
-2.0000 - 1.6061i
```

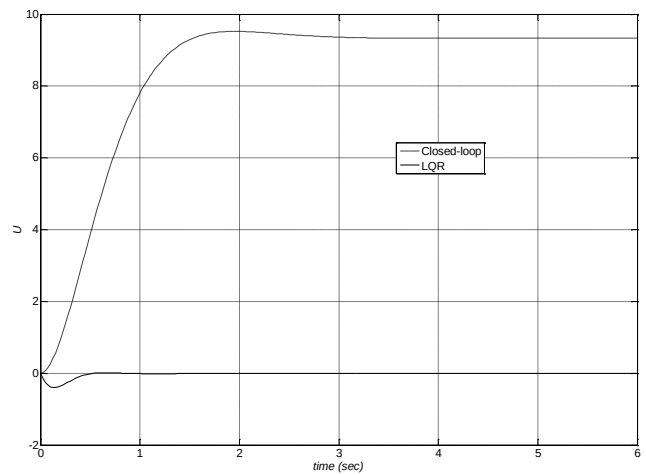
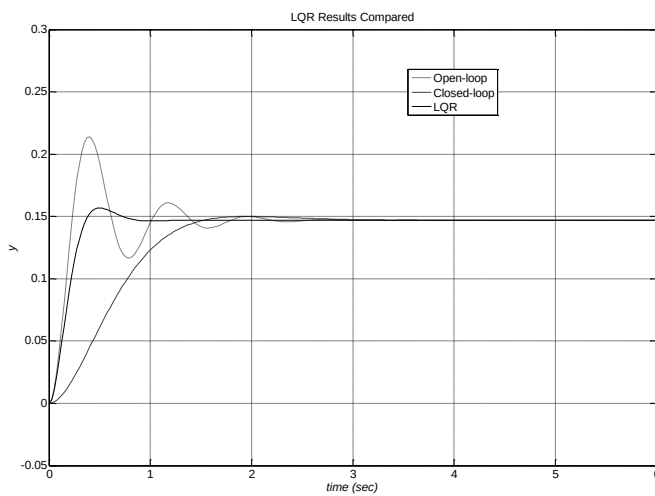
```
K =    -61.42      0
```



```
ObsPoles =
-20.0000 +16.0612i
-20.0000 -16.0612i
```

```
L =
3.6
44.6
```

```
K_LQR =
472.91    0.73
 0.73    6.84
```



Continuing Exercises Solutions

Note: continuing Exercises 1-3 have many options. This solutions manual presents one complete solution for one option each as stated below:

1. Three-Mass Translational Mechanical System
 - Case iii. Single-input, single-output: input $u_2(t)$ and output $y_3(t)$.
2. Inverted Pendulum
 - i. Single-input, single-output: input $f(t)$ and output $\theta(t)$.
 - (b) zero input $f(t)$ and an initial condition of $\theta(0) = 0.1$ rad.
3. Robot Joint/Link Control
 - i. Single-input, single-output: input armature voltage $v_A(t)$ and output robot load angle $\theta_L(t)$.
 - (a) Unit step input armature voltage $v_A(t)$; zero initial state.
4. Ball/Beam System
5. Proof-Mass Actuator

Each Continuing Exercise solution is presented as a whole, i.e. for all chapters in a serial fashion.

Also note: we have found that MATLAB changes their MIMO controller and observer design algorithms substantially between versions. There are infinite solutions to the controller and observer gain matrices to place a given set of eigenvalues in the closed-loop A matrices. These solutions were obtained in Fall 2006 with MATLAB version R14. If your solutions are different, they may be entirely acceptable – just check the resulting matrices for the correct eigenvalues. The SISO cases have unique solutions that should not vary with MATLAB version.

Continuing Exercise 1 – Three-Mass Translational Mechanical System

Case iii. Single-input, single-output: input $u_2(t)$ and output $y_3(t)$.

Open-Loop System

```

A =
    0         1         0         0         0         0
   -30      -1.2        20        0.8         0         0
    0         0         0         1         0         0
    10        0.4       -25        -1        15        0.6
    0         0         0         0         0         1
    0         0        10        0.4       -23.33      -0.93

B =
    0
    0
    0
   0.5
    0
    0

C =    0  0  0  0  1  0

d =    0

num/den =
              0.2 s^3 + 5.24 s^2 + 12 s + 150
-----
s^6 + 3.1333 s^5 + 81.0267 s^4 + 135.2 s^3 + 1723.3333 s^2 + 1000 s + 8333.3333

CharPoly = 1.0e+003 * [0.0010    0.0031    0.0810    0.1352    1.7233    1.0000    8.3333]

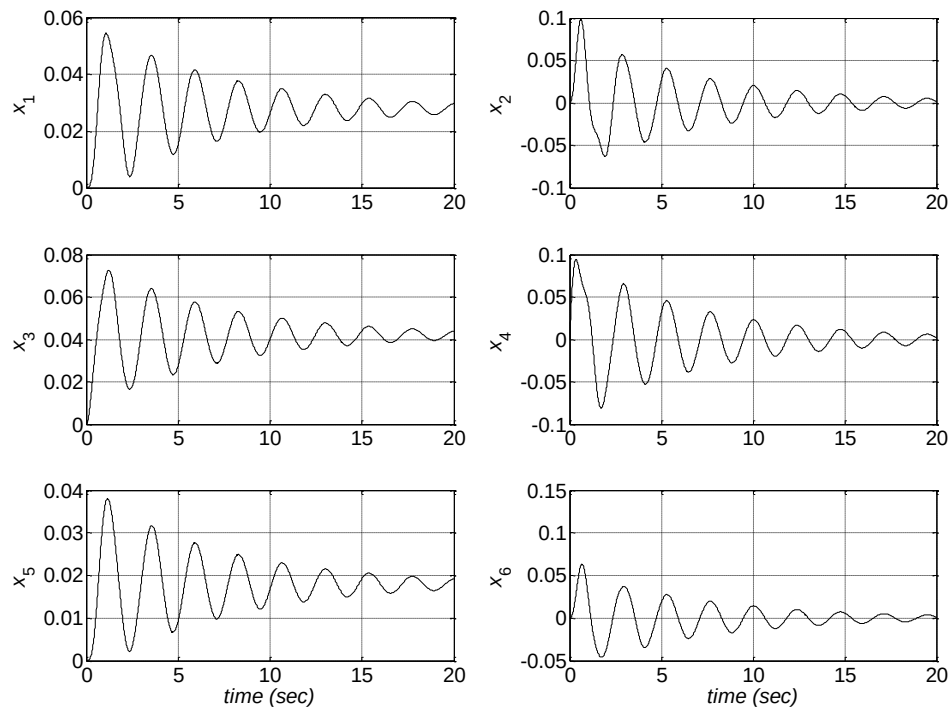
```

Open-Loop Response

```

Xsso =
    0.0280
         0
    0.0420
         0
    0.0180
         0

```



Controllability & Observability

$P =$

0	0	0.4	9.12	-42.32	-427.11
0	0.4	9.12	-42.32	-427.11	2844.96
0	0.5	-0.5	-11.72	37.64	387.64
0.5	-0.5	-11.72	37.64	387.64	-2317.91
0	0	0.2	4.61	-18.66	-192.37
0	0.2	4.61	-18.66	-192.37	1146.37

rank of $P = 6$; $n = 6$; System is fully state-controllable

$Q =$

0	0	0	0	1	0
0	0	0	0	0	1
0	0	10	0.4	-23.33	-0.93
4	0.16	-19.33	9.23	27.78	-22.22
87.47	7.5	-449.69	-37.32	656.92	54.05
-598.17	63.54	1623.54	-384.75	-1821.09	584.08

rank of $Q = 6$; $n = 6$; System is fully state-observable

Canonical Realizations

$T_c =$

233.3333	18.6667	10.3733	0.4000	0	0
0.0000	233.3333	18.6667	10.3733	0.4000	0
350.0000	28.0000	27.2267	1.0667	0.5000	0
0.0000	350.0000	28.0000	27.2267	1.0667	0.5000
150.0000	12.0000	5.2400	0.2000	0	0
0	150.0000	12.0000	5.2400	0.2000	0

$A_c =$
 $1.0e+003 *$

0.0000	0.0010	0.0000	0	0.0000	0.0000
0.0000	0.0000	0.0010	0.0000	0.0000	0.0000
0.0000	0.0000	0.0000	0.0010	0.0000	0.0000
0.0000	0.0000	0.0000	0.0000	0.0010	0.0000
0.0000	0	0	0.0000	0.0000	0.0010
-8.3333	-1.0000	-1.7233	-0.1352	-0.0810	-0.0031

Bc =

0.0000
0.0000
0.0000
0.0000
0.0000
1.0000

Cc = 150.0000 12.0000 5.2400 0.2000 0 0

Dc = 0

To =

-0.0008	0.0095	0.0120	-0.3000	0.0000	11.0000
0.0095	0.0120	-0.3000	0.0000	11.0000	-17.6000
0.0000	0.0002	-0.0040	0.1000	0.0000	-5.5000
0.0002	-0.0040	0.1000	0.0000	-5.5000	10.5000
0	0	0	0	0	1.0000
0	0	0	0	1.0000	-3.1333

Ao =

1.0e+003 *

0.0000	0.0000	0.0000	0.0000	0.0000	-8.3333
0.0010	0.0000	0.0000	0.0000	0.0000	-1.0000
0.0000	0.0010	0.0000	0.0000	0.0000	-1.7233
0.0000	0	0.0010	0.0000	0.0000	-0.1352
0	0	0	0.0010	0.0000	-0.0810
0	0	0	0	0.0010	-0.0031

Bo =

150.0000
12.0000
5.2400
0.2000
0
0

Co = 0 0 0 0 0 1

Do = 0

Tm =

0.0000	0.5252	0.0000	-0.7939	0.0000	0.5467
-3.5600	-0.0712	5.3815	0.1076	-3.7057	-0.0741
0.0000	0.5061	0.0000	0.1946	0.0000	-1.0346
-2.6017	-0.0520	-1.0004	-0.0200	5.3189	0.1064
0.0000	0.3499	0.0000	0.8025	0.0000	0.7398
-0.9311	-0.0186	-2.1357	-0.0427	-1.9688	-0.0394

Am =

-0.9023	6.6560	0	0	0	0
-6.6560	-0.9023	0	0	0	0
0	0	-0.5231	5.0873	0	0
0	0	-5.0873	-0.5231	0	0
0	0	0	0	-0.1412	2.6537
0	0	0	0	-2.6537	-0.1412

```

Bm =
    -0.3970
     0.0538
     0.0973
    -0.0100
     0.4013
    -0.0214

Cm =    -0.0053   -0.0389    0.0109    0.1062   -0.0079   -0.1478

Dm =      0

```

Stability Analysis

```

P =

    9.41    -0.12    -0.97    -0.42    -0.23     0.1
   -0.12     0.45     0.35     0.42    -0.12     0.27
  -0.97     0.35    15.76     0.13     1.24    -0.42
  -0.42     0.42     0.13     1.32     0.52     0.88
 -0.23    -0.12     1.24     0.52    21.34     0.35
    0.1     0.27    -0.42     0.88     0.35     1.48

```

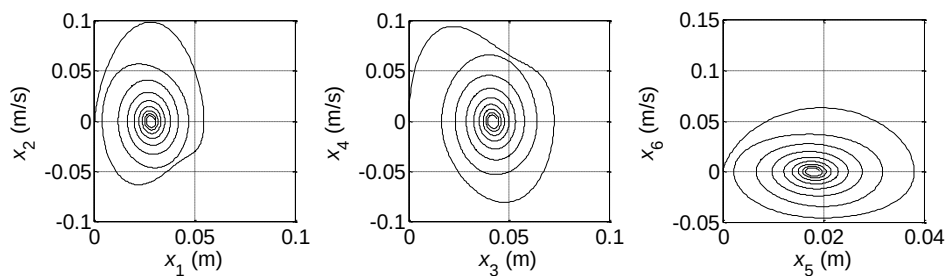
Sylvesters Criterion; the six principal minors:

```

9.41
4.24
65.39
59.88
1242.97
1071.62

```

All principal minors are positive, therefore the system is asymptotically stable! Phase portraits:



Dynamic Shaping

```

DomPoles =
    -1.3333 + 1.1946i
    -1.3333 - 1.1946i

```

```

Poles6 =
    -1.3333 - 1.1946i
    -1.3333 + 1.1946i
   -13.3333
   -14.3333
   -15.3333
   -16.3333

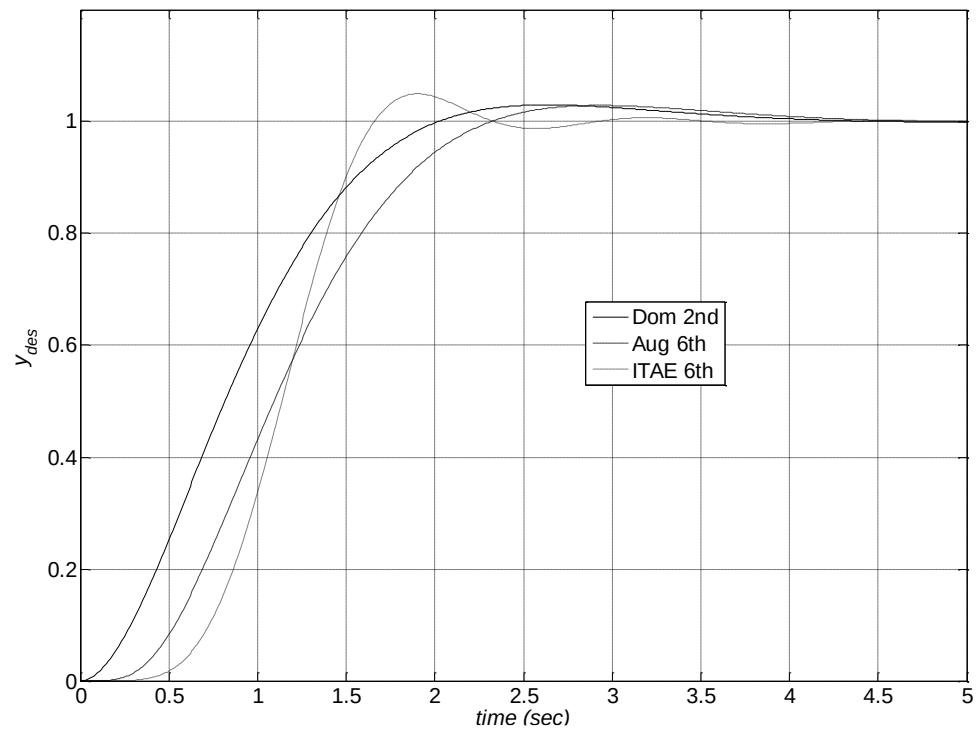
```

```

Poles6i =

```

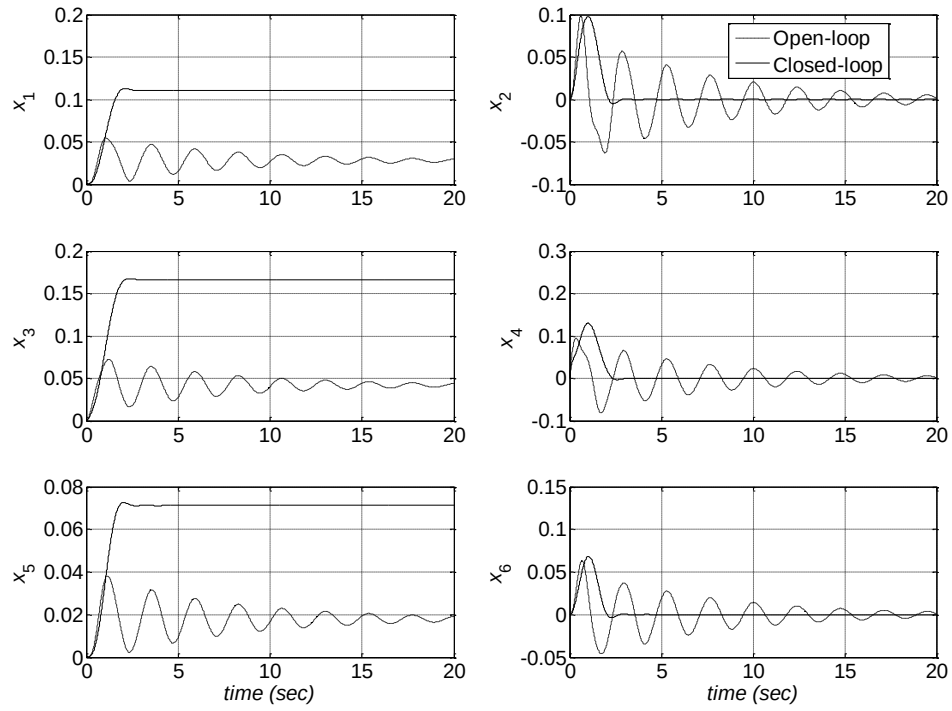

$-1.1094 + 4.5233i$
 $-1.1094 - 4.5233i$
 $-2.0784 + 2.8028i$
 $-2.0784 - 2.8028i$
 $-2.6303 + 1.0285i$
 $-2.6303 - 1.0285i$



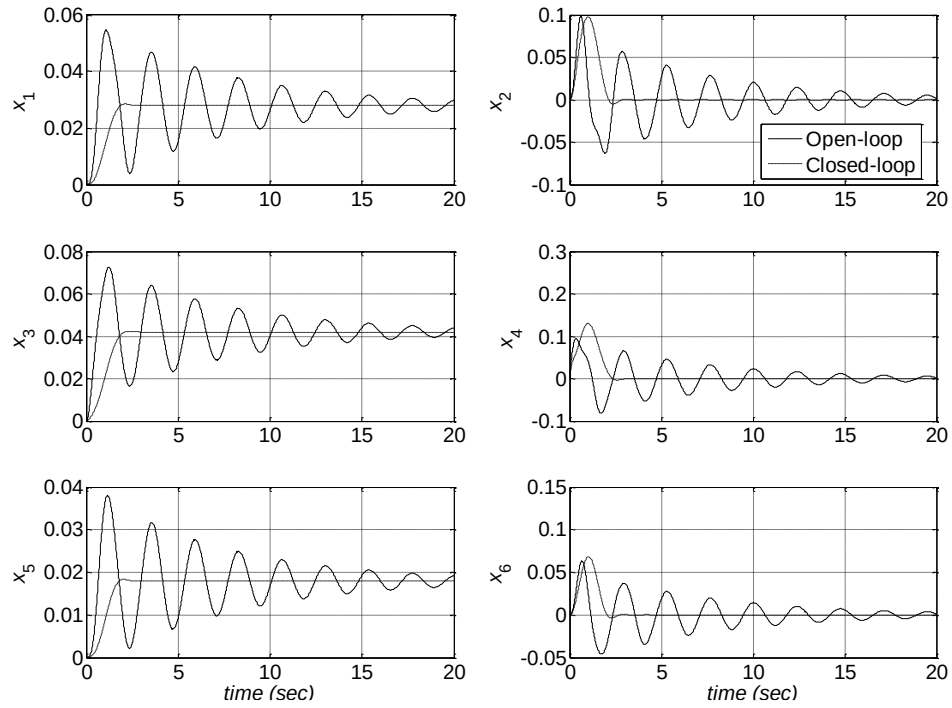
Controller Design

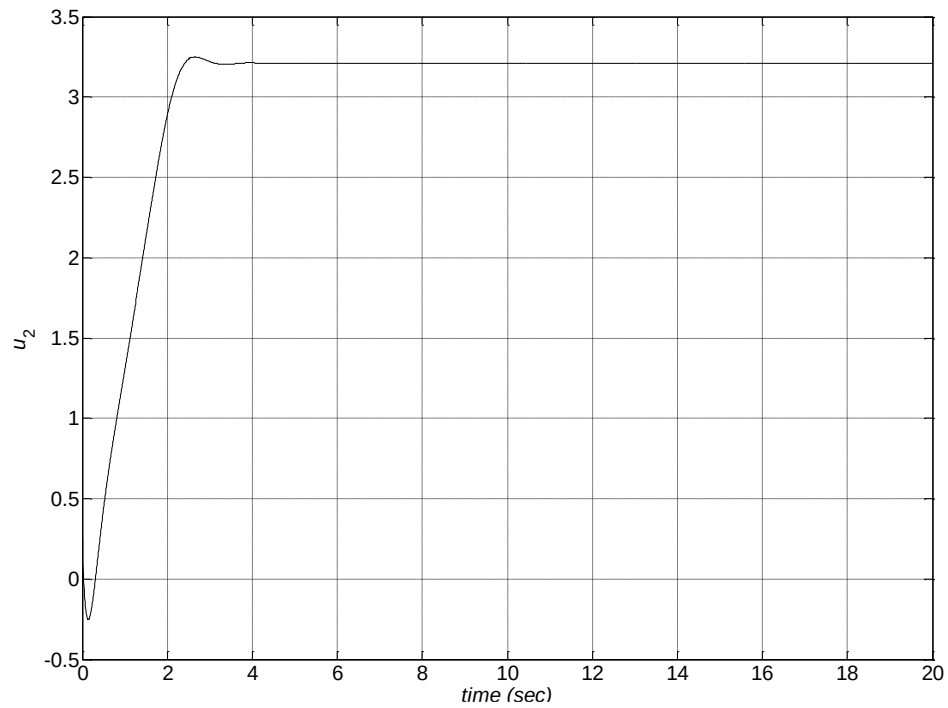
K for ITAE 6th order desired poles:

K = -88.34 -15.22 -15.4 17.01 131.85 -3.85



CORR = 0.25 0.25 0.25

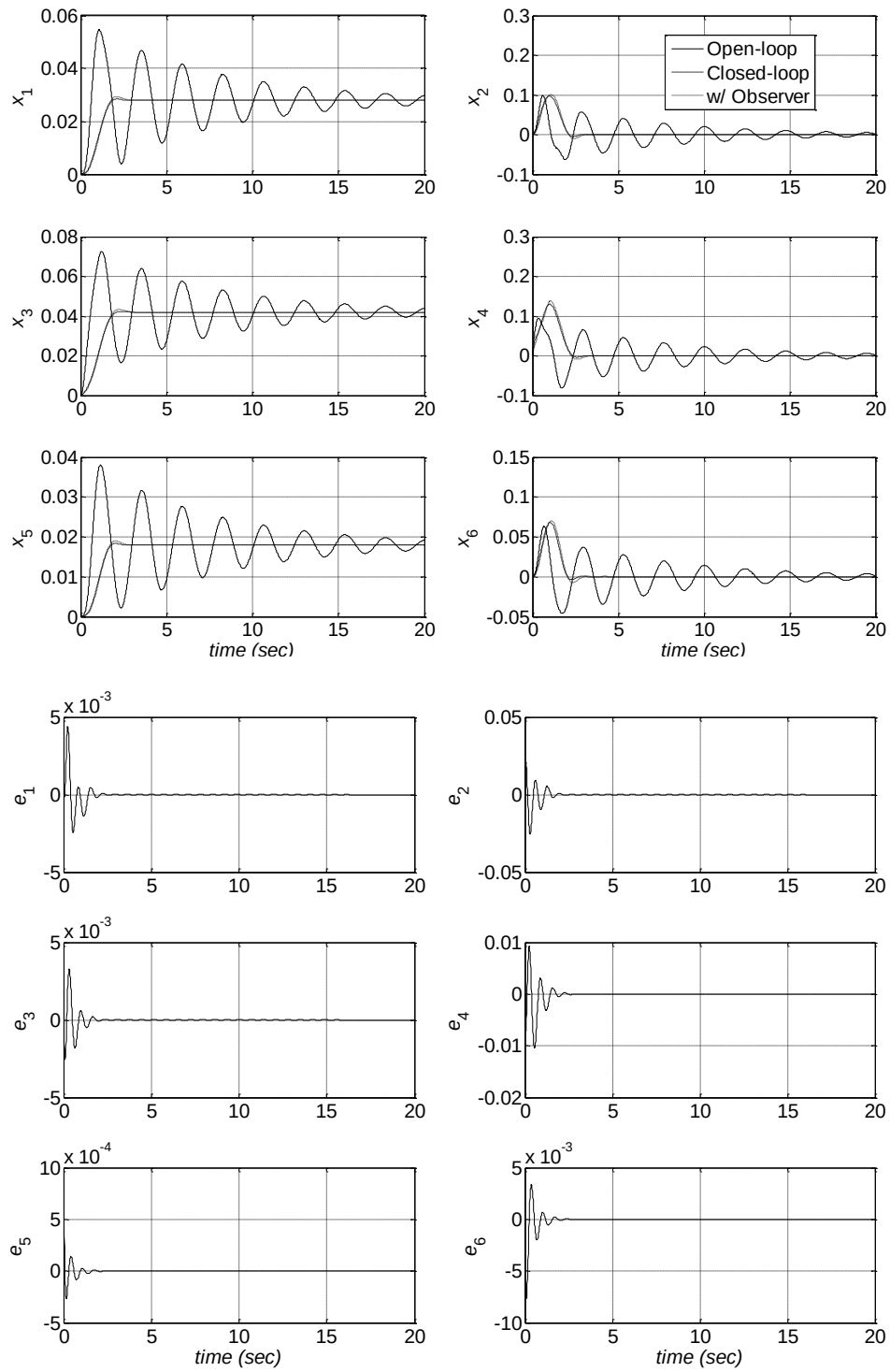




Observer Design

```
PolesObs =
-2.2188 + 9.0466i
-2.2188 - 9.0466i
-4.1569 + 5.6056i
-4.1569 - 5.6056i
-5.2605 + 2.0569i
-5.2605 - 2.0569i
```

```
L =
128.85
-797.9
130.96
  309
  20.14
194.29
```



Continuing Exercise 2 – Inverted Pendulum

- i. Single-input, single-output: input $f(t)$ and output $\theta(t)$.
 (b) zero input $f(t)$ and an initial condition of $\theta(0) = 0.1$ rad.

Open-Loop System

```

A =
    0         1         0         0
    0         0         4.91        0
    0         0         0         1
    0         0        19.62        0

B =
    0
    0.5
    0
    0.67

C =    0    0    1    0

D =    0

poleso =
    0
    0
    4.4294
   -4.4294

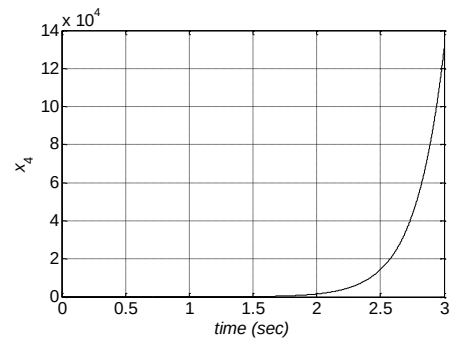
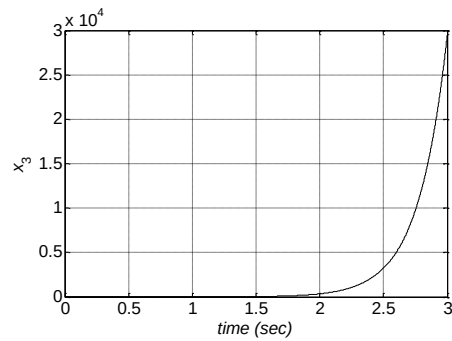
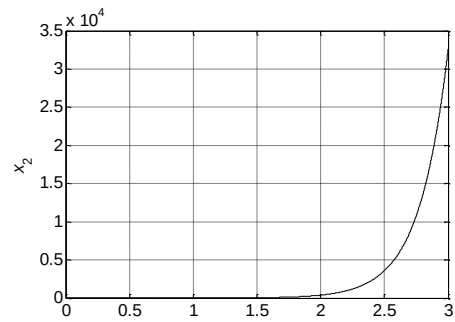
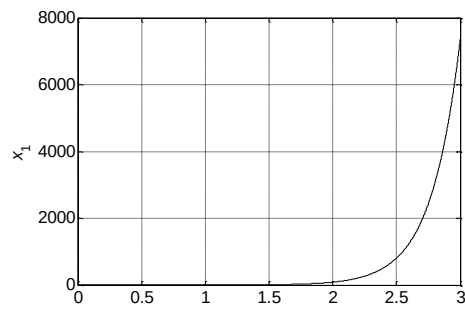
num/den =
    0.66667 s^2
    -----
    s^4 - 19.62 s^2

CharPoly =    1.0000        0   -19.6200        0        0

Eigenvalue    Damping    Freq. (rad/s)
  4.43e+000    -1.00e+000    4.43e+000
  0.00e+000    -1.00e+000    0.00e+000
  0.00e+000    -1.00e+000    0.00e+000
 -4.43e+000     1.00e+000    4.43e+000

```

Open-Loop Response



Controllability & Observability

$P =$

0	0.5	0	3.27
0.5	0	3.27	0
0	0.67	0	13.08
0.67	0	13.08	0

rank of $P = 4$; $n = 4$; System is fully state-controllable

$Q =$

0	0	1	0
0	0	0	1
0	0	19.62	0
0	0	0	19.62

rank of $Q = 2$; $n = 4$; System is NOT fully state-observable

Canonical Realizations

$T_c =$

-6.5400	0	0.5000	0
0	-6.5400	0	0.5000
0	0	0.6667	0
0	0	0	0.6667

$A_c =$

0	1.0000	0	0
0	0	1.0000	0
0	0	0	1.0000
0	0	19.6200	0

```
Bc =
      0
    0.0000
      0
    1.0000
```

```
Cc =      0      0    0.6667      0
```

```
Dc =      0
```

```
To =
      0    3.0000      0   -0.7500
    3.0000      0   -0.7500      0
      0   -0.1529      0    0.1147
   -0.1529      0    0.1147      0
```

```
Ao =
      0    0.0000      0    0.0000
    1.0000      0      0      0
      0    1.0000      0   19.6200
      0      0    1.0000      0
```

```
Bo =
      1
      0
      0
      0
```

```
Co =      0    0.6667      0   13.0800
```

```
Do =      0
```

Cannot find DCF since the A matrix is singular (The two zero poles cause this).

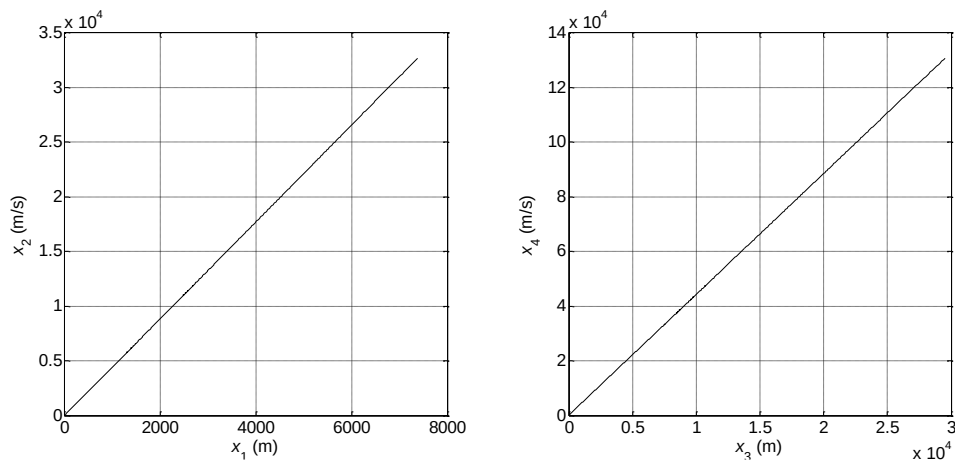
Stability Analysis

??? Error using ==> lyap
Solution does not exist or is not unique.

Use simple eigenvalue analysis instead:

```
poleso =
      0
      0
    4.4294
   -4.4294
```

Unstable! Phase portraits:

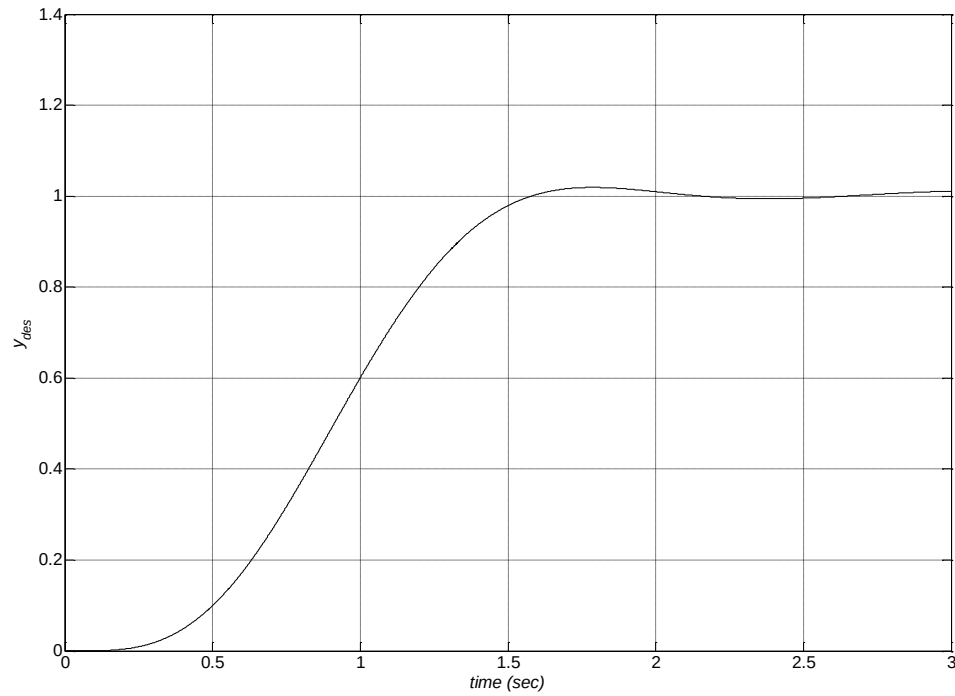


Dynamic Shaping

ITAE4 =
1.0000 6.3000 30.6000 72.9000 81.0000

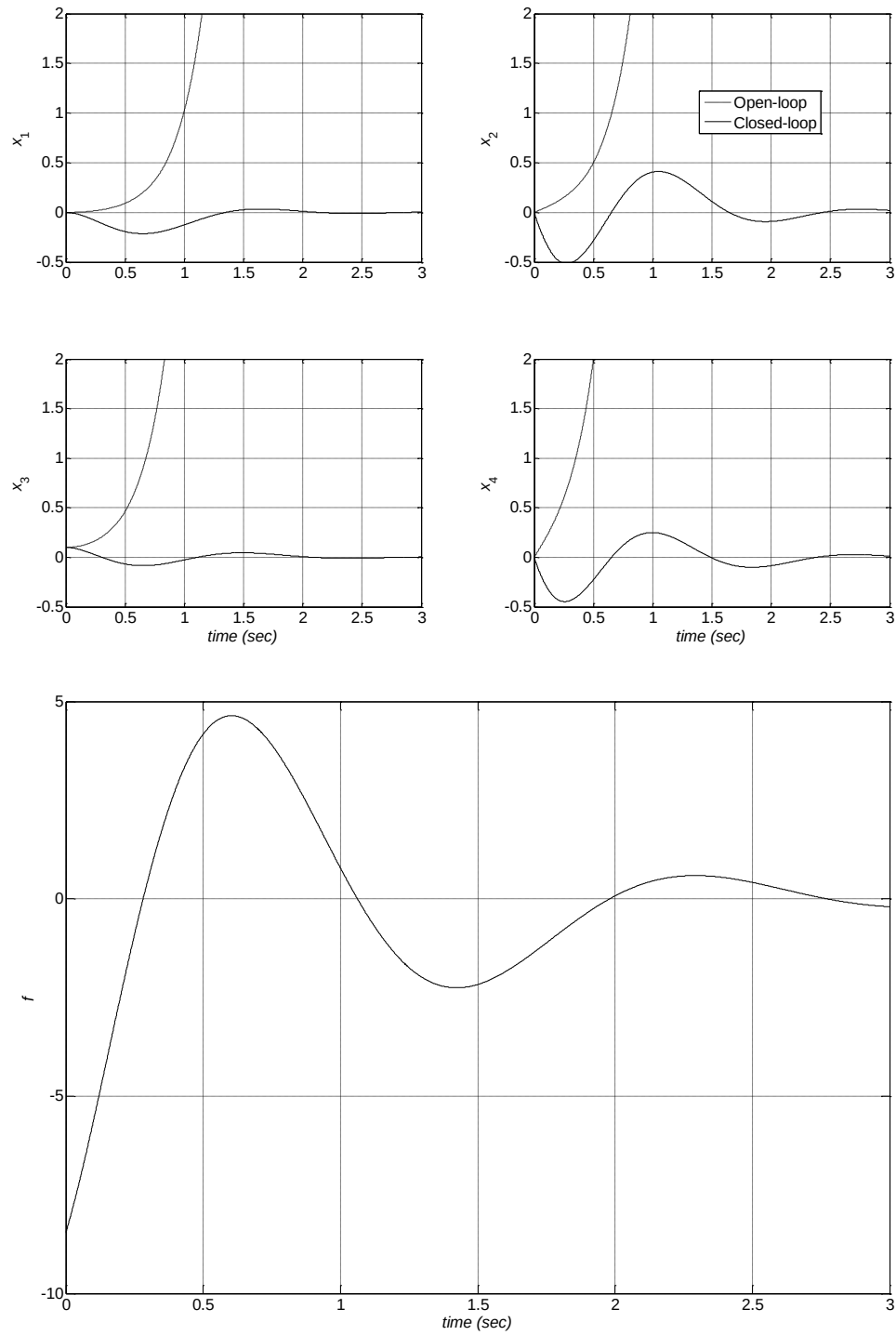
Poles4i =
-1.2719 + 3.7890i
-1.2719 - 3.7890i
-1.8781 + 1.2424i
-1.8781 - 1.2424i

Eigenvalue	Damping	Freq. (rad/s)
-1.27e+000 + 3.79e+000i	3.18e-001	4.00e+000
-1.27e+000 - 3.79e+000i	3.18e-001	4.00e+000
-1.88e+000 + 1.24e+000i	8.34e-001	2.25e+000
-1.88e+000 - 1.24e+000i	8.34e-001	2.25e+000



Controller Design

kay = -12.39 -11.15 84.62 17.81

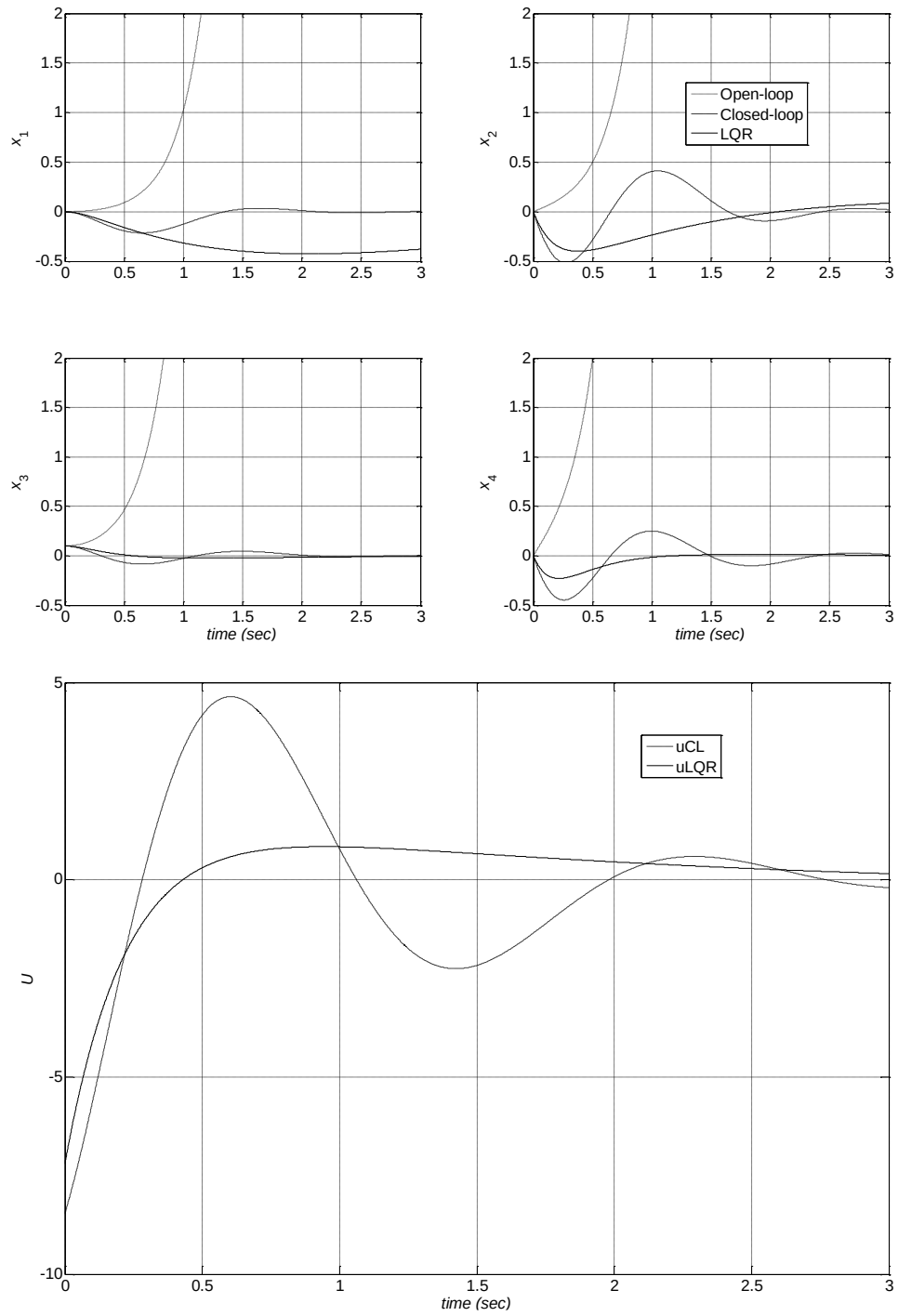


LQR Design

```

kayLQR =
    3.11      4.34    -17.01     -4.75
    4.34     11.77   -48.15    -13.49
   -17.01   -48.15   591.55    144.11
    -4.75   -13.49   144.11     35.63

```



Continuing Exercise 3 – Robot Joint/Link Control

- i. Single-input, single-output: input armature voltage $v_A(t)$ and output robot load angle $\theta_L(t)$.
 (a) Unit step input armature voltage $v_A(t)$; zero initial state.

Open-Loop System

```

A =
    0         1         0
    0         0         1
    0   -7507.53   -2333.35

B =
    0
    0
  4306.95

C =    1    0    0

D =    0

num/den =
           4306.9502
-----
s^3 + 2333.3502 s^2 + 7507.531 s

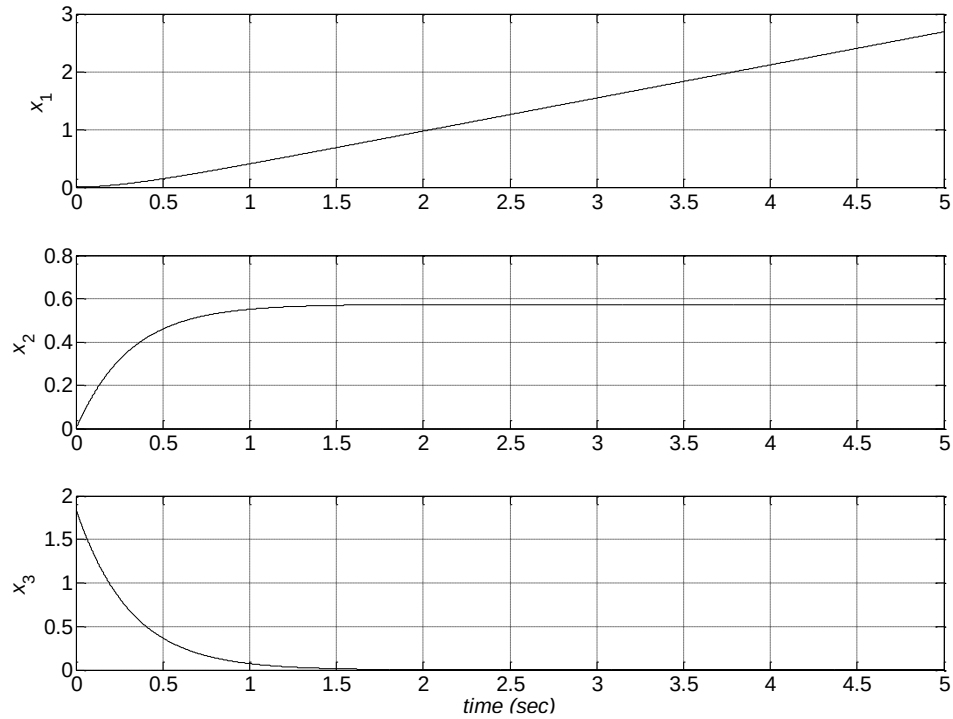
poles =
  1.0e+003 *
    0
   -2.3301
   -0.0032

CharPoly =    [1.0000    2333.3502    7507.531    0]

Eigenvalue      Damping      Freq. (rad/s)
  0.00e+000     -1.00e+000     0.00e+000
 -3.22e+000      1.00e+000     3.22e+000
 -2.33e+003      1.00e+000     2.33e+003

```

Open-Loop Response



Controllability & Observability

```
P =
    0          0      4306.95
    0      4306.95  -10049622.94
  4306.95 -10049622.94 23416954806.2
```

rank of P = 3; n = 3; System is fully state-controllable

```
Q =
  1  0  0
  0  1  0
  0  0  1
```

rank of Q = 3; n = 3; System is fully state-observable

Canonical Realizations

```
Tc =
  1.0e+003 *
    4.3070      0      0
      0    4.3070      0
      0      0    4.3070
```

```
Ac =
  1.0e+003 *
      0    0.0010      0
      0      0    0.0010
      0   -7.5075   -2.3334
```

```
Bc =
      0
      0
```

1

```
Cc =      1.0e+003 * [      4.3070      0      0]
```

```
Dc =      0
```

```
To =
      1.0e+006 *
      0.0000      0.0000      0.0000
      0.0000      0.0000     -0.0023
      0.0000     -0.0023      5.4370
```

```
Ao =
      1.0e+003 *
      0.0000      0.0000      0.0000
      0.0010      0.0000     -7.5075
      0.0000      0.0010     -2.3334
```

```
Bo =
      1.0e+003 *
      4.3070
      0.0000
      0.0000
```

```
Co =      0.0000      0.0000      1.0000
```

```
Do =      0
```

```
Td =
      1.0000     -0.0916      0.0000
           0      0.2952     -0.0004
           0     -0.9510      1.0000
```

```
Ad =
      1.0e+003 *
           0      0.0000      0.0000
           0     -0.0032      0.0000
           0           0     -2.3301
```

```
Bd =
      1.0e+003 *
      0.0006
      0.0063
      4.3129
```

```
Cd =      1.0000     -0.0916      0.0000
```

```
Dd =      0
```

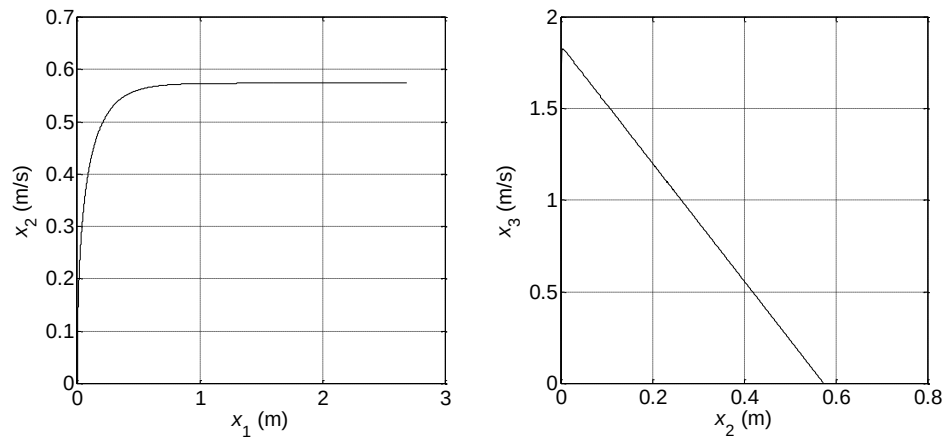
Stability Analysis

```
??? Error using ==> lyap
Solution does not exist or is not unique.
```

Use simple eigenvalue analysis instead:

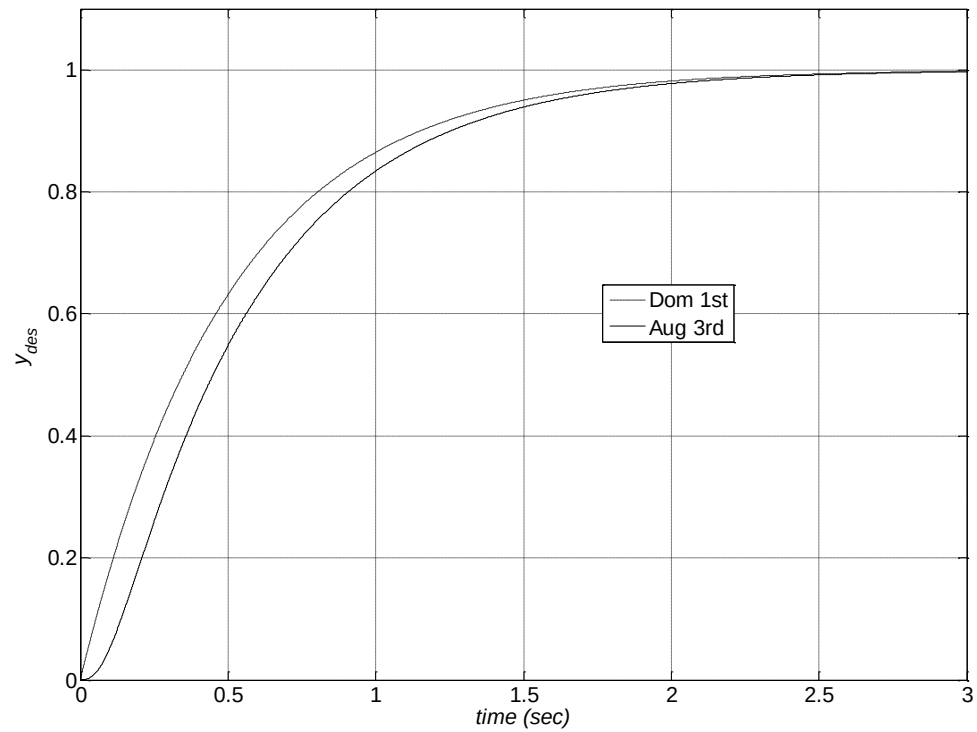
```
poleso =
      1.0e+003 *
           0
      -0.0032
      -2.3301
```

Marginally stable! Phase portraits:



Dynamic Shaping

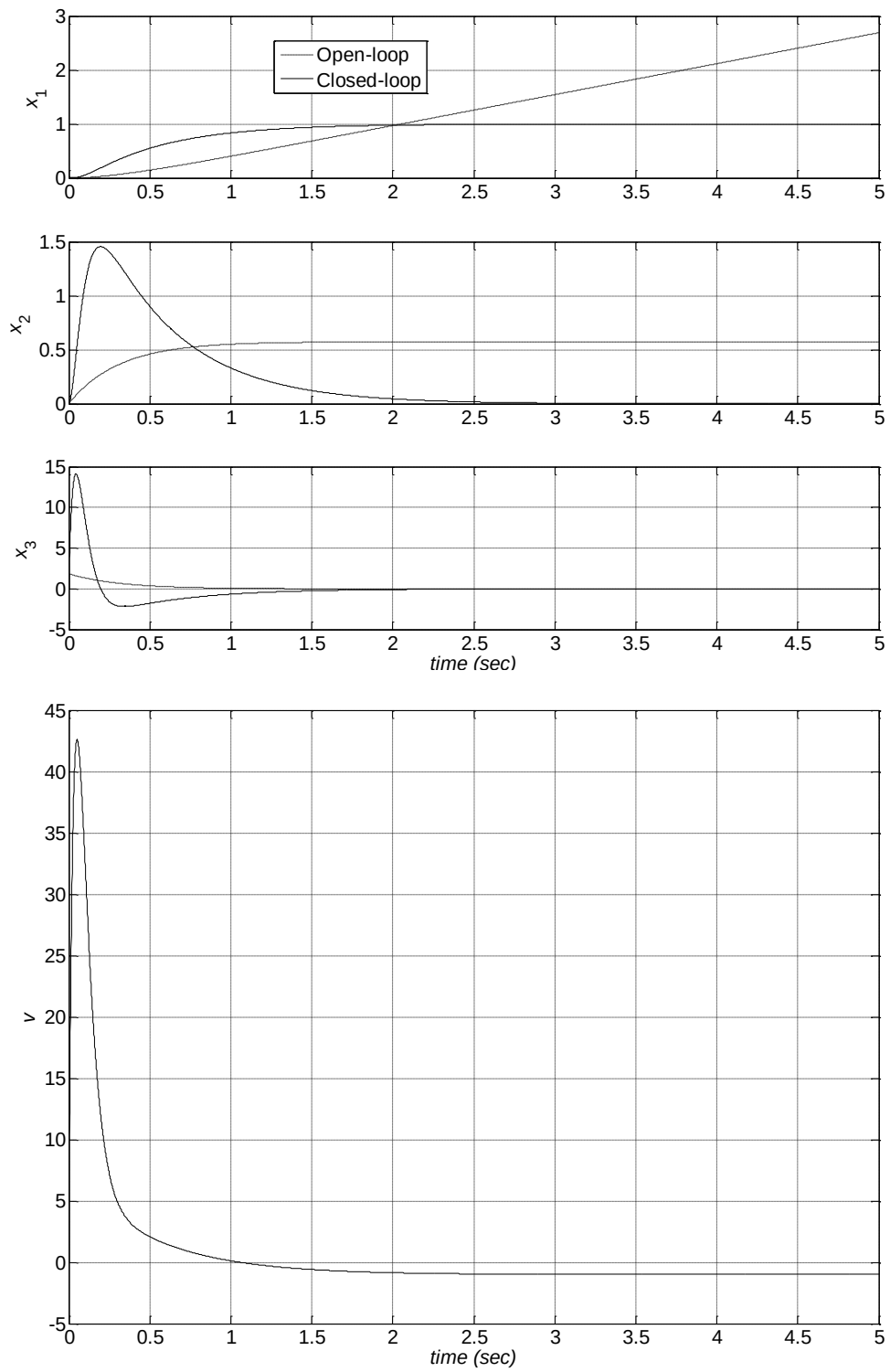
```
DomPole =      -2
Poles3 =      -2   -20   -21
```



Controller Design

K for augmented 3rd-order poles:

```
K =      0.2      -1.63      -0.53
```



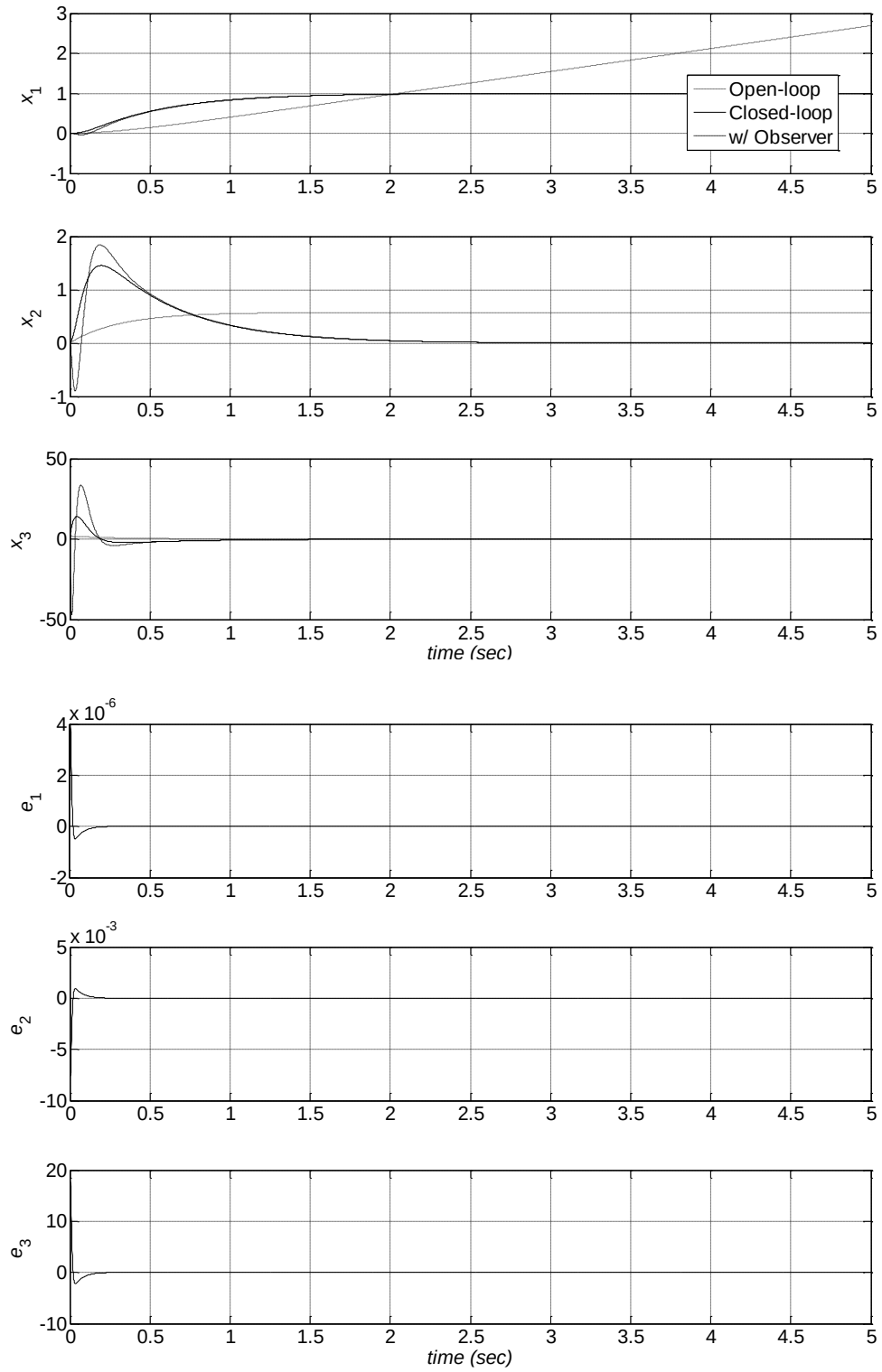
Observer Design

$L =$

```

-1903.35
 4483874.96
-10447320937.24

```



Continuing Exercise 4 – Ball/Beam System

Open-Loop System

```

A =
    0         1         0         0
    0         0    -7.01         0
    0         0         0         1
   -71.31      0         0         0

B =
    0
    0
    0
    8.03

C =    1    0    0    0

D =    0

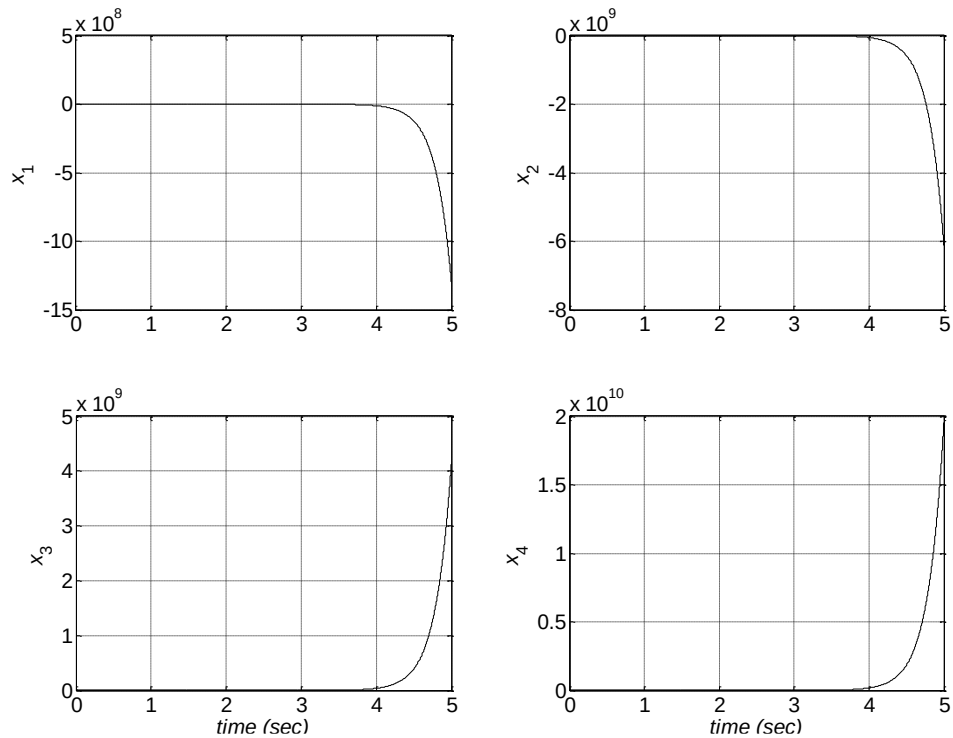
Transfer function:
   -56.28
   -----
  s^4 - 499.5

CharPoly =    [1.0000    0.0000    0.0000    0.0000 -499.5435]

Eigenvalue      Damping      Freq. (rad/s)
4.73e+000        -1.00e+000    4.73e+000
1.37e-015 + 4.73e+000i  -2.83e-016    4.73e+000
1.37e-015 - 4.73e+000i  -2.83e-016    4.73e+000
-4.73e+000        1.00e+000    4.73e+000

```

Open-Loop Response



Controllability & Observability

$P =$

0	0	0	-56.28
0	0	-56.28	0
0	8.03	0	0
8.03	0	0	0

rank of $P = 4$; $n = 4$; System is fully state-controllable

$Q =$

1	0	0	0
0	1	0	0
0	0	-7.01	0
0	0	0	-7.01

rank of $Q = 4$; $n = 4$; System is fully state-observable

Canonical Realizations

$T_c =$

-56.2797	0	0	0
0.0000	-56.2797	0	0
0.0000	0.0000	8.0337	0
0.0000	0.0000	0.0000	8.0337

$A_c =$

0.0000	1.0000	0	0
0.0000	0	1.0000	0
0.0000	0	0	1.0000
499.5435	0.0000	0.0000	0.0000

$B_c =$

```

0
0
0
1
Cc =    -56.2797      0      0      0
Dc =      0
To =
      0      0      0      1.0000
      0      0      1.0000      0.0000
      0    -0.1427      0.0000      0.0000
    -0.1427      0.0000      0.0000      0.0000
Ao =
    0.0000      0.0000      0.0000    499.5435
    1.0000      0      0      0.0000
      0      1.0000      0      0.0000
      0      0      1.0000      0.0000
Bo =
   -56.2797
      0
      0
      0
Co =      0      0      0      1
Do =      0
Tm =
    4.0391    -0.8544    -1.2660      0.2678
    0.0000     1.7087      0.0000      0.5356
   -8.0782      0.0000    -2.5320      0.0000
    4.0391     0.8544    -1.2660    -0.2678
Am =
   -4.7276      0      0      0
      0      0.0000      4.7276      0
      0    -4.7276      0.0000      0
      0      0      0      4.7276
Bm =
    2.1513
    4.3027
    0.0000
   -2.1513
Cm =      0.0619      0.0000    -0.0619      0.0619
Dm =      0

```

Stability Analysis

??? Error using ==> lyap
Solution does not exist or is not unique.

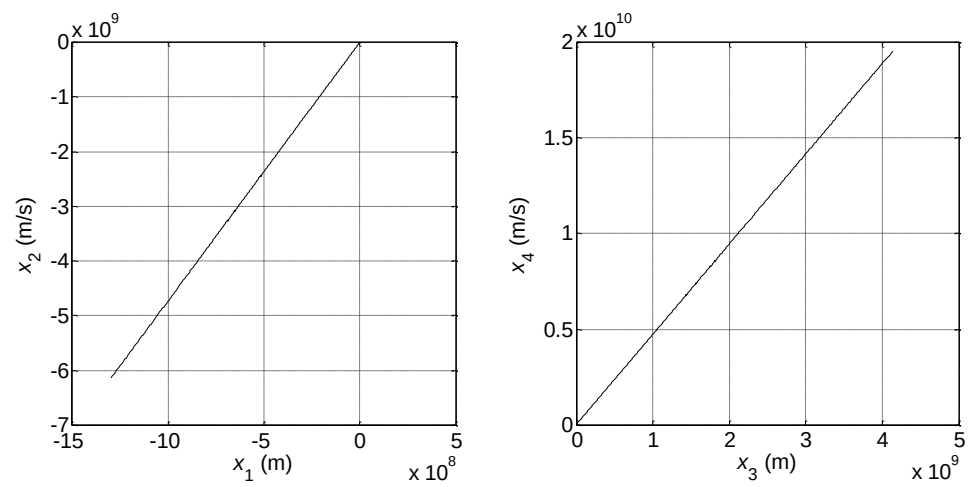
Use simple eigenvalue analysis instead:

```

poleso =
   -4.7276
    0.0000 + 4.7276i
    0.0000 - 4.7276i
     4.7276

```

Unstable! Phase portraits:

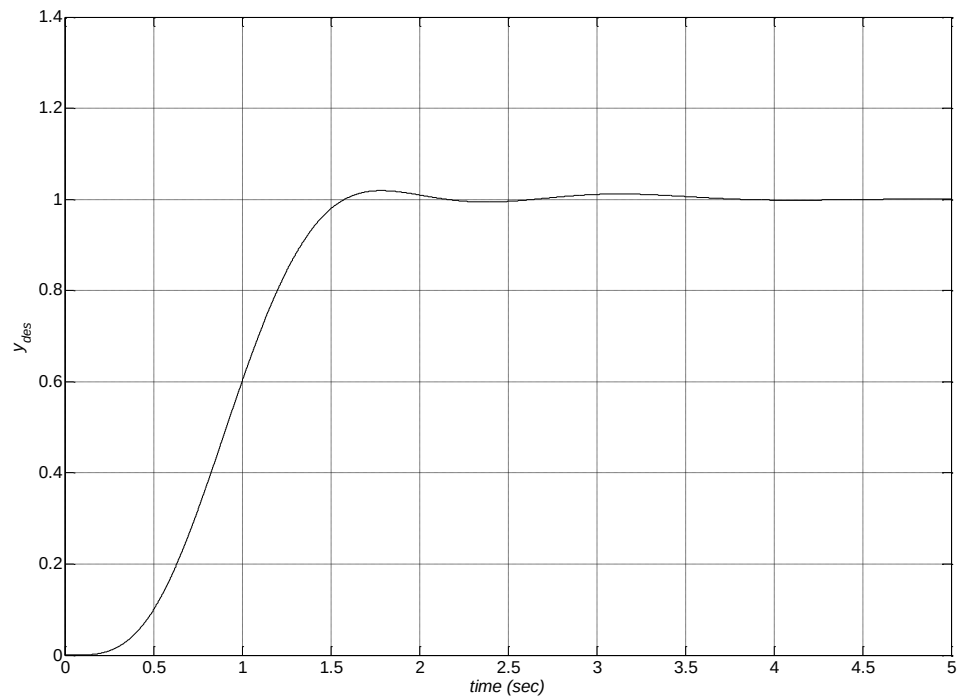


Dynamic Shaping

ITAE4 = 1.0000 6.3000 30.6000 72.9000 81.0000

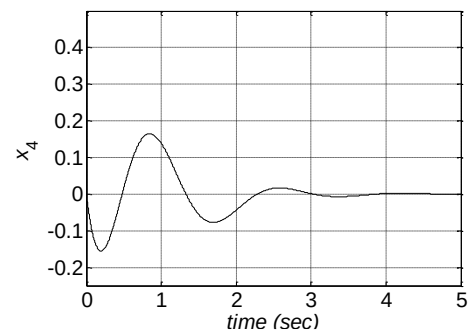
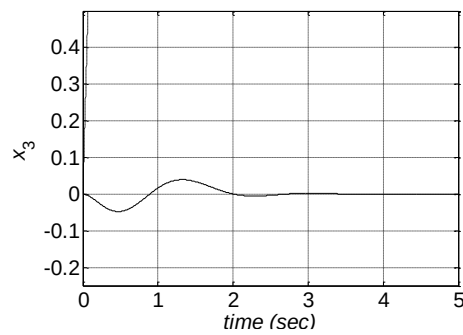
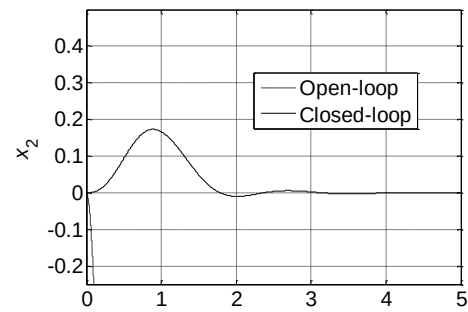
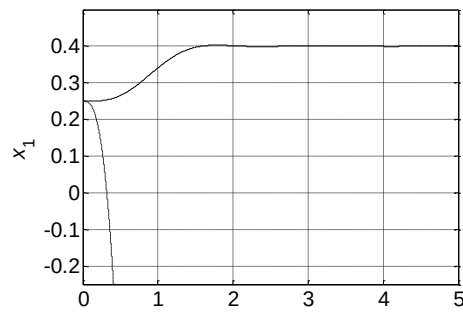
Poles4i =
-1.2719 + 3.7890i
-1.2719 - 3.7890i
-1.8781 + 1.2424i
-1.8781 - 1.2424i

Eigenvalue	Damping	Freq. (rad/s)
-1.27e+000 + 3.79e+000i	3.18e-001	4.00e+000
-1.27e+000 - 3.79e+000i	3.18e-001	4.00e+000
-1.88e+000 + 1.24e+000i	8.34e-001	2.25e+000
-1.88e+000 - 1.24e+000i	8.34e-001	2.25e+000



Controller Design

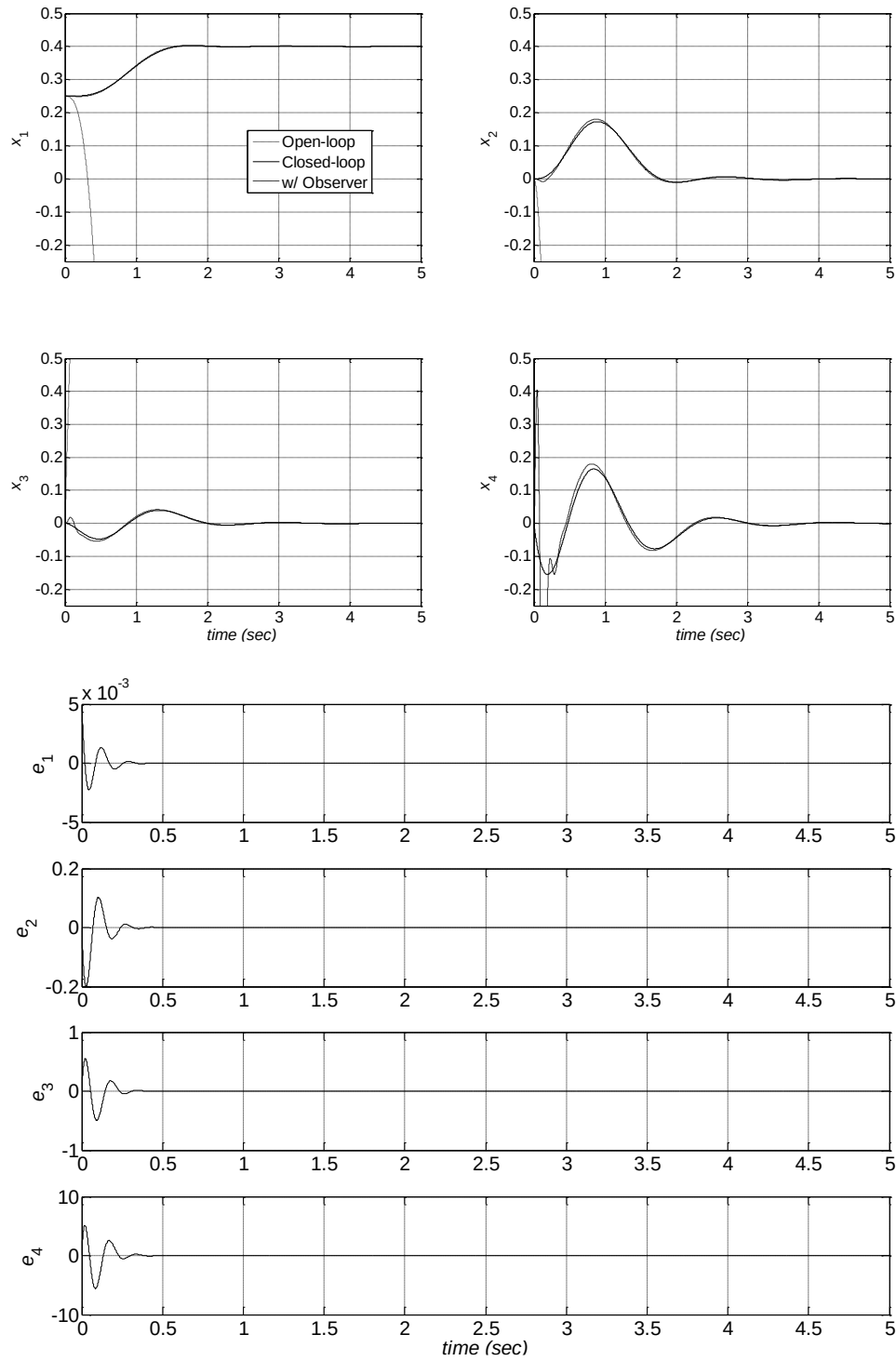
K = -10.32 -1.3 3.81 0.78



Observer Design

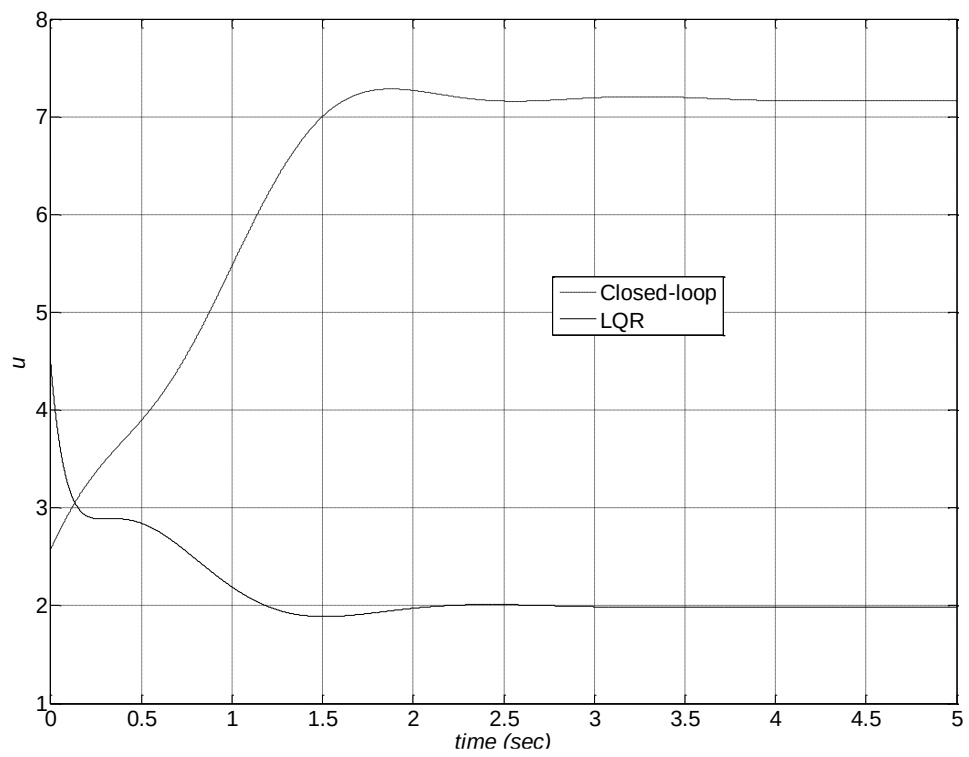
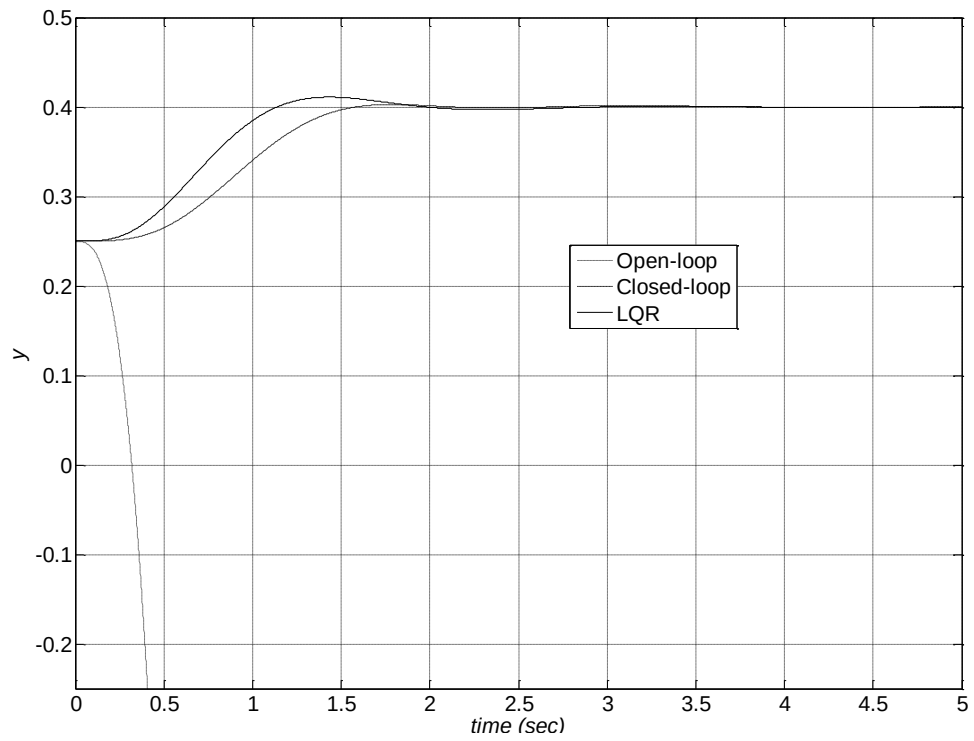
Observer Poles =
 -12.7195 +37.8898i
 -12.7195 -37.8898i
 -18.7805 +12.4242i
 -18.7805 -12.4242i

L =
 63
 3060
 -10406.15
 -115695.22



LQR Design

$K_{LQR} =$			
52.4	15.99	-20.3	-2.22
15.99	7.47	-10.94	-0.73
-20.3	-10.94	23.07	1.55
-2.22	-0.73	1.55	0.28



Continuing Exercise 5 – Proof-Mass Actuator System

Open-Loop System

```

A =
    0         1         0         0
  -130.49      0         0         0
    0         0         0         1
   946.68      0         0         0

B =
    0
   -5.08
    0
  1842.1

C =    1    0    0    0

D =    0

poleso =
    0
    0
    0 +11.4234i
    0 -11.4234i

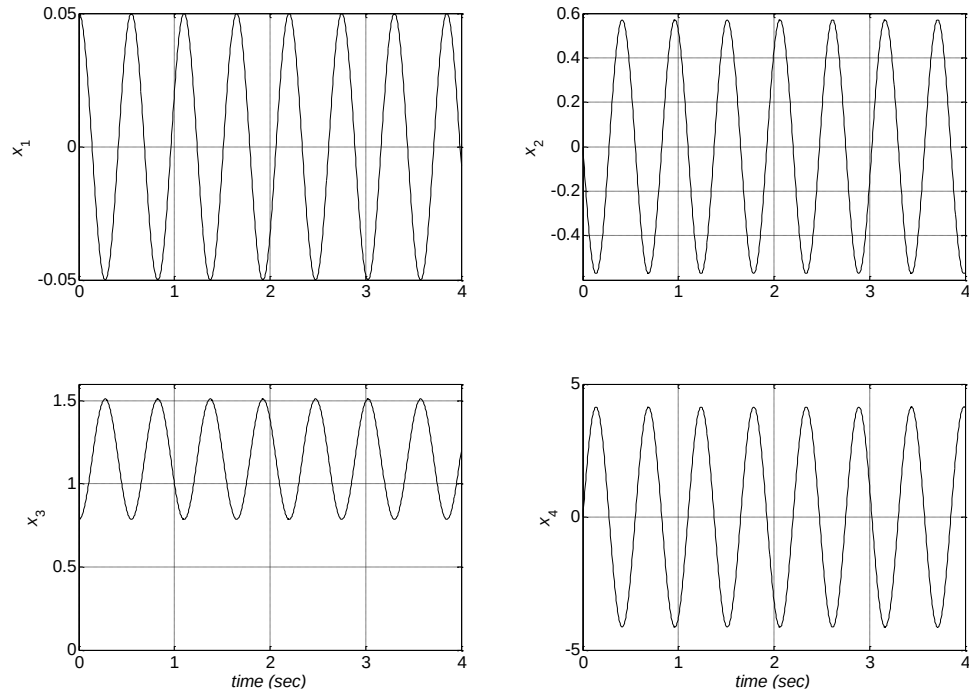
Transfer function:
   -5.081
  -----
 s^2 + 130.5

CharPoly =    [1.0000         0   130.4945         0         0]

      Eigenvalue      Damping      Freq. (rad/s)
0.00e+000 + 1.14e+001i  -6.12e-017   1.14e+001
0.00e+000 - 1.14e+001i  -6.12e-017   1.14e+001

```


Open-Loop Response



Controllability & Observability

P =

0	-5.08	0	663.11
-5.08	0	663.11	0
0	1842.1	0	-4810.53
1842.1	0	-4810.53	0

rank of P = 4; n = 4; System is fully state-controllable

Q =

1	0	0	0
0	1	0	0
-130.49	0	0	0
0	-130.49	0	0

rank of Q = 2; n = 4; System is NOT fully state-observable

Canonical Realizations

Tc =

1.0e+005 *	0	0	-0.0001	0
	0	0	0	-0.0001
2.3557	0	0.0184	0	0
0	2.3557	0	0.0184	0

Ac =

0	1.0000	0	0
0	0	1.0000	0
0	0	0	1.0000
0	0	-130.4945	0

Bc =

```

0
0
0
1
Cc =      0      0  -5.0815      0
Dc =      0
To =
      0      0.0040      0      0.0006
0.0040      0      0.0006      0
      0      0.0015      0      0.0000
0.0015      0      0.0000      0
Ao =
      0      0      0      0
1.0000      0      0      0
      0      1.0000      0 -130.4945
      0      0      1.0000      0
Bo =
1.0000
      0
0.0000
      0
Co =      0  -5.0815      0  663.1051
Do =      0

```

Cannot find DCF since the A matrix is singular (The two zero poles cause this).

Minimality

Minimal realization:

2 states removed.

a =

```

      x1      x2
x1      0      1
x2 -130.49      0

```

b =

```

      u1
x1      0
x2 -5.0815

```

c =

```

      x1      x2
y1      1      0

```

d =

```

      u1
y1      0

```

Stability Analysis

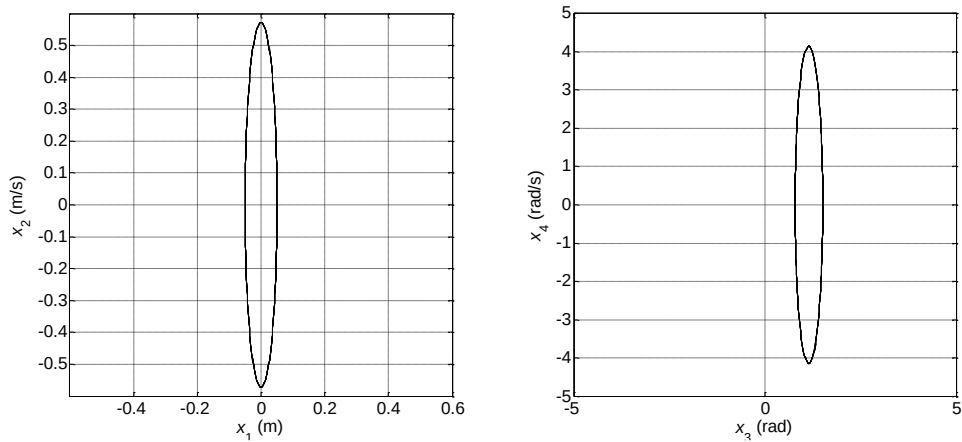
??? Error using ==> lyap

Solution does not exist or is not unique.

Use simple eigenvalue analysis instead:

```
poleso =  
    0  
    0  
    0 +11.4234i  
    0 -11.4234i
```

Marginally stable! Phase portraits:

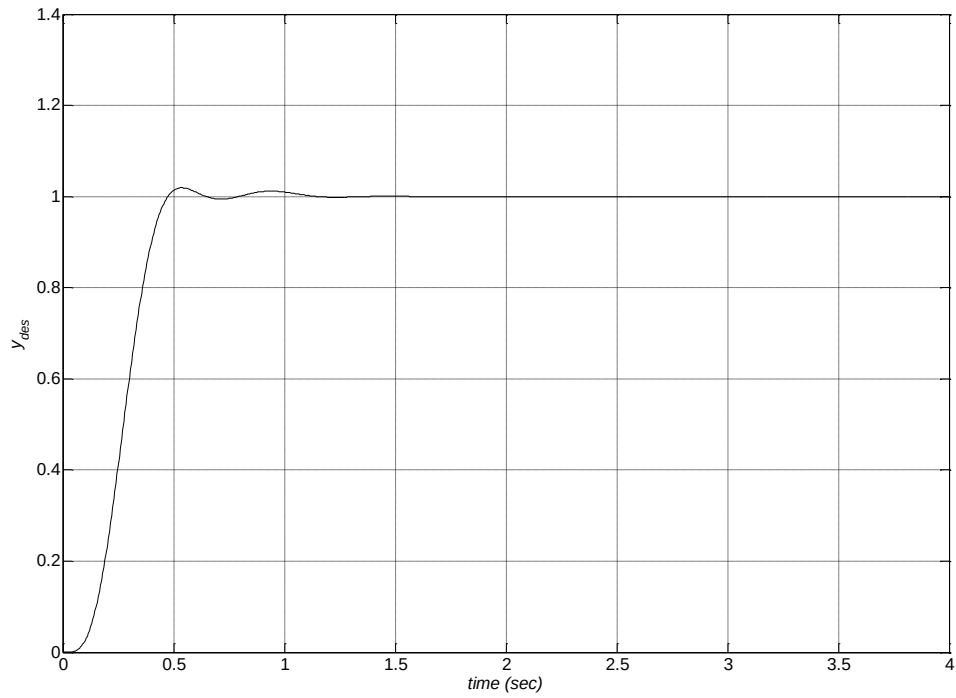


Dynamic Shaping

```
ITAE4 =      1          21          340          2700          10000
```

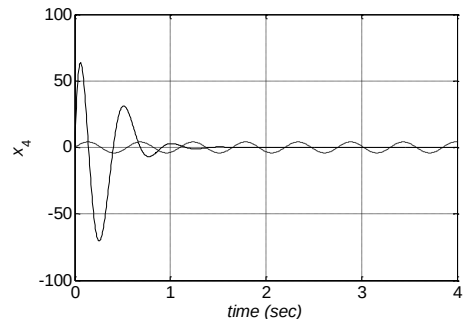
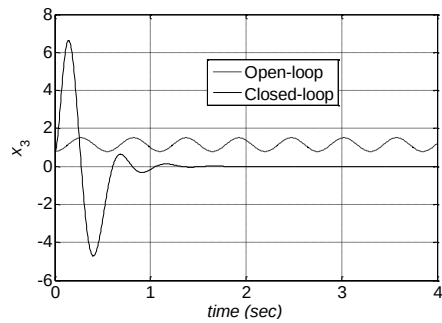
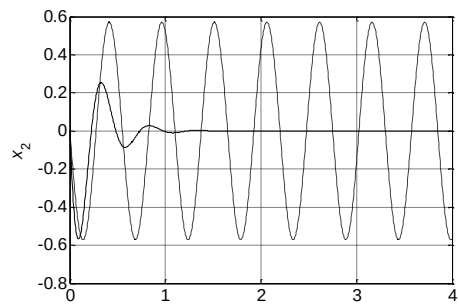
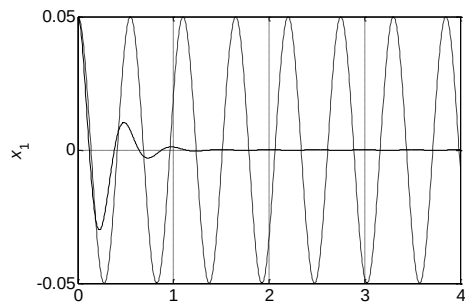
```
Poles4i =  
-4.2398 +12.6299i  
-4.2398 -12.6299i  
-6.2602 + 4.1414i  
-6.2602 - 4.1414i
```

Eigenvalue	Damping	Freq. (rad/s)
-4.24e+000 + 1.26e+001i	3.18e-001	1.33e+001
-4.24e+000 - 1.26e+001i	3.18e-001	1.33e+001
-6.26e+000 + 4.14e+000i	8.34e-001	7.51e+000
-6.26e+000 - 4.14e+000i	8.34e-001	7.51e+000



Controller Design

$K =$ -25.84 0.02 0.04 0.01



Observer Design

Observer Poles =
 $1.0e+002 * \begin{bmatrix} -0.4240 + 1.2630i \\ -0.4240 - 1.2630i \\ -0.6260 + 0.4141i \\ -0.6260 - 0.4141i \end{bmatrix}$

L =
 $\begin{bmatrix} 104.94 & 84.21 \\ 7783.88 & 6102.76 \\ -85.57 & 105.06 \\ -5252.17 & 7855.32 \end{bmatrix}$

